

Доказательное исследование и DSS для проектирования игр

Игровые функции, методы реализации, движки, зависимости, оборудование и календарный план

Выпуск: 2026-09-11. Ревизия: e04a4c3. Формат: локальная PC-модель с прозрачными claims и сценарными диапазонами.

Принцип: ни одно число не публикуется как измерение без опубликованного источника с локатором или явной формулы с входными параметрами. Отсутствие данных даёт статус unknown, а не ноль и не «совместимо».

1. Область применимости и ограничения

Система — локальное PC-ориентированное приложение поддержки инженерных решений. Оно связывает игровые функции, варианты реализации, движки и инструменты, конфликты, технологические зависимости, цели качества, нагрузочный профиль, референсный класс оборудования и план работ. Исследование универсальное: реальные игры используются как проверяемые кейсы механизма, но не как паспорта производительности для чужого проекта.

Количественная модель ограничена Windows PC и Linux PC. Целевой FPS, разрешение, quality, RAM/VRAM, streaming pool, draw-call budget, simulation radius, physics tick, сетевой latency/tick и storage задаются профилем. Если runtime-профиля проекта нет, система не обещает точный FPS, точную latency или календарную дату.

Вне текущего результата: прямое подключение к Unreal Insights/Unity Profiler, импорт runtime-метрик, денежный бюджет, multi-user авторизация и автоматическое изменение проекта движка.

1.1 Что система утверждает и чего не утверждает

Утверждение	Статус	Основание
Механизм метода подтверждён	documented / case_evidence	Документация, книга, спецификация или материал проекта с локатором
Число является измерением	measured	Опубликованное измерение с условиями и контекстом
Число вычислено из входов	derived	Явная формула и сохранённые входные параметры
Сценарный балл для ранжирования	expert_estimate	Помечается как сценарный, не как измерение
Точный FPS/latency/дата чужого проекта	не утверждается	Требуется runtime-профиль и трассы проекта
Совместимость без источника	не утверждается	Даёт `unknown`, а не `supported`

2. Методология источников

Источник — любой проверяемый носитель, где можно установить механизм, ограничение, числовой результат или контекст. Реестр не ограничен статьями или интервью: книги, отдельные издания, стандарты, спецификации, академические статьи, материалы конференций и видео, инженерные postmortem, исходный код, открытые проекты, бенчмарки, учебники и технические анализы допустимы.

Книга хранится с автором, издателем, изданием и главой. Спецификация — с версией и разделом. Видео или доклад — с датой и timestamp/слайдами. Исходный код — с репозиторием, commit/tag и путём. Для любого типа URL сам по себе недостаточен: claim обязан иметь локатор.

2.2 Типы источников в реестре

Тип источника	Записей
official_documentation	317
secondary	311
hardware_benchmark	129
conference_talk	39
open_source	33
book	26
engineering_blog	21
vendor_press_release	14
engineering_article	7
interview	5
api_specification	5
academic_paper	5
standard	4
studio_engineering	2
research	2
benchmark	2
technical_report	1
technical_guidance	1
studio_information	1
studio_engineering_article	1
practice_standard	1
postmortem	1
official_game_material	1
official_developer_resource	1
official_case_study	1
official_case_index	1
engineering_talk	1
conference_paper	1
benchmark_methodology	1

2.3 Основание (basis) утверждений

basis	Claims
documented	1580
unknown	262
case_evidence	244
derived	208
expert_estimate	164
measured	104

- measured — опубликовано измерение с условиями;
- documented — механизм или ограничение прямо указаны в материале;
- derived — значение вычислено из явно сохранённых входов и формулы;
- case_evidence — подтверждено материалом реального проекта;
- expert_estimate — сценарный балл для ранжирования или планирования;
- unknown — данных недостаточно.

2.4 Методы с наименьшим числом claims

Метод	Claims
path_tracing_sample_denoiser_budget	6
agent_update_budget	7
behaviour_tree_update_budget	7
crowd_instancing_impostors	7
dynamic_light_priority_budget	7
dynamic_resolution_scaling	7
froxel_volumetric_fog	7
full_path_tracing_pipeline	7
gpu_procedural_placement	7
light_range_attenuation_lod	7
ml_frame_generation	7
npc_perception_budget	7
post_effect_selective	7
screen_space_gi	7
splitscreen_render_budget	7

2.5 Сила доказательства: прямой пример, перекрёстный пример, объявленный пробел

Класс доказательства	Сущностей	Комментарий
Инструмент движка: прямой пример (тот же движок)	52	из 70: shipped-тайтл на той же платформе, что и инструмент
Инструмент движка: только перекрёстный пример	18	возможность подтверждена на другом движке, adoption не подтверждён
Инструмент движка: adoption не подтверждён, пробел объявлен	21	поле adoption_evidence_gap, basis=unknown — пробел виден, а не заполнен
Технологические узлы с объявленным пробелом	4	исследовательские техники без shipped-тайтла
Методы с объявленным пробелом	2	нет второго независимого shipped-примера

Покрытие «70 из 70» по игровым примерам означает лишь то, что к каждому инструменту приложен хотя бы один shipped-тайтл, где эта возможность работает. Это более слабое утверждение, чем «инструмент применён в показанной игре». Разделение ведётся явно: • прямой пример — тайтл на том же движке, что и инструмент (52 сущностей);

- перекрёстный пример — тайтл на другом движке: подтверждает возможность, но не adoption (18 сущностей);
- объявленный пробел — shipped-подтверждения нет, и это зафиксировано полем adoption_evidence_gap с basis=unknown (21 сущностей). Сущностей, у которых нет прямого примера и при этом пробел не объявлен: 0. Это целевое значение — ноль: молчаливая дыра недопустима, потому что она indistinguishable от проверенного факта.

3. Классификация игровых функций

3.1 Категории и состав

Категория	Функций
Рендер	7
Освещение	6
Мир и загрузка	5
Симуляция	5
Технические подсистемы	4
Визуальные эффекты	3
Персонажи	2
Производство	2
Геймплей	1
Звук	1
Прогресс и данные	1
Производительность	1
Производство контента	1
Сеть	1

3.2 Полный перечень функций

Код	Название	Категория	Форматы	Типы мира	Опорный источник
open_world_streaming	Потоковая загрузка открытого мира	Мир и загрузка	["3D", "2.5D", "2D"]	["open_world", "procedural", "sandbox"]	Unreal Engine: World Partition
large_scale_terrain	Крупномасштабный ландшафт	Мир и загрузка	["3D", "2.5D"]	["open_world", "procedural", "sandbox", "hub"]	Unreal Engine: Nanite Virtualized Geometry
procedural_terrain	Процедурная генерация ландшафта и мира	Мир и загрузка	["3D", "2.5D", "2D"]	["procedural", "open_world", "sandbox"]	Procedural generation
procedural_vegetation	Растительность и объекты окружения	Мир и загрузка	["3D", "2.5D", "2D"]	["open_world", "procedural", "sandbox", "linear"]	Unreal Engine: Instanced Static Mesh
storage_streaming	Потоковая работа с накопителем	Мир и загрузка	["3D", "2.5D", "2D"]	["open_world", "procedural", "sandbox", "linear", "hub"]	Unity Manual: Texture Streaming
dynamic_global_illumination	Динамическое глобальное освещение	Освещение	["3D", "2.5D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Unreal Engine: Lumen Global Illumination and Reflections
path_tracing	Трассировка пути	Освещение	["3D"]	["linear", "hub", "arena", "open_world"]	Path tracing
ray_traced_effects	Трассировочные эффекты (тени, отражения, AO)	Освещение	["3D"]	["linear", "hub", "arena", "open_world"]	Ray tracing (graphics)
dynamic_lighting	Множественные динамические источники света	Освещение	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "sandbox"]	Global illumination
baked_lighting	Запечённое освещение	Освещение	["3D", "2.5D", "2D"]	["linear", "hub", "arena"]	Lightmap
dynamic_shadows	Динамические тени	Освещение	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Shadow mapping
particle_systems	Системы частиц	Визуальные эффекты	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Particle system
physics_simulation	Физическая симуляция	Симуляция	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Unreal Engine: Chaos Physics
vehicle_simulation	Транспорт и физика движения	Симуляция	["3D", "2.5D"]	["open_world", "arena", "sandbox", "linear"]	Vehicle dynamics
character_animation	Анимация персонажей	Персонажи	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Skinning
gameplay_ability_system	Геймплейные способности и модификаторы	Геймплей	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "sandbox"]	Game Programming Patterns: State
crowd_simulation	Толпы NPC	Персонажи	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "sandbox"]	Unreal Engine: Mass Entity
ai_pathfinding	ИИ и поиск пути	Симуляция	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Navigation mesh
advanced_npc_ai	Поведенческий ИИ NPC	Симуляция	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "sandbox"]	Video game artificial intelligence
water_simulation	Водные поверхности	Визуальные эффекты	["3D", "2.5D", "2D"]	["open_world", "procedural", "sandbox", "linear"]	Gerstner wave
volumetric_effects	Объёмные эффекты (туман, облака, дымка)	Визуальные эффекты	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Volumetric rendering
post_processing	Постобработка изображения	Рендер	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Temporal anti-aliasing
rendering_architecture	Архитектура рендера	Рендер	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Deferred shading

Код	Название	Категория	Форматы	Типы мира	Опорный источник
render_scalability	Масштабирование качества	Рендер	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Unreal Engine: Scalability Reference
geometry_pipeline	Конвейер геометрии	Рендер	["3D", "2.5D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	meshoptimizer: mesh optimization library
upscaling_frame_generation	Масштабирование и генерация кадров	Рендер	["3D", "2.5D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Deep learning super sampling
mesh_shaders	Меш-шейдеры и генерация геометрии на GPU	Рендер	["3D", "2.5D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	DirectX mesh shader specification
multiplayer_netcode	Сетевой код мультиплеера	Сеть	["3D", "2.5D", "2D"]	["open_world", "arena", "hub", "sandbox", "procedural"]	Client-side prediction
save_system	Система сохранений	Прогресс и данные	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Unreal Engine: Saving and Loading Your Game
audio_system	Аудиосистема	Звук	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Hunt: Showdown — Audio readability, realism and consistency
build_delivery	Сборка и доставка контента	Производство	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Unity Manual: Addressables
runtime_memory	Память и сборка мусора	Производительность	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Unity Manual: Garbage collection best practices
destruction_simulation	Разрушения и геометрические кэши	Симуляция	["3D", "2.5D"]	["open_world", "linear", "hub", "arena", "sandbox"]	Rendering the Hellscape of Doom Eternal (SIGGRAPH 2020)
project_architecture	Архитектура проекта	Производство	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Data-oriented design
art_pipeline	Арт-пайплайн	Производство контента	["3D", "2.5D", "2D"]	["open_world", "linear", "hub", "arena", "procedural", "sandbox"]	Lightmap
split_screen_rendering	Split-screen рендеринг	Рендер	["3D", "2.5D"]	["linear", "hub", "arena", "sandbox"]	It Takes Two tech analysis (Digital Foundry)
portal_rendering	Дополнительные виды и порталы	Технические подсистемы	["3D", "2.5D", "2D"]	[]	Epic: Scene Capture
hair_rendering	Волосы: пряди или карточки	Технические подсистемы	["3D", "2.5D", "2D"]	[]	Epic: Hair Rendering and Simulation
cloth_simulation	Симуляция ткани	Технические подсистемы	["3D", "2.5D", "2D"]	[]	Unity 6: Cloth
runtime_security	Проверки целостности и античит	Технические подсистемы	["3D", "2.5D", "2D"]	[]	Epic: Anti-Cheat Interfaces

4. Варианты реализации по подсистемам

4.1 Подсистемы и что измерять

Подсистема	Варианты	Что измерять/проверять	Опорные источники
World and streaming	World Partition / HLOD / tile streaming / procedural generation	active cells, geometry, IO, decompression, memory and traversal hitches	S01, S18
Rendering	Nanite or conventional geometry, raster, RT, VRS, upscaling, frame generation	main thread, draw/mesh preparation, raster/RT, VRAM and quality trade-offs	S02-S06, S19-S21
Simulation and AI	fixed-step physics, ECS/jobs, relevance, budgets, director	simulation tick, parallel work, memory layout and content authoring	S07-S09, S11, S18, S20
Networking	authoritative client/server, replication relevance, delta compression, sub-tick	server tick, serialization, bandwidth, latency and divergence validation	S12-S14, S22
Audio	occlusion, material filtering, HRTF/spatial audio, mix readability	rays/queries, CPU, middleware cost, device variability and perception	S17, S18
Build and delivery	asset bundles, patching, compression, cache and storage streaming	cook time, disk space, IO latency and release regression	S01, S07, S18

4.2 Implementation и optimization по категориям

Категория	Implementation	Optimization	Всего
Освещение	15	7	22
Мир и загрузка	11	9	20
Рендер	5	14	19
Симуляция	10	6	16
Персонажи	6	5	11
Визуальные эффекты	4	5	9
Сеть	5	3	8
Технические подсистемы	6	0	6
Звук	1	2	3
Производство	0	3	3
Геймплей	2	0	2
Прогресс и данные	2	0	2
Производство контента	0	2	2
Производительность	0	1	1

В каталоге 67 implementation и 57 optimization методов. Модель не выбирает один «лучший движок» глобально: TOPSIS сравнивает сопоставимые альтернативы внутри функции, а применимость проверяется до ранжирования. Уверенность оценки — отдельный критерий и не является процентом ускорения.

5. Сравнение движков и инструментов

Движок/подход	Инструменты и механизмы	Проверка применимости	Доказательства
Unreal	World Partition, Nanite, Lumen, VSM, Mass, Replication Graph	Версия, RHI, SM6/RT, streaming range and asset compatibility require a prototype.	S01-S04
Unity	Job System, Entities/DOTS, Addressables, Netcode	Package versions and migration boundary are part of the dependency contract.	S07-S09
Godot	scene/visibility optimization, high-level multiplayer, custom scripts	Open-source flexibility does not remove project-specific profiling.	S10
CryEngine	renderer, audio and streaming capabilities	The case evidence is stronger for audio than for a universal hardware budget.	S17
Source / Source 2	network simulation, tick/sub-tick, established multiplayer workflows	Valve case materials document mechanisms, not a transferable project estimate.	S11-S14
HeroEngine	shared-world/editor-oriented workflows from the existing catalog	Public evidence and current-version compatibility must be reviewed per tool.	catalog claims
Custom	own runtime, network, renderer or service	Every mechanism needs a project-owned design record and measurement plan.	expert boundary

6. Версия и платформенная совместимость

Платформа / ветка	Что доступно	Ограничение проверки	Источники
Windows PC (D3D12, SM6)	Nanite, Lumen HW path, VSM, VRS, mesh shaders, RT tiers	D3D12 feature level задаёт функциональность, но не производительность; опции проверяются через CheckFeatureSupport	S04, S38, S39, S40
Windows PC (D3D12, SM5)	часть методов недоступна или требует fallback	Отсутствие возможности не считается совместимостью: метод либо помечается недоступным, либо требует явного fallback-плана	S04
Windows PC (Vulkan)	RT, mesh shaders, VRS через расширения Vulkan	Расширения и очереди объявляются при создании устройства; нужен отдельный capability-профиль	S22
Linux PC (Vulkan)	RT и compute-пути, отличные от D3D12-ветки	Количественная граница модели ограничена Windows/Linux PC; результаты не переносятся на консоли и мобильные	S22
Non-PC (console/mobile/cloud)	остаются в профиле для полноты	Не получают совместимый аппаратный ориентир; помечаются `not_modeled`, а не приближённой оценкой	expert boundary
Server / dedicated	replication, tick, relevance, serialization	Server-only стоимость не уменьшает требования client PC и не суммируется с ними	S26, S27

Поддержка движка — не бинарный engine=yes. Она включает version, package/plugin, target platform/API, scope (runtime, editor, build, server, development), tool relation и свежесть доказательства. HeroEngine оставлен в каталоге из-за фиксированного охвата, но его public applicability — legacy/needs_review, а не подтверждённая совместимость.

7. Зависимости, конфликты и альтернативы

Граф состоит из методов, движков, инструментов, API, SDK и библиотек. Ребро направлено и хранит mandatory, min/max version, platform, scope, severity, источник, описание и workaround.

7.1 Типы связей между решениями

Тип связи	Записей
complement	183
dependency	80
risk	75
overlap	67
alternative	41
unknown	8
hard_conflict	4

7.2 Состав узлов графа

Тип узла	Узлов
method	124
tool	70
plugin	8
engine	7
lib	6
api	4
sdk	2

7.3 Обязательные и условные рёбра (выборка)

Источник	Цель	Тип	Обязательность	Мин. версия	Платформа	Severity	Источник
method:cloth_constraint_simulation	method:fixed_timestep_physics	dependency	обязательная	—	—	3	Physics Sub-Stepping in Unreal Engine
method:runtime_fracture_budget	method:physics_lod_sleeping	dependency	обязательная	—	—	3	Godot Engine - RigidBody3D class reference
method:runtime_fracture_budget	method:broadphase_spatial_partitioning	dependency	обязательная	—	—	3	GPU Gems 3, Chapter 32: Broad-Phase Collision Detection with CUDA
method:flipbook_particles	method:sprite_particle_atlas	dependency	обязательная	—	—	3	Niagara Flipbook Baker Quick Start Guide in Unreal Engine
method:fixed_timestep_physics	method:raycast_vehicle_physics	dependency	обязательная	—	—	3	Godot Engine - VehicleBody3D class reference
method:multithreaded_physics_jobs	method:fixed_timestep_physics	dependency	обязательная	—	—	3	Physics Sub-Stepping in Unreal Engine
method:composition_bootstrap_architecture	method:ecs_data_oriented_crowd	dependency	обязательная	—	—	3	Unity DOTS product page
method:vehicle_simulation_lod	method:physics_lod_sleeping	dependency	обязательная	—	—	3	Godot Engine - RigidBody3D class reference
method:animation_compression	method:motion_matching	dependency	обязательная	—	—	3	Learned Motion Matching (project page + abstract)
method:animation_lod_budget	method:gpu_skinning_compute	dependency	обязательная	—	—	3	нет публичного источника
method:motion_matching	method:animation_lod_budget	dependency	обязательная	—	—	3	нет публичного источника
method:normal_bake_retotology_pipeline	method:gpu_skinning_compute	dependency	обязательная	—	—	3	Skeletal Mesh Rendering Paths in Unreal Engine (Unreal Engine 5.8 Documentation)
method:art_direction_stylization	method:ability_visual_effect_budget	dependency	обязательная	—	—	3	Scalability and Best Practices for Niagara (Unreal Engine 5.8 Documentation)
method:tilemap_chunk_streaming	method:tilemap_layer_culling	dependency	обязательная	—	—	3	Unity Manual - Tilemap Renderer component reference
method:sprite_atlas_batching	method:sprite_sheet_compression	dependency	обязательная	—	—	3	Block Compression (Direct3D 10) - Microsoft Learn
method:sprite_atlas_batching	method:tilemap_layer_culling	dependency	обязательная	—	—	3	Unity Manual - Tilemap Renderer component reference
method:ability_visual_effect_budget	method:data_driven_ability_system	dependency	обязательная	—	—	3	Understanding the Unreal Engine Gameplay Ability System (Unreal Engine 5.8 Documentation)
method:client_prediction_reconciliation	method:tickrate_budgeting	dependency	обязательная	—	—	3	Source Multiplayer Networking (Valve Developer Community wiki)
method:client_prediction_reconciliation	method:headless_dedicated_server	dependency	обязательная	—	—	3	нет публичного источника
method:delta_compression_state	method:headless_dedicated_server	dependency	обязательная	—	—	3	Setting Up Dedicated Servers in Unreal Engine (UE 5.8 Documentation)
method:delta_compression_state	method:network_relevancy_priority	dependency	обязательная	—	—	3	нет публичного источника
method:headless_dedicated_server	method:tickrate_budgeting	dependency	обязательная	—	—	3	VALORANT's 128-Tick Servers (Riot Games tech blog)
method:headless_dedicated_server	method:network_relevancy_priority	dependency	обязательная	—	—	3	Replication Graph in Unreal Engine (UE 5.8 Documentation)

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method:subtick_net_working	method:client_prediction_reconciliation	dependency	обязательная	—	—	3	нет публичного источника
method:lag_compensation_rewind	method:subtick_networking	dependency	обязательная	—	—	3	Lag Compensation (Valve Developer Community wiki)
method:lag_compensation_rewind	method:client_prediction_reconciliation	dependency	обязательная	—	—	3	нет публичного источника
method:lag_compensation_rewind	method:network_relevancy_priority	dependency	обязательная	—	—	3	нет публичного источника
method:audio_convolution_reverb	method:audio_occlusion_propagation	dependency	обязательная	—	—	3	Steam Audio - Programmer's Guide (C API)
method:runtime_security_budget	method:headless_dedicated_server	dependency	обязательная	—	—	3	Using the Anti-Cheat Interfaces (Epic Online Services / Easy Anti-Cheat)
method:managed_gc_alloc_budget	method:tickrate_budgeting	dependency	обязательная	—	—	3	нет публичного источника
method:hardware_raytraced_gi	method:deferred_forward_plus_choice	dependency	обязательная	—	—	3	нет публичного источника
method:hardware_raytraced_gi	method:temporal_upscaling	dependency	обязательная	—	—	3	нет публичного источника
method:hardware_raytraced_gi	method:selective_ray_traced_effects	dependency	обязательная	—	—	3	нет публичного источника
method:sdf_global_illumination	method:hardware_raytraced_gi	dependency	обязательная	—	—	3	нет публичного источника
method:screen_space_gi	method:deferred_forward_plus_choice	dependency	обязательная	—	—	3	нет публичного источника
method:screen_space_gi	method:temporal_upscaling	dependency	обязательная	—	—	3	нет публичного источника
method:irradiance_volume_probes	method:lightmap_atlas_baking	dependency	обязательная	—	—	3	нет публичного источника
method:temporal_irradiance_cache	method:hardware_raytraced_gi	dependency	обязательная	—	—	3	нет публичного источника
method:temporal_irradiance_cache	method:deferred_forward_plus_choice	dependency	обязательная	—	—	3	нет публичного источника
method:voxel_cone_tracing	method:deferred_forward_plus_choice	dependency	обязательная	—	—	3	нет публичного источника
method:dynamic_light_priority_budget	method:tiled_clustered_light_culling	dependency	обязательная	—	—	3	Clustered Deferred and Forward Shading
method:light_range_attenuation_lod	method:dynamic_light_priority_budget	dependency	обязательная	—	—	3	нет публичного источника
method:distance_field_shadows	method:sdf_global_illumination	dependency	обязательная	—	—	3	нет публичного источника
method:screen_space_contact_shadows	method:deferred_forward_plus_choice	dependency	обязательная	—	—	3	нет публичного источника
method:static_shadow_caching	method:virtual_shadow_maps	dependency	обязательная	—	—	3	Virtual Shadow Maps (Unreal Engine 5.8 Documentation)
method:lightmap_atlas_baking	method:lightmap_compression_streaming	dependency	обязательная	—	—	3	Real-Time Rendering, 4th edition
method:gpu_lightmap_baking	method:lightmap_atlas_baking	dependency	обязательная	—	—	3	Lightmap (Wikipedia)
method:froxel_volumetric_fog	method:deferred_forward_plus_choice	dependency	обязательная	—	—	3	нет публичного источника
method:volumetric_half_resolution	method:froxel_volumetric_fog	dependency	обязательная	—	—	3	нет публичного источника
method:rt_effect_resolution_budget	method:selective_ray_traced_effects	dependency	обязательная	—	—	3	Exploring Ray Traced Future in Metro Exodus (GTC 2019)

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method:full_path_tracing_pipeline	method:path_tracing_sample_denoiser_budget	dependency	обязательная	—	—	3	Path Tracer (Unreal Engine 5.8 Documentation)
method:full_path_tracing_pipeline	method:selective_ray_traced_effects	dependency	обязательная	—	—	3	нет публичного источника
method:full_path_tracing_pipeline	method:temporal_upscaling	dependency	обязательная	—	—	3	нет публичного источника
method:deferred_forward_plus_choice	method:dYNAMIC_light_priority_budget	dependency	обязательная	—	—	3	Rendering the Hellscape of Doom Eternal
method:depth_prepass_early_z	method:deferred_forward_plus_choice	dependency	обязательная	—	—	3	Real-Time Rendering, 4th edition
method:dynamic_resolution_scaling	method:temporal_upscaling	dependency	обязательная	—	—	3	нет публичного источника
method:post_effect_selective	method:quality_tier_scalability	dependency	обязательная	—	—	3	Scalability (Unreal Engine 5.8 Documentation)
method:screenspace_light_shafts	method:post_effect_selective	dependency	обязательная	—	—	3	Volumetric Fog: Unified, compute shader based solution to atmospheric scattering
method:variable_rate_shading	method:deferred_forward_plus_choice	dependency	обязательная	—	—	3	нет публичного источника
method:ml_frame_generation	method:temporal_upscaling	dependency	обязательная	—	—	3	нет публичного источника
method:portal_scene_capture_budget	method:quality_tier_scalability	dependency	обязательная	—	—	3	Scalability (Unreal Engine 5.8 Documentation)
method:splitscreen_renderer_budget	method:quality_tier_scalability	dependency	обязательная	—	—	3	Scalability (Unreal Engine 5.8 Documentation)
method:splitscreen_renderer_budget	method:post_effect_selective	dependency	обязательная	—	—	3	Scalability (Unreal Engine 5.8 Documentation)
method:gpu_meshlet_culling_budget	method:meshlet_pipeline_adoption	dependency	обязательная	—	—	3	VK_EXT_mesh_shader proposal (Vulkan Documentation Project)
method:world_partition_streaming	method:async_loading_pipeline	dependency	обязательная	—	—	3	Asynchronous Asset Loading in Unreal Engine
method:hierarchical_lod	method:world_partition_streaming	dependency	обязательная	—	—	3	World Partition - Hierarchical Level of Detail in Unreal Engine
method:gpu_compute_culling	method:mesh_index_optimization	dependency	обязательная	—	—	3	нет публичного источника
method:terrain_clipmap	method:heightmap_compression	dependency	обязательная	—	—	3	C-BDAM - Compressed Batched Dynamic Adaptive Meshes for Terrain Rendering
method:virtual_texturing	method:async_loading_pipeline	dependency	обязательная	—	—	3	Nanite: A Deep Dive (SIGGRAPH 2021, Advances in Real-Time Rendering in Games)
method:heightmap_compression	method:build_size_startup_budgets	dependency	обязательная	—	—	3	Oodle Data (Unreal Engine Documentation)
method:virtual_geometry_clusters	method:gpu_compute_culling	dependency	обязательная	—	—	3	Nanite: A Deep Dive (SIGGRAPH 2021, Advances in Real-Time Rendering in Games)
method:virtual_geometry_clusters	method:mesh_index_optimization	dependency	обязательная	—	—	3	нет публичного источника
method:gpu_instancing_vegetation	method:gpu_compute_culling	dependency	обязательная	—	—	3	GPU-Driven Rendering Pipelines (SIGGRAPH 2015, Advances in Real-Time Rendering in Games)

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method:gpu_procedural_placement	method:gpu_instancing_vegetation	dependency	обязательная	—	—	3	GPU-Based Procedural Placement in Horizon Zero Dawn (GDC 2017)
method:impostors_billboards	method:vegetation_atlas_lod	dependency	обязательная	—	—	3	GPU Gems 3, Chapter 4: Next-Generation SpeedTree Rendering
method:vegetation_atlas_lod	method:gpu_instancing_vegetation	dependency	обязательная	—	—	3	Instanced Static Mesh in Unreal Engine
method:terrain_generation_streaming_budget	method:async_loading_pipeline	dependency	обязательная	—	—	3	Asynchronous Asset Loading in Unreal Engine
method:build_size_startup_budgets	method:directstorage_io	dependency	обязательная	—	—	3	DirectStorage (Win32 apps) - Microsoft Learn
method:directstorage_io	method:async_loading_pipeline	dependency	обязательная	—	—	3	Asynchronous Asset Loading in Unreal Engine
method:directstorage_io	method:terrain_generation_streaming_budget	dependency	обязательная	—	—	3	DirectStorage (Win32 apps) - Microsoft Learn
tool:ue_vsm	api:directx12	runtime_api	обязательная	SM6	pc_windows	2	Supported Features by Rendering Path for Desktop
tool:ue_replication_graph	engine:unreal	engine_tool	обязательная	—	—	2	Unreal Engine: Replication Graph
tool:u_dots	engine:unity	engine_tool	обязательная	—	—	2	Unity Manual: Entities (DOTS)
tool:u_netcode	tool:u_dots	package	обязательная	1.0	pc_windows,pc_linux	2	Unity Manual: Netcode
tool:ue_nanite	api:vulkan	runtime_api	обязательная	SM6	pc_linux	2	Supported Features by Rendering Path for Desktop
tool:u_dots	plugin:unity.entities	package	обязательная	1.0	pc_windows,pc_linux	2	Unity Manual: Entities (DOTS)
tool:u_dots	plugin:unity.burst	package	обязательная	1.0	pc_windows,pc_linux	2	Unity Manual: Entities (DOTS)
tool:u_netcode	plugin:unity.entities	package	обязательная	1.0	pc_windows,pc_linux	2	Unity Manual: Netcode
tool:u_netcode	plugin:unity.transport	package	обязательная	1.0	pc_windows,pc_linux	2	Unity Manual: Netcode
tool:u_addressables	plugin:unity.addressables	package	обязательная	1.0	pc_windows,pc_linux	2	Unity Manual: Addressables
tool:u_srp	plugin:unity.srp	package	обязательная	1.0	pc_windows,pc_linux	2	Unity Manual: Optimizing Graphics Performance
method:directstorage_io	sdk:directstorage	sdk	обязательная	—	pc_windows	2	DirectStorage API Now Available on PC (DirectX Developer Blog)
tool:ue_nanite	api:directx12	runtime_api	обязательная	SM6	pc_windows	1	Supported Features by Rendering Path for Desktop
tool:u_addressables	engine:unity	engine_tool	обязательная	—	—	1	Unity Manual: Addressables
tool:ue_nanite	engine:unreal	engine	обязательная	—	—	1	Nanite Virtualized Geometry Overview (Unreal Engine 5.8 Documentation)
tool:ue_lumen	engine:unreal	engine	обязательная	—	—	1	Lumen Global Illumination and Reflections (Unreal Engine 5.8 Documentation)
tool:ue_world_partition	engine:unreal	engine	обязательная	—	—	1	World Partition (Unreal Engine 5.8 Documentation)
tool:ue_vsm	engine:unreal	engine	обязательная	—	—	1	Virtual Shadow Maps (Unreal Engine 5.8 Documentation)
tool:ue_niagara	engine:unreal	engine	обязательная	—	—	1	Unreal Engine: Niagara Visual Effects
tool:ue_chaos	engine:unreal	engine	обязательная	—	—	1	Unreal Engine: Chaos Physics

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tool:ue_hlod	engine:unreal	engine	обязательная	—	—	1	Hierarchical Level of Detail in Unreal Engine (Epic official documentation)
tool:ue_virtual_texturing	engine:unreal	engine	обязательная	—	—	1	Virtual Texturing in Unreal Engine
tool:ue_insights	engine:unreal	engine	обязательная	—	—	1	Unreal Insights
tool:ue_navmesh	engine:unreal	engine	обязательная	—	—	1	Unreal Engine: Navigation Mesh
tool:ue_replication_graph	engine:unreal	engine	обязательная	—	—	1	Replication Graph in Unreal Engine
tool:ue_significance	engine:unreal	engine	обязательная	—	—	1	Significance Manager
tool:ue_anim_budget	engine:unreal	engine	обязательная	—	—	1	Animation Budget Allocator
tool:ue_ism	engine:unreal	engine	обязательная	—	—	1	Instanced Static Mesh in Unreal Engine
tool:ue_mass	engine:unreal	engine	обязательная	—	—	1	MassEntity in Unreal Engine
tool:ue_lwc	engine:unreal	engine	обязательная	—	—	1	Unreal Engine: Large World Coordinates
tool:ue_lod	engine:unreal	engine	обязательная	—	—	1	Unreal Engine: Level of Detail
tool:ue_gas	engine:unreal	engine	обязательная	—	—	1	Gameplay Ability System in Unreal Engine (Epic official documentation)
tool:ue_behavior_tree	engine:unreal	engine	обязательная	—	—	1	Behavior Trees
tool:ue_vehicles	engine:unreal	engine	обязательная	—	—	1	нет публичного источника
tool:u_dots	engine:unity	engine	обязательная	—	—	1	Unity Entities package manual
tool:u_jobs	engine:unity	engine	обязательная	—	—	1	Unity Manual - Write multithreaded code with the job system
tool:u_addressables	engine:unity	engine	обязательная	—	—	1	Unity Manual: Addressables
tool:u_srp	engine:unity	engine	обязательная	—	—	1	Unity Manual: Optimizing Graphics Performance
tool:u_instancing	engine:unity	engine	обязательная	—	—	1	Introduction to GPU instancing
tool:u_occlusion	engine:unity	engine	обязательная	—	—	1	Occlusion Culling (Unity official documentation)

В выборке обязательных рёбер: 120; с публичным источником: 89. Рёбра без публичного источника несут явную пометку «нет публичного источника», а не молчаливую совместимость.

7.4 Проверки графа

Проверка	Что обнаруживает	Реакция системы
missing_node	ребро ссылается на узел, которого нет в графе	ребро помечается unresolved
cyclic_mandatory	цикл обязательных зависимостей	цикл не скрывается; одно ребро детерминированно понижается до complement с записью resolution
unsupported_version	версия ниже/выше поддерживаемой	метод помечается неприменимым для этой версии
version_unknown	версия не указана	статус `unknown`, а не «совместимо»
api_incompatibility	требуется API/feature, которого нет на цели	метод отфильтровывается до ранжирования
basket_hard_conflict	в корзине есть hard conflict	hard conflict исключается из рабочей корзины
unresolved_dependency	транзитивная зависимость не закрыта	корзина помечается неполной
unknown_relation	связь типа unknown	показывается пользователю как открытый вопрос
user_defined_tool	пользовательский движок/инструмент	помечается user_defined; публичный источник не требуется, но нужен project-owned design record

8. Масштаб сцены

Параметр	Единицы/входы	Что определяет	Источники
Размер мира	km ² , число cell/grid, loading range	поточные запросы, IO-байты, activation time, RAM/VRAM residency	S01, S24
Плотность контента	instances/клетка, плотность PCG-вывода	cook/build time, draw/mesh preparation, проху-несоответствие	S25
Дальность видимости	view distance, HLOD-уровни	GPU raster/RT, raster-pass cost, VRAM потоковой подкачки	S24, S19
Число активных агентов	NPC, physics bodies, dynamic lights	simulation tick, navigation, animation, parallel CPU	S07, S31
Плотность эффектов	particles, decals, post-проходы	GPU raster, overdraw, transient memory	S19, S21

Размер мира сам по себе не является числом для FPS. Управляемые переменные — размер ячейки, loading range, число источников, HLOD-представление, размер чанка, скорость чтения, декомпрессия и время активации. Поэтому в DSS они отдельные поля, а не один scalar open_world=true.

9. CPU, GPU, RAM, VRAM и накопитель

Ось	Состав	Правило моделирования	Источники
CPU main-thread	game logic, render-thread, IO completion, activation	никогда не суммируется с parallel и не складывается в один scalar	S18, S47
CPU parallel	jobs/ECS, physics, animation, culling, worker threads	верхняя граница задаётся Amdahl, а не числом ядер	S31, S37
GPU raster	geometry pass, shading, decals, particles, post	зависит от внутреннего разрешения и shading-rate	S19, S05
GPU RT	RT GI/reflections/shadows, BVH build/refit	отдельная ось от raster; tier проверяется перед выбором	S03, S40
RAM	textures, geometry, audio, streaming pool, transient	не складывается с VRAM в одну величину	S18, S54
VRAM	residency текстур/геометрии, render targets, RT structures	ограничивается одновременно raster, RT и streaming pool	S54, S55
Storage / IO	read throughput, decompression, cold/warm cache	быстрый носитель не компенсирует плохую приоритизацию ассетов	S23, S52
Network	bandwidth, tick, relevance set, serialization	моделируется отдельно от render FPS и input latency	S12, S26

9.1 Пример композиции памяти

Компонент	Входы	Результат	Основание
Textures / geometry	2.4 + 1.1	3.5 GiB	scenario input
Render targets / transient	0.8 + 0.6	1.4 GiB	scenario input
Audio / other	0.2	0.2 GiB	scenario input
Subtotal	3.5 + 1.4 + 0.2	5.1 GiB	sum, no headroom
20% safety/headroom	5.1 × 1.20	6.12 GiB	derived planning allowance

RAM и VRAM не складываются в одну величину. GPU подбирается одновременно по raster, RT, VRAM, API и feature flags; CPU — по single-thread и multi-thread. Physics и AI пересчитываются по собственному fixed tick и не считаются как дополнительный FPS. Апскейлинг изменяет внутреннее разрешение; frame generation добавляет собственный проход.

10. Сетевые режимы

Режим	Модель	Ограничение	Источники
Single-player / local	нет сети; стоимость кадра только локальная	не получает сетевого бюджета и не вычитает его	expert boundary
Local co-op / split-screen	несколько видов на одном устройстве	несколько камер/видов умножают raster-работу; FPS не является нормативом	S15
Listen / host-authoritative	хост считает симуляцию и рендерит	server-нагрузка и client-нагрузка складываются на одном устройстве	S26, S27
Dedicated authoritative	выделенный сервер, клиент только предсказывает	server frame и client frame моделируются отдельно	S12, S14
Competitive high-tick	128 Hz-класс симуляции, prediction/rewind	высокий tick — свойство сервиса, а не универсальная рекомендация	S13, S14
Sub-tick / event-timed	точный момент события внутри такта	sub-tick, prediction и rewind — дополняющие части протокола, а не три скидки	S12
P2P / relay	relay-маршрутизация без выделенного сервера	latency и loss зависят от маршрута; требуется измерение, а не оценка	S30, S53

10.1 Примеры воспроизводимых сетевых расчётов

Scenario	Inputs	Derived result	Limits
Server → 9 clients	C=9, f=20/s, payload=120 B, IPv4+UDP H=28 B	$9 \times 20 \times (120 + 28) = 26,640 \text{ B/s} \approx 26.0 \text{ KiB/s} \approx 213 \text{ kbps}$	derived; excludes encryption, retransmits and other traffic
128 Hz server frame	1 / 128 s	7.8125 ms per tick	published VALORANT case S14; not universal target
Riot 3 games/core target	7.8125 / 3	$2.6042 \text{ ms}; \times 0.90 = 2.3438 \text{ ms}$	published case S14; host/game-specific

Server tick, render FPS, input latency, buffering, relevance set, serialization, packet loss и bandwidth моделируются отдельно. Пример Epic с Fortnite — документация механизма, а не лимит нового проекта.

11. Целевые показатели качества и производительности

11.1 Статусы оценки цели

Статус	Значение	Правило	Основание
meets	сценарий укладывается в frame-time budget при заданных условиях	применяется к p95/p99 frame-time, а не к одной средней частоте кадров	derived
at_risk	сценарий близок к границе или зависит от непроверенного условия	показывает, какой именно ресурс на границе и что нужно измерить	derived
unknown	данных недостаточно для вывода	не заменяется нулём, средним или «похожей картой»	expert_estimate
not_modeled	цель вне количественной границы модели (non-PC, latency, 1% low без измерения)	остаётся в профиле, но без числового ориентира	expert boundary

11.2 Бюджет кадра

Target FPS	Frame budget, ms	Интерпретация
30	33.333	low/quality-first target
60	16.667	common responsiveness target
90	11.111	high-refresh baseline
120	8.333	high-refresh target
144	6.944	competitive/high-refresh case
240	4.167	very high-refresh case; content-dependent

11.3 Пиксельная нагрузка разрешения

Разрешение	Pixel ratio	База
1280x720	0.444	relative to 1920x1080
1920x1080	1.000	baseline
2560x1440	1.778	relative to 1080p
3840x2160	4.000	relative to 1080p; 2.25x 1440p

$\text{frame_budget_ms} = 1000 / \text{target_fps}$. Сценарные цели показываются как meets, at_risk, unknown или not_modeled. Для 1% low FPS, startup, streaming latency, save time, network latency, server tick и traffic не создаётся значение без входного значения или измерения. Критерий meets применяется к frame-time budget.

12. Стадии и work packages

Тип пакета	Пакетов	Мин. P50	Сред. P50	Сред. P80	Основание
content	248	0.38	1.87	3.23	expert_estimate
design	248	0.38	1.77	3.08	expert_estimate
documentation	124	0.38	2.45	4.54	expert_estimate
feasibility	182	0.38	2.31	4.05	expert_estimate
integration	248	0.38	2.47	4.14	expert_estimate
optimization	248	0.38	1.82	3.15	expert_estimate
qa	248	0.38	1.79	3.12	expert_estimate
release	248	0.37	1.52	2.71	expert_estimate

План строится из work packages: design, feasibility/prototype, integration, content/assets, optimization, QA/regression, release stabilization, documentation/maintenance. Каждый пакет имеет min, P50, P80, роль, параллелизуемость, стадию, late factor, зависимости и основание. Legacy implementation_cost сохранён для совместимости, но не является основой плана.

13. Трудоёмкость P50/P80 и календарь для разных команд

13.1 Профили команд

Код	Название	Размер	Потоки	Коммуникации	Непредвиденное
solo	Solo	1	1	5%	25%
small_2_5	Малая команда (2-5)	4	2	12%	18%
custom	Собственный состав	6	3	15%	16%
mid_6_15	Средняя команда (6-15)	10	5	18%	15%
large_16_plus	Большая команда (16+)	24	12	25%	12%

P50/P80 выражены в человеко-днях и не являются отраслевым нормативом. Команда влияет на календарную ёмкость, но не уменьшает сумму person-days.

13.2 Трёхточечная оценка

Inputs	Calculation	Статус
O=2, M=4, P=10 человеко-дней	$(2 + 4 \times 4 + 10) / 6 = 4.667$	beta/PERT expected duration
Тот же диапазон	$\sigma = (10 - 2) / 6 = 1.333$	approximate spread; not a calibrated percentile
P80 proxy, only if explicitly assumed normal	$4.667 + 0.8416 \times 1.333 = 5.789$	derived scenario, not a measurement

13.3 Предел параллелизации (Amdahl)

Workers	Speedup	Допущение
N=1	1.000	baseline
N=2	1.429	s=0.40, idealized
N=4	1.818	s=0.40, idealized
N=8	2.105	s=0.40, idealized
N=16	2.353	s=0.40, idealized
N→∞	2.500	serial ceiling 1/s

13.4 Critical path (пример)

Task	Duration	Predecessors	Path result
A Design	3	—	A-B-D-E / A-C-D-E
B Prototype	5	A	A-B-D-E = 18
C Content/asset prep	8	A	A-C-D-E = 21; critical
D Integration	6	B,C	merge waits for max(B,C)
E QA/release gate	4	D	longest path ends at 21

Календарь вычисляется по dependency DAG и доступности роли. Независимые задачи могут идти параллельно; integration, QA, release gates и узкие роли формируют critical path. Поздняя стадия добавляет rework/late-risk note, но не удаляет метод из выдачи.

14. Профили нагрузки

Профиль	Состав	Доминирующие оси	Источники
Открытый мир	streaming + HLOD + большая дальность	IO, decompression, VRAM residency, traversal hitches	S01, S24, S28
Арена / линейный	ограниченный набор ячеек, много эффектов	GPU raster, overdraw, transient memory	S16, S19
Симуляция/стратегия	много агентов, ECS/jobs, navigation	parallel CPU, simulation tick, cache locality	S07, S31
Соревновательный сетевой	высокий tick, prediction/rewind, relevance	server frame, serialization, bandwidth, divergence	S12, S14
Кооперативный кампанийный	несколько видов, адаптивный pacing	несколько камер, AI director, physics	S11, S15
Аудио-насыщенный	много источников, occlusion, DSP	ray/query count, DSP time, output latency	S17, S29

15. Оборудование

15.1 Покрытие и основание

Тип	basis	Записей
CPU	derived	51
GPU	derived	79

Тип	Класс	Записей
CPU	1	2
CPU	2	9
CPU	3	23
CPU	4	11
CPU	5	6
GPU	1	18
GPU	2	21
GPU	3	22
GPU	4	13

Тип	Класс	Записей
GPU	5	5

Всего: 51 CPU / 79 GPU. Индекс CPU/GPU — нормализованный anchor, а не FPS. В карточке оборудования хранятся benchmark name, raw value, context, normalization note, evidence basis и source link. Если каталог не содержит устройства, одновременно покрывающего условия, результат сообщает exceeds_catalog и не подставляет похожую карту молча.

15.2 CPU (верхняя часть по single-thread)

Модель	Single-thread	Multi-thread	Класс	Benchmark	Basis
Core i9-14900K	1.00	0.94	5	PassMark CPU Benchmarks — Intel Core i9-1490	derived
Core i9-13900K	0.98	0.93	5	PassMark CPU Benchmarks — Intel Core i9-1390	derived
Core i7-14700K	0.95	0.84	5	PassMark Intel Core i7-14700K Benchmark — CP	derived
Ryzen 7 9800X3D	0.94	0.64	5	PassMark AMD Ryzen 7 9800X3D Benchmark — CPU	derived
Core i7-13700K	0.92	0.73	4	PassMark CPU Benchmarks — Intel Core i7-1370	derived
Ryzen 9 7950X	0.91	1.00	5	PassMark CPU Benchmarks — AMD Ryzen 9 7950X	derived
Core i5-14600K	0.91	0.62	4	PassMark Intel Core i5-14600K Benchmark — CP	derived
Ryzen 9 7900X	0.90	0.82	5	PassMark CPU Benchmarks — AMD Ryzen 9 7900X	derived
Ryzen 7 7700X	0.89	0.57	4	PassMark CPU Benchmarks — AMD Ryzen 7 7700X	derived
Core i5-13600K	0.88	0.60	4	PassMark CPU Benchmarks — Intel Core i5-1360	derived
Core i9-12900K	0.88	0.66	4	PassMark CPU Benchmarks — Intel Core i9-1290	derived
Ryzen 5 7600X	0.88	0.46	4	PassMark CPU Benchmarks — AMD Ryzen 5 7600X	derived

15.3 GPU (верхняя часть по raster)

Модель	Raster	RT	VRAM, GB	Класс	Benchmark
GeForce RTX 4090	1.00	1.00	24	5	PassMark Video Card Benchmarks — GeForce RTX
GeForce RTX 5090	1.00	1.00	32	5	PassMark Video Card Benchmarks — GeForce RTX
GeForce RTX 5080	0.94	0.94	16	5	PassMark Video Card Benchmarks — GeForce RTX
GeForce RTX 4080	0.87	0.91	16	5	PassMark Video Card Benchmarks — GeForce RTX
Radeon RX 7900 XTX	0.86	0.83	24	5	PassMark Video Card Benchmarks — Radeon RX 7
GeForce RTX 5070 Ti	0.85	0.85	16	4	PassMark Video Card Benchmarks — GeForce RTX
Radeon RX 7900 XT	0.78	0.77	20	4	PassMark Video Card Benchmarks — Radeon RX 7
GeForce RTX 5070	0.75	0.75	12	4	PassMark Video Card Benchmarks — GeForce RTX
Radeon RX 9070 XT	0.71	0.71	16	4	PassMark Video Card Benchmarks — Radeon RX 9
GeForce RTX 3090	0.70	0.70	24	4	PassMark Video Card Benchmarks — GeForce RTX
Radeon RX 6900 XT	0.70	0.70	16	4	PassMark Video Card Benchmarks — Radeon RX 6
GeForce RTX 3080	0.68	0.66	10	4	PassMark Video Card Benchmarks — GeForce RTX

16. Реальные игровые кейсы

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Age of Empires	Ensemble Studios	Genie Engine / 1997	·	A completely different anti-cheat model with no client trust problem at all: because every peer must run an identical simulation, any divergence is tagged 'out of sync' and the game stops, which Bettner says made hacking a client or its communication stream ex	This only works because lockstep has no server authority and no hidden information by design. You cannot add out-of-sync detection to a client-server shooter - and 'stop the match' is a very different product response to cheating than 'ban
Age of Empires (and Age of Empires II)	Ensemble Studios	Genie Engine / 1997	·	Shipped a peer-to-peer synchronous (lockstep) simulation on a 28.8 kbps modem with 8 players: 200 ms turns, commands scheduled two turns ahead, 2-byte speed control, out-of-sync detection as anti-cheat, and deterministic recordings used as bug repros.	The 200 ms turn, the 250-unit wall and the 250-500 ms latency tolerance were all derived from 28.8 kbps modems, 15 fps and 1997 CPUs. None of them is a modern constant - but the relationship (bandwidth per unit vs bandwidth per command) and
Alan Wake 2	Remedy Entertainment	Northlight / 2023	·	Remedy states AW2's GPU-driven mesh-shader pipeline does occlusion culling down to single-pixel precision and uses everything in a scene as an occluder.	Do not copy the single-pixel occlusion claim to hardware without mesh shaders and a GPU-driven pipeline.
Alan Wake 2	Remedy Entertainment	Northlight / 2023	·	Remedy's own write-up states the shipped game uses mesh shaders and culls meshlets, showing individual meshlets around Cauldron Lake fed into the renderer.	Do not copy the meshlet sizing or culling scheme without Remedy's mesh-shader-capable minimum hardware.
Alan Wake 2	Remedy Entertainment	Northlight / 2023	·	Ships 'Path Traced Indirect Lighting' alongside DLSS Frame Generation and DLSS Ray Reconstruction on PC, i.e. RT GI layered on top of a raster pipeline rather than replacing it.	Remedy paired PT indirect lighting with a specific upscaler/denoiser stack and specific content. No frame-time figures were published, so no performance number can be transferred.
Ashes of the Singularity	Oxide Games / Stardock	Nitrous / 2016	·	AMD's GPU trace of the benchmark scene shows it 'makes heavy use of the compute queue': 'The asynchronous compute queue is used for most of the frame. It can be seen from the trace that the graphics queue is not stalled while waiting on the compute queue.'	AMD's trace shows queue behaviour only - no frame-time or FPS number is published, and nothing about this title's async strategy transfers to another engine.
Assassin's Creed IV: Black Flag	Ubisoft	AnvilNext / 2013	·	Bart Wronski's SIGGRAPH 2014 presentation ships the froxel volumetric fog solution in AC4 and reports: 'Total cost was surprisingly small, around 1.1ms. Calculating it in double resolution had a cost of 1.6ms. Most costly part was building density and lighting	Measured on PS3/PS4-era hardware and AC4 content; the ms values must not be transferred, only the ratio between resolution and cost.
Assassin's Creed Unity	Ubisoft	AnvilNext (Ubisoft in-house) / 2014	·	Ships a crowd system with AI level-of-detail and pooling that swaps low-res NPCs to high-res ones; with a cap of 40 real AIs and 120 high-res models the team reports a scene with 10,000 crowd NPCs on screen simultaneously.	10,000 on-screen NPCs with 40 real AIs is that game's crowd budget for a Paris street scene on PS4/Xbox One. It is not a performance number and does not transfer: the acceptable ratio of 'real' to 'fake' NPCs depends entirely on how closely
Assassin's Creed Unity	Ubisoft Montreal	AnvilNext / 2014	·	The GDC 2015 session describes an AI level-of-detail recycling/pooling system that let the shipped game put 10,000 crowd NPCs on screen using 40 real AIs and 120 high-resolution models.	Do not copy the 40-real-AI / 120-high-res-model split: it is specific to Unity's crowd scale and hardware targets.

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Baldur's Gate 3	Larian Studios	Divinity 4.0 (in-house) / 2023	·	Listed by the meshoptimizer maintainer among commercial games that use the library. The listing documents use of the library; it does not say which optimisations are enabled, at what settings, or what was measured.	A usage credit, not a measurement. No ACMR, bytes-per-triangle, decode throughput or frame-time figure is published for this title, and a credit tells you nothing about which parts of the library were used or how much they bought.
Battlefield V	DICE (EA)	Frostbite / 2018	open_world · dedicated_server	DICE's GDC session is documented as covering the nitty-gritty details of the studio's ray-traced reflection implementation, and the NVIDIA GTC 2019 deck states Battlefield V was the first DXR game released.	Reflections only, at 1080p with a limited effect list, on first-generation RT hardware; that scope does not transfer to a full ray-traced lighting pipeline.
Battlefield V	DICE (EA)	Frostbite / 2018	open_world · dedicated_server	DICE's GDC session is documented as a technical presentation on ray-traced reflections in a large multiplayer shooter, establishing only the dedicated-server context.	No anti-cheat fact is evidenced by this source; it is listed only to make the gap explicit.
BeamNG.drive	BeamNG	In-house (Torque 3D-derived) / 2015	·	BeamNG documents a soft-body, node-and-beam vehicle simulation in which the vehicle structure itself deforms, with named beam parameter pairs for suspension, structural members and steering - the high-fidelity pole of vehicle simulation rather than a raycast m	This is the opposite approach to a raycast vehicle, and it is a studio marketing/documentation n page rather than a technical paper. The soft-body approach is affordable because the whole product is about vehicle simulation; it does not trans
Black Myth: Wukong	Game Science	Unreal Engine 5 / 2024	·	Digital Foundry reports 'a very distinct shadow cascade line' near the main character when Virtual Shadow Maps are excised without Full RT, with distant shadows blobbing out.	Do not copy the VSM-excised fallback configuration: it produces visible cascade seams that reviewers flagged.
Black Myth: Wukong	Game Science	Unreal Engine 5 / 2024	·	Digital Foundry reports the game forgoes Virtual Shadow Maps unless Full RT is enabled because VSMs are expensive on non-Nanite foliage, producing more shadow issues than other UE5 titles.	Do not copy the decision to disable VSMs in a foliage-heavy game unless you accept the shadow artifacts it caused here.
Black Myth: Wukong	Game Science	Unreal Engine 5 / 2024	·	Digital Foundry reports that without Full RT the shipped game uses software Lumen on PC for diffuse GI and specular reflections, i.e. distance-field-based tracing.	Do not copy software Lumen as equivalent to hardware RT here; the review notes reflections differ most.
CMU motion capture database (academic corpus, not a shipped game)	Carnegie Mellon University	n/a / 2003	·	Open benchmark for the same codec: 2534 clips at 24 FPS, 9h 49m 37.58s, 1429.38 MB raw -> 65.66 MB (21.77:1), max error 0.0495 cm, 99.96% of samples below threshold.	Academic mocap at 24 FPS with no game-specific authoring (no root-motion cleanup, no exotic clips). Its max error is 80x lower than Paragon's, which is a statement about the content, not about the codec.
Call of Duty (SIGGRAPH 2020 talk, cited by The Coalition)	Infinity Ward / Activision	Call of Duty engine / 2020	·	The Coalition's write-up points to 'Software-Based Variable Rate Shading in Call of Duty' (SIGGRAPH 2020) which emulates VRS via MSAA on platforms without hardware VRS and uses ExecuteDispatchIndirect for compute passes.	Do not copy software VRS emulation unless you can afford the de-blocking pass; it is a workaround for hardware without VRS support.

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Chunked LOD reference implementation (Puget Sound demo)	Thatcher Ulrich (independent)	Public-domain reference implementation / 2002	·	Flew a 16K x 16K USGS Puget Sound dataset (512 MB raw) from a preprocessed 214 MB chunk file at 4 m error tolerance, and generated a 32K x 32K quadtree-tiled texture (9 levels of 128x128 JPEG nodes, ~61 MB) to surface it, at ~30 fps on a 1 GHz Pentium 3 with a	2002 hardware and a public-domain research implementation. The 512 MB -> 214 MB ratio applies to that dataset at that tolerance on that codec; it is not a general compression ratio. The ~30 fps figure is a 1 GHz P3/GeForce2Go number and mus
Counter-Strike 2	Valve	Source 2 / 2023	соревновательные арены · authoritative multiplayer	Sub-tick architecture records the instant of relevant input events independently of the old tick-only framing.	Официальное описание не задаёт переносимый бюджет трафика или задержки для другой игры.
Counter-Strike 2	Valve Corporation	Source 2 / 2023	·	A competitive title receiving frequent, small client updates where download size directly affects how quickly the player base can join a match.	No patch-size or delta-ratio figures are published, so this does not quantify patch efficiency.
Counter-Strike 2	Valve Corporation	Source 2 / 2023	·	The only shipped game I could find with a sub-tick architecture: Valve claims tick rate no longer matters for moving, shooting or throwing, and that grenades will always land the same way - while the VDC wiki records the server as hard-coded 64 tick with sub-t	Everything. Valve published the intent and nothing else: no wire format, no measurement, no error budget. You cannot copy this - you can only re-derive it and measure it yourself. Also note that a 64+sub-tick server is not automatically che
Cuphead	StudioMDHR	Unity / 2017	·	A fully hand-drawn style executed as an asset pipeline problem: '50,000 frames of amazing hand-drawn animation', drawn and inked on paper and coloured in Photoshop, authored at '24 frames-per-second' while 'the game runs at a sparkling 60 frames-per-second'. U	The 24fps-animation/60fps-game split is an authored art decision, not a performance result. 50,000 frames is a statement about StudioMDHR's art budget across the whole game - it does not scale down to a smaller team, and it says nothing abo
Cuphead	StudioMDHR	Unity / 2017	·	The deliberate opposite choice, included because it defines the boundary of this method: '50,000 frames of amazing hand-drawn animation', authored at '24 frames-per-second' while the game runs at 60. This is the art direction that skeletal 2D deformation canno	A counter-example, not a comparable implementation. The frame count is a studio art-budget fact, not a performance result, and it tells you nothing about the CPU cost of a skeletal 2D system.
Cuphead	StudioMDHR	Unity / 2017	·	The extreme end of the atlas problem: '50,000 frames of amazing hand-drawn animation', with Unity's 'Sprite Renderer, Sprite Packer, 2D Physics, and particle effects' named as the tools that 'help[ed] StudioMDHR process the massive amount of their exquisite ar	50,000 frames is StudioMDHR's art budget for one game, not a performance figure. Their animation runs at 24fps while the game runs at 60fps, so frame reuse rather than per-frame draw cost is the interesting property here. Nothing about thei
Cyberpunk 2077	CD Projekt Red	REDengine 4 / 2020	·	A shipped open-city title with a very large and heterogeneous material set, and a later path-traced mode that stresses shader variety across the whole scene.	REDengine 4 is a bespoke engine; its material system and permutation strategy are not transferable, and this source does not state that bindless binding is used.
Cyberpunk 2077	CD Projekt Red	REDengine 4 / 2020	·	A shipped title with a very large install footprint, representing the high end of the same range.	Install size here reflects a specific content scope and audio localisation set; it is an anchor point, not a target.

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Cyberpunk 2077	CD PROJEKT RED	REDengine 4 / 2023	open_world · single	NVIDIA documents the Ray Tracing: Overdrive technology preview for Cyberpunk 2077 as shipping with DLSS 3, letting GeForce RTX 40-series players run the mode at 4K.	The combination of full path tracing with DLSS 3 frame generation is a PC-only showcase; the resulting latency and quality trade-offs do not transfer to console targets.
DOOM Eternal	id Software	id Tech 7 / 2020	сегментированные боевые уровни · single-player with online modes	Доклад SIGGRAPH 2020 описывает geometry caches, gore, decals, material compositing и workflow вокруг целевого frame rate.	Собственный движок, ассеты и платформенный порт делают детали непереносимыми без повторного измерения.
DOOM Eternal	id Software	id Tech 7 / 2020	·	The shipped game bins lights and decals with a hybrid cluster+tile scheme because artists wanted more dynamic lights and decals while the binning budget stayed under 500us.	Do not copy the <500us binning budget without id Tech 7's compute-raster pipeline and bindless resource model.
DOOM Eternal	id Software	id Tech 7 / 2020	·	Doom Eternal's GPU triangle culling is budgeted for 3M visible triangles with only ~1M reaching the rasterizer, removing ~70% of submitted triangles.	Do not treat id Tech 7's triangle-culling budget as a meshlet budget: Doom Eternal is not a meshlet renderer.
Dawn (NVIDIA real-time tech demo)	NVIDIA	custom Cg/HLSL renderer / 2003	·	The canonical published GPU-skinning implementation: a skeleton of 98 bones driving a mesh of more than 180,000 triangles with four bone matrices per vertex, bone palette in vertex-shader constants.	A single hero character on 2004 hardware, no crowd, no multi-pass reuse, no cache budget. It demonstrates the algorithm, not the economics of a modern 100-character scene.
Diplomacy is Not an Option	Door 407	Unity (ECS for Unity / DOTS) / 2022	·	Unity documents that the studio uses DOTS 'almost everywhere' in this real-time strategy game and finds it 'especially useful for pathfinding and optimizing our gameplay logic' - the data-oriented foundation that makes large-unit-count 2D/RTS crowd rendering t	This is a vendor marketing page quoting the developer. It confirms DOTS usage for gameplay/pathfinding, not specifically 2D crowd instancing, and gives no unit counts or framerate numbers.
Doom Eternal	id Software	id Tech 7 / 2020	·	The SIGGRAPH 2020 presentation states the game holds 60 FPS on consoles, and its results section reports up to 5 ms of GPU savings in dense scenes from triangle culling and merging - evidence of an explicit, measured frame budget culture rather than a physics-	The 5 ms figure is for triangle culling/merging, not for physics timestep, and the 60 FPS target is that game's own target on its own hardware. Neither number transfers to another project; the transferable observation is that a shipped AAA
F.E.A.R.	Monolith Productions	F.E.A.R. engine (LithTech Jupiter EX) / 2005	·	The shipped game's AI uses a 3-state FSM plus A* planning over data-driven actions/SmartObjects, so behaviour is composed from database entries rather than a large hand-written FSM.	Do not copy F.E.A.R.'s 3-state FSM as an ability system: it is a goal-oriented action planner, not a cooldown/effect ability framework.
F.E.A.R.	Monolith Productions	LithTech Jupiter EX / 2005	linear · single	The GDC 2006 paper states A* is used to find paths through the navigational mesh as well as to plan action sequences, and that when an AI crawls under an obstacle it first searches for a navigational path and then for a plan to overcome the obstacle.	F.E.A.R.'s navmesh and cover-node density are tuned for small indoor combat spaces; the same node budget does not transfer to a large open world.
For Honor	Ubisoft Montreal	Ubisoft in-house (AnvilNext-derived pipeline; animation system custom) / 2017	·	Motion matching was presented by Ubisoft Montreal's Simon Clavet at GDC 2016 as the studio's next-gen animation approach. The published pipeline is: 'Mocap is tweaked, imported, and marked up'; 'At runtime Gameplay makes a request (desired trajectory and event	The talk publishes no database size, no search time, and no CPU budget - the optimisation slide explicitly defers to 'a future talk'. Also, this is a melee-combat game where weapon position and stance matching matter; a shooter or a third-p

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Forspoken	Luminous Productions	Luminous Engine (in-house) / 2023	·	In the DirectX blog post announcing DirectStorage availability on PC, Microsoft points readers to 'the GDC talk presented by Luminous Productions about their integration of DirectStorage in Forspoken' - the only shipped title named in the material I verified.	I verified only that Microsoft points to the talk; I did not access the talk itself, so no load time, throughput or CPU figure is asserted. Any number from Forspoken's integration would be its own hardware, its own asset layout and its own
Fortnite	Epic Games	Unreal Engine 4/5 / 2017	·	Epic's own documentation uses Fortnite's modular characters as the motivating case: characters are 'comprised of several Skeletal Meshes (heads, bodies, backpacks, etc.) all of which can be ticking at a given time', and the budgeter shows per-mesh rate numbers	Modular characters multiply component count per character, which is why budgeting mattered there. A monolithic-mesh character project has a much smaller per-character component count and may not need the allocator at the same crowd size. Ep
Fortnite Battle Royale	Epic Games	Unreal Engine 4 / 2017	·	Epic built and shipped the Replication Graph plugin specifically because Fortnite starts each match with 100 players and about 50,000 replicated actors, and documents the tree/dormancy/always-relevant node split using Fortnite as the worked example.	The 100-player / 50,000-actor figures describe Fortnite's content density and actor granularity. A game with 100 players but 2,000 replicated actors has a different problem and may not need a replication graph at all; a game with 5,000 acto
GPU geometry clipmap reference implementation (NVIDIA / Microsoft Research)	NVIDIA / Microsoft Research (Arul Asirvatham, Hugues Hoppe)	Direct3D 9 sample / 2005	·	Rendered a ~20.2-billion-sample US terrain dataset compressed >100x into 355 MB at 130 fps (60M triangles/sec) at 1024x768 on a GeForce 6800 GT - the canonical measured demonstration of the method.	2005 hardware (Pentium 4 2.4 GHz, 1 GB RAM, GeForce 6800 GT) at 1024x768. Neither the fps nor the ms update figures are usable as a modern budget. The transferable content is the structural result: ~71 draw calls for 11 levels, ~5-pixel tri
Gears 5 / Gears Tactics	The Coalition	Unreal Engine 4 / 2020	·	The Coalition states the games use Dynamic Resolution Scaling to hit a smooth 60 FPS, rendering the next frame at lower resolution when near budget.	Do not copy the DRS thresholds or the 10% figure to a different engine/hardware target.
Gears 5 / Gears Tactics	The Coalition	Unreal Engine 4 / 2020	·	VRS texture generation runs on the Async Compute Queue, overlapping the next frame's depth pass; the studio had already moved the post-process chain to overlap the next frame's depth pass.	Do not copy the overlap window: it depends on where VRS texture generation sits in the post chain (tonemapping here).
Gears 5 / Gears Tactics (Hivebusters)	The Coalition	Unreal Engine 4 / 2020	·	The Coalition's own VRS write-up states the Xbox Series X S launch of Gears 5/Tactics and the Hivebusters DLC 'added new rendering features including contact shadows and screen space global illumination'.	Do not copy the pass list blindly: Gears applied VRS to contact-shadow-bearing passes under a specific budget that depends on its own target framerate.
General shipped practice (Game Programming Patterns)	Robert Nystrom (pattern catalogue)	engine-agnostic / 2014	·	The Object Pool pattern is documented as a standard optimisation pattern for games, motivated by memory fragmentation and allocation cost on consoles and in languages with garbage collection.	This is a design-pattern reference, not evidence of a specific shipped game's results. It establishes that the technique is standard practice; it provides no measurement and no shipped-title confirmation.

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Godot 4 (engine feature)	Godot Engine contributors	Godot 4.x / 2024	·	Godot offers a double-precision build for physics/rendering and documents origin shifting as the alternative path for platforms where doubles are too expensive - an independent engine making the same trade-off Epic documents.	Cross-engine confirmation of the design space only. Godot's double build, its performance cost, and its origin-shifting helper are not transferable to Unreal or to a custom engine; do not copy any constant.
Godot-based shipped titles (engine capability)	Godot contributors (engine)	Godot 4.x / 2023	·	Godot exposes 32 physics layers with per-object layer and mask bitmasks as the standard, documented mechanism for filtering both collision simulation and queries.	Engine capability, not a shipped-game result. No pair counts, timings, or layer taxonomy transfers to another project or engine.
Godot-based titles using VehicleBody3D	Godot contributors (engine)	Godot 4.x / 2023	·	Godot ships a raycast vehicle system as a built-in node and documents plainly that it is not designed for realistic vehicle physics, directing advanced users to write their own integration.	Engine capability with a published scope limit, not a shipped-game result. No handling metrics or cost figures transfer; the transferable content is the scope limit itself.
Godot-based titles using the documented save pattern	Godot contributors (engine documentation)	Godot 4.x / 2023	·	Godot documents a complete snapshot save/load pattern (FileAccess to a named user:// path, JSON or binary serialisation, staged loading of nested objects) as the engine's reference approach.	Engine tutorial, not a shipped-game result. No save sizes or load times transfer, and the tutorial's simplified model (no nested Persist children) is explicitly called out as a limitation that real projects must solve.
Hi-Fi RUSH	Tango Gameworks	Customized Unreal Engine 4 / 2023	·	A shipped stylized 3D game held to 60fps at native resolution, built on a deferred toon renderer composed from customized UE4 passes plus added passes, combined in deferred post-process volumes; the session covers comic shaders, toon lights, dynamic and static	The 60fps-at-native-resolution result is a property of this game's scene complexity, content density and target hardware. It does not predict the cost of a deferred toon renderer in a different scene or on different hardware, and no per-pas
Hitman: Absolution	IO Interactive	Glacier 2 / 2012	·	IO Interactive's GDC talk describes the techniques and optimisations used to achieve 1200-character crowds in a real production level while running at 30 fps on current-gen consoles, with every character individually interactable and influenceable.	'1200 characters at 30fps on current-gen consoles' is a 2012 PS3/Xbox 360 target for that game's specific crowd density and camera. Neither the agent count nor the frame rate transfers to other hardware or to a different crowd/interaction d
Horizon Zero Dawn	Guerrilla Games	Decima / 2017	·	Guerrilla's own talk description states the shipped game uses low-level and high-level streaming systems with dedicated memory management and scheduling.	Do not copy Decima's streaming tiers without its asset conversion pipeline and memory allocators, which the talk presents as a coupled system.
Horizon Zero Dawn	Guerrilla Games	Decima / 2017	·	A large open world with dense vegetation and long view distances, requiring distant objects to be merged and simplified to stay within the draw budget.	Decima's streaming and LOD pipeline is bespoke; the is not documented in enough detail to copy.
Hunt: Showdown	Crytek	CryEngine / 2019	большие PvPvE-карты · competitive multiplayer	Команда Hunt описывает occlusion rays, material-dependent filtering, HRTF/CrySpatial и аудио как источник игровой читаемости.	Число emitters, лучей и стоимость CPU/аудио-потока зависят от карты и middleware.

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Hunt: Showdown	Crytek	CryEngine (not stated in the cited post) / 2019	·	Crytek published a dedicated developer-insight post on Hunt's audio ('Did you hear that?') covering how players should read audio cues - evidence that propagation/occlusion is treated as a core competitive mechanic rather than polish.	The post publishes NO ray counts, CPU cost or memory budget, so nothing about Hunt's implementation can be copied or used as a performance target. Only the product stance (audio is a competitive information channel) transfers.
Hunt: Showdown	Crytek	CryEngine (not stated in the cited post) / 2019	·	Crytek treats audio - including how spaces colour sound - as a competitive information channel, publishing a dedicated developer post on reading audio cues.	The post is about players reading cues, not about implementation. It publishes no reverb configuration, IR data or budget, and must not be cited as evidence for any technical choice.
IXION	Kasedo Games	Unity (ECS for Unity / DOTS) / 2022	·	Unity documents that Kasedo Games used ECS for Unity 'to power heavy NPC simulation' for IXION, a city-builder/survival /space-exploration title - the closest documented match to a crowd-scale ECS workload in this pack.	Vendor page, no numbers. 'Heavy NPC simulation' is not quantified, so no agent count or framerate transfers.
It Takes Two	Hazelight	Unreal Engine 4 / 2021	линейные кооперативные сцены · local/online co-op	Технический анализ Digital Foundry рассматривает split-screen и две точки зрения как часть рендер-пайплайна.	Результат зависит от сцены, разрешения, качества и конкретной реализации; FPS не переносится.
It Takes Two	Hazelight Studios	Unreal Engine 4 / 2021	linear · coop	The Hazelight interview and the EA press release both identify It Takes Two as the studio's previous co-op title (23 million units sold, 2021 Game of the Year) whose workflow and tooling the team built on for Split Fiction. Neither source states the word split	This is co-op lineage evidence, not a verified split-screen implementation statement for It Takes Two; it must not be cited as proof that It Takes Two renders two viewports, and no cost may be inferred.
Just Cause 4	Avalanche Studios	Avalanche engine / 2018	·	A large open world with many vehicles, aircraft and physics props, where only nearby vehicles can be fully simulated.	The talk covers full-detail vehicle physics, not the LOD policy; the degradation strategy is not documented.
Just Cause 4	Avalanche Studios	Avalanche engine / 2018	open_world · single	Avalanche's GDC talk is documented as covering how Just Cause 4 copes with a wide range of vehicles and player styles through the design of its tire dynamics, and how those relate to the overall vehicle gameplay experience.	The talk is about vehicle handling models, not general rigidbody simulation; its tire model does not transfer to non-vehicle physics.
Justice (Ni Shui Han)	NetEase	Unreal Engine (TressFX patch) /	open_world · dedicated_server	AMD's TressFX page credits NetEase's Justice (Ni Shui Han) with a screenshot and states TressFX is a GPU-based hair/fur rendering and simulation library with an LOD system and an Unreal Engine patch. The source does not state the game's release year, so it is	The source proves GPU strand hair, not cloth; it is included only as evidence that GPU-simulated strands ship in a game, and no cloth figure may be inferred.
Justice (Ni Shui Han)	NetEase	Unreal Engine (TressFX patch) / 2019	open_world · dedicated_server	AMD's TressFX page shows a screenshot credited to NetEase's Chinese martial-arts game Justice (Ni Shui Han) as a shipped use of the GPU strand hair library.	Strand hair at this fidelity is GPU-budget-heavy and was shipped with vendor library support; the strand counts are not published and must not be assumed.
Left 4 Dead	Valve	Source / 2008	линейные кооперативные кампании · 4-player co-op	AI Director модулирует драматический темп по оценке интенсивности команды.	Не переносит количество заражённых, FPS или сложность кампании в другой проект.

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Left 4 Dead	Valve	Source / 2008	·	Ships the AI Director: an adaptive dramatic pacing controller with Build Up / Sustain Peak / Peak Fade / Relax phases, plus procedural population driven by navmesh flow distance, the Active Area Set and potentially-visible sets.	The 3-5 s / 30-45 s phase timings, the 20-30 mob size, the 90-180 s interval and the 75%-from-behind rule are L4D's tuned constants for a 4-player co-op survival-horror loop. They are not portable budgets; a different team size, TTK or map
Left 4 Dead	Valve	Source / 2008	·	Uses the contrasted approach - A* through a navigation mesh plus reactive path following toward a look-ahead point with local obstacle avoidance - and documents that this yields a path 'reasonably close to optimized' while making (re)pathing cheap.	This is the alternative, not the method. It shows the trade-off space (per-agent A* + local avoidance instead of shared flow fields) in a 4-player co-op game with tens of agents, not thousands. No cost figure transfers.
Lords of the Fallen	Hexworks	Unreal Engine 5 / 2023	·	Digital Foundry notes Virtual Shadow Maps add a lot to the visuals of Lords of the Fallen and questions why VSM cannot be toggled in the game's menu.	Do not copy the shipped configuration that omits a user-facing VSM toggle; that is a menu-policy choice, not a technical requirement.
Lords of the Fallen	Hexworks	Unreal Engine 5 / 2023	·	Digital Foundry measures the shipped PS5/Series X build at a dynamic 1152p with a 648p lower bound in performance mode.	Do not copy the 1152p/648p bounds; they are specific to this title's quality mode and hardware.
Lords of the Fallen	Hexworks	Unreal Engine 5 / 2023	·	Digital Foundry observes that at 30fps the shipped game's volumetric lighting runs at a higher resolution, reducing stair-stepping across streaks of light (the game's visible light-shaft-like effect).	Do not treat volumetric lighting as equivalent to a screen-space post-process light-shaft pass; the review describes volumetrics, not a radial-blur god-ray pass.
Lumen in the Land of Nanite (Epic demo)	Epic Games	Unreal Engine 5 (early access) / 2021	·	The demo's geometry (~882M Nanite triangles, 4.61 GB compressed on disk with Kraken level 5) is streamed through a fixed-page, GPU-request-driven pipeline whose root page is always resident and whose GPU transcode runs at ~50 GB/s on PS5 - i.e. the whole scene	4.61 GB on disk and 5.6 bytes per Nanite triangle are measurements of one Epic demo scene compressed with a specific codec at a specific level. They are not a general 'bytes per triangle' constant. The 50 GB/s transcode is a PS5 figure. Do
Lumen in the Land of Nanite (Epic technology demo)	Epic Games	Unreal Engine 5 (early access) / 2021	·	The demo that produced the published Nanite numbers: 433M source triangles -> 882M Nanite triangles, 25.90 GB raw -> 7.67 GB memory format -> 4.61 GB compressed disk, i.e. 5.6 bytes per Nanite triangle with Kraken level 5, rendered with no manual LOD authoring	One demo scene, one codec, one compression level. '5.6 bytes per Nanite triangle' is NOT a universal constant - it depends on topology regularity, attribute count and the LZ backend. Do not size a project's disk budget from it without measu
Lyra (Epic sample game)	Epic Games	Unreal Engine 5 / 2022	·	Epic points at Lyra as the worked example for migrating an existing animation system to the thread-safe/Fast Path model: 'For a workflow example of optimizing an Unreal Engine project to use Thread Safe functions, see the Adapting Lyra Animation to your UE5 Ga	Lyra is a small-scale sample; it demonstrates the migration technique, not the crowd budget that makes it necessary.
Lyra (starter game sample)	Epic Games	Unreal Engine 5 / 2022	·	Lyra is the shipped reference for Systems as a Service: 'Lyra impacts are managed by the B_WeaponImpacts class ... Having a separate object owning impacts solved an issue where impacts would despawn when switching weapons. B_WeaponImpacts has a default system	No measurement is published: there is no impact-per-second figure, no live-instance count and no frame-time cost for Lyra's impacts. The sample proves the pattern and names the owning class and the surface-template-per-type design; it does

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Lyra Starter Game (Unreal sample)	Epic Games	Unreal Engine 5 / 2022	·	Epic ships a full worked example of building, cooking and testing a dedicated server plus two clients, including the separate Server/Client Target.cs files and separate cook targets - the reference implementation for the build-side work.	Lyra is a small-scale shooter sample. It demonstrates the build mechanics, not the density economics - a working Lyra server says nothing about whether your game is affordable at scale.
Metro Exodus	4A Games	4A Engine (DXR) / 2019	·	Metro Exodus sorts ray-tracing instances by accumulated per-frame priority and throttles BLAS updates, keeping the queue in a 'balanced' state of about 5k-6k outdated instances out of 20k+.	Do not copy the instance-priority throttling numbers: they budget ray-tracing BLAS work, not dynamic punctual lights.
Metro Exodus	4A Games	4A Engine (DXR) / 2019	·	The shipped game had a 'super-lazy-realtime grid of probes for GI' and voxel GI before RTGI, and its RTGI path stores per-pixel irradiance as L1 spherical harmonics.	Do not copy the 96-bit/pixel SH irradiance encoding without Metro's two-pass denoiser and its HDR range assumptions.
Metro Exodus	4A Games	4A Engine (DXR) / 2019	·	BLAS/TLAS management runs as async compute hidden behind the pre-trace and SSR compute shaders, with both a compute-queue and a parallel-to-GBuffer mode implemented.	Do not copy the 'statistically insignificant difference' between Metro's two async modes without re-measuring on your hardware.
Minecraft	Mojang Studios	Java Edition (in-house) / 2011	·	The world is persisted per region/chunk rather than as one monolithic save file, so saving an arbitrarily large world only rewrites the regions that actually changed.	Minecraft's chunk format is a bespoke, simple record layout. It does not model Unity/Unreal serialized objects, and its write amplification behaviour will not match a GameObject or UObject save graph.
No Man's Sky	Hello Games	No Man's Sky engine / 2016	·	The GDC 2017 session abstract states the game transitions seamlessly from space to an interactive populated terrain and generates planets continuously in real time.	Do not copy a procedural-streaming architecture to a hand-authored world; NMS's approach is procedural by necessity.
No Man's Sky	Hello Games	Hello Games in-house / 2016	·	Planetary-scale rendering where an entire planet must be representable from orbit down to surface detail, forcing many levels of distance-based reduction.	Procedural planets generate LODs differently from authored meshes; the HLOD tooling is not transferable.
No shipped title documented	n/a - research prototype plus NVIDIA RTXNTC SDK	NVIDIA RTXNTC SDK (beta) / 2023	·	There is no shipped game to cite. NTC is a SIGGRAPH 2023 research result with an NVIDIA SDK; the strongest published claim is the project page teaser - 16x the texels of BC high at 30% less memory - and no runtime decode cost is published in the material I cou	Nothing can be transferred from a non-existent shipped title. The paper's numbers are the authors' own measurements on their own material set, and the per-texel reduction above is my arithmetic on two summary figures, not a published rate.
Overwatch	Blizzard Entertainment	Blizzard in-house / 2016	·	Blizzard's GDC 2017 talk states Overwatch uses an Entity Component System architecture, and that the team 'leverages ECS to curtail complexity, even as they continue to add new crazy features'; the same talk covers how the networked simulation uses determinism	Confirms ECS shipped in a hero shooter, but Overwatch's entity count is small (dozens of actors) - the motivation there was complexity management and determinism for netcode, not crowd scale. That motivation does not transfer to a project w

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Overwatch	Blizzard Entertainment	In-house (Overwatch engine) / 2016	·	Blizzard presented the game's gameplay architecture and netcode at GDC 2017, documenting a component-based entity administration for a shipped game with dozens of heroes and heavily recombining ability behaviour.	This confirms the approach was used and was worth a GDC talk for a networked hero shooter. It does not transfer Overwatch's specific component taxonomy, replication model, or performance results - those are bound to that game's networking m
Paragon	Epic Games	Unreal Engine 4 / 2018	·	ACL's reference shipped-game corpus: 6558 clips / 7h 00m 45.27s / 4276.11 MB raw, compressed to 181.25 MB (23.59:1) at a max error of 3.9834 cm. The data is used for research under NDA and is not publicly available.	The ratio is a property of Paragon's clip mix (including very short clips, 2 FPS clips, world-space clips and heavy scale animation). A project with mostly short looping cycles will not reproduce 23.59:1, and the 3.98cm max error is concent
PhysX-integrated titles (engine middleware)	NVIDIA (middleware)	PhysX SDK (used by Unreal, Unity and many in-house engines) / 2018	·	PhysX exposes an explicitly selectable broadphase (sweep-and-prune, multibox pruning, parallel variants, and a GPU broadphase), so broadphase selection is a documented, tunable production decision in shipped engines.	Confirms the mechanism is exposed in shipping middleware, not that any particular game chose any particular type. No game-specific body counts or timings transfer.
Portal	Valve	Source engine / 2007	·	Valve's own commentary states the shipped game renders the through-portal view with a virtual camera, switched to recursive frame-buffer rendering for antialiasing and memory reasons, and supports up to nine recursive portal views.	Do not copy the nine-view recursion cap or the offscreen-texture decision; the commentary presents them as trade-offs specific to Portal's hardware era.
Portal 2	Valve Corporation	Source engine / 2011	·	A shipped puzzle game built entirely around recursive portal rendering, where each visible portal requires re-rendering part of the scene.	Source's portal rendering is integrated into the engine's vis/rendering pipeline; a render-target-based approach in a modern engine has a different cost profile.
Quake II RTX	NVIDIA Lightspeed Studios	Q2VKPT (Vulkan, VKRay) / 2019	·	A fully ray-traced remaster in which reflections are resolved globally rather than by re-rendering a mirrored view per surface.	Q2VKPT is a research-grade renderer requiring RTX hardware; it is not a substitute for a planar reflection path on mid-range GPUs.
Quake II RTX	NVIDIA Lightspeed Studios	Q2VKPT (Vulkan, VKRay) / 2019	·	'Quake II RTX is a pure ray-traced game. That means all lighting, reflections, shadows and VFX are ray-traced, with no traditional effects or techniques utilized.' NVIDIA's newsroom calls it 'the world's first game that is fully path-traced'.	The content is a 1997-era game with very low geometric density; it proves viability, not affordability for a modern open-world asset budget.
Quake III Arena	id Software	id Tech 3 / 1999	·	The reference implementation of snapshot/delta state sync against a per-client acknowledged baseline is public: id's GPL release contains the server-side snapshot builder (sv_snapshot.c) that diffs entity state and writes a client snapshot.	The code is the right thing to read to understand the mechanism, but its field-level encoding, baseline retention policy and packet sizes were designed for 1999 modems. Do not copy the wire format; copy the bookkeeping model.
RAGE	id Software	id Tech 5 / 2011	·	id Software published the software virtual texture architecture it shipped - page table, physical page cache, GPU feedback/page-ID pass, anisotropic trilinear filtering with page borders, transcode on load and cache eviction - the productionisation of the mega	This is an architecture reference, not a performance result. No frame time, no VRAM figure and no texture resolution from RAGE is asserted here because I did not extract a verified number from the paper. Do not quote a 'megatexture size' fo

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Rainbow Six Siege	Ubisoft Montreal	AnvilNext + RealBlast / 2015	·	AI visibility is explicitly coupled to destruction state ('AI visibility through partially broken walls'), and the destruction event system warns that listeners must be asynchronous.	Shows the integration requirement (perception must be invalidated by world change) in a destructible multiplayer shooter. It does not transfer a perception query budget, and the coupling to RealBlast is specific to that engine.
Rainbow Six Siege	Ubisoft Montreal	AnvilNext + RealBlast / 2015	·	Procedural destruction is integrated with AI navigation: the game updates navlinks for trapdoors, breachable walls and floors as the environment changes.	Confirms the requirement (nav must follow destruction), not a tiling or streaming implementation. R6 Siege is a small-map multiplayer game, so its navigation footprint is not comparable to an open world; no memory or bake-time figure transf
Ratchet & Clank: Rift Apart	Insomniac Games / Nixxes Software	Insomniac Engine / 2023	·	The Windows port of a title built around instantaneous world transitions, delivered on PC with an SSD-oriented streaming path.	The specific IO API used by the port is not documented in this source; this evidences the pattern, not the API call.
Rayman Legends	Ubisoft Montpellier	UbiArt Framework / 2013	·	A counter-example that is useful for scoping: UbiArt is a 2D engine whose GDC 2014 session focuses on 'the design team's approach' and on 'the engine's role in improving prototyping and level design pre-production processes', i.e. art and design iteration, not	The session overview confirms tooling and prototyping benefits only. It does NOT confirm any deformation, bake or streaming technique, and a 2D/vector art pipeline is not evidence about a PBR character pipeline - it is evidence that one cla
Rayman Legends	Ubisoft Montpellier	UbiArt Framework / 2013	·	Very high on-screen sprite counts including effect frames, requiring atlased sources.	Same limitation: bespoke 2D pipeline, no published batching figures.
Red Faction: Guerrilla	Volition	Geo-Mod 2.0 / Havok / 2009	·	A former Volition physics lead states the shipped Geo-Mod 2.0 system fragments buildings into thousands of independent debris pieces at runtime, which precluded pre-baked lighting and forced navmesh rebuilds.	Do not copy fully dynamic destruction into a production that depends on baked lighting or precomputed navigation; this title had to give both up.
Red Faction: Guerrilla	Volition	Geo-Mod 2.0 / 2009	open_world · dedicated_server	Volition's GDC session is documented as describing how the multiplayer level design process had to account for destructible environments, including how to avoid being abused by them.	Geo-Mod 2.0 allows destruction of almost all structures; that level of destructibility forced bespoke level-design rules that do not transfer to a game with selective destructibility.
Remnant 2	Gunfire Games	Unreal Engine 5 / 2023	·	Digital Foundry's UE5 survey states Remnant 2 does not use Lumen and instead falls back to SDFAO (signed-distance-field ambient occlusion) and other raster techniques.	Do not copy the SDFAO fallback as a visual target: it is the cheaper non-Lumen path, not an equivalent GI solution.
Remnant 2	Gunfire Games	Unreal Engine 5 / 2023	·	Digital Foundry's UE5 survey states Remnant 2 does not use Lumen and falls back to SDFAO and other raster techniques.	Do not treat SDFAO fallback as GI: it is ambient occlusion, not a global-illumination solution.
Returnal	Housemarque	Unreal Engine 4 / 2021	·	A bullet-hell shooter with extremely dense particle effects, where the number of simultaneous emitters makes per-frame simulation of every effect unaffordable.	The talk covers Housemarque's own VFX pipeline; it does not state which effects are flipbooks versus simulated.
Returnal	Housemarque	Unreal Engine 4 / 2021	·	Dense atmospheric particle and light-shaft effects layered on top of heavy combat VFX.	The talk does not separate the cost of light shafts from other VFX.

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Roco Kingdom: World	Tencent Games / MoreFun Studios	Unreal Engine 4 (custom) / 2026	·	The shipped 8km x 8km cross-platform title uses baked PRT lightmaps for static lighting and dynamic indirect lighting, with Forward+ reserved for dynamic local lighting.	Do not copy the PRT/lightmap split to a game with fully dynamic lighting or destructive environments; this pipeline assumes static local lighting.
Roco Kingdom: World	Tencent Games / MoreFun Studios	Unreal Engine 4 (custom) / 2026	·	The shipped game stores baked PRT probe lighting in a Cascade Lighting Volume with 4 LODs, 4 m outdoor / 2 m indoor probe intervals, streamed and interpolated on the GPU.	Do not copy the 4 m / 2 m probe intervals or the 4-LOD CLV without re-deriving them from your world scale and memory budget.
Senua's Saga: Hellblade 2	Ninja Theory	Unreal Engine 5 / 2024	·	Digital Foundry states the game's Nanite geometry 'is nicely enhanced by the use of virtual shadow maps', a key UE5 feature used to raise shadow resolution for dense geometry.	Do not assume the VSM cost model of Hellblade 2 (narrow linear scenes) transfers to open worlds with heavy non-Nanite foliage.
Senua's Saga: Hellblade 2	Ninja Theory	Unreal Engine 5 / 2024	·	Digital Foundry reports Hellblade 2 supports dynamic resolution scaling, but only when using UE5 TSR upscaling, not with DLSS, XeSS or FSR2 in the menu.	Do not copy the menu restriction that couples DRS to TSR only; the upscaler SDKs support DRS independently.
Senua's Saga: Hellblade 2	Ninja Theory	Unreal Engine 5 / 2024	·	Digital Foundry praises Hellblade 2's graphics menu for exposing multiple reconstruction and frame-generation options and per-setting frame-time trade-offs.	Do not copy the menu's restriction that locks DRS to TSR; that was a UX decision flagged as a nitpick.
Senua's Saga: Hellblade II	Ninja Theory	Unreal Engine 5 / 2024	·	Character fidelity is a primary deliverable, requiring high-detail source sculpts reduced to a shippable real-time mesh.	Ninja Theory used photogrammetry and bespoke tooling; the pipeline cost is not transferable and no mesh budgets are published.
Senua's Saga: Hellblade II	Ninja Theory	Unreal Engine 5 (not stated in the cited source) / 2024	·	Ninja Theory presented a GDC 2025 talk specifically on ambience design and acoustic systems, i.e. the acoustic space itself is a designed, authored system in a shipped game.	Only the session record was verified; no slide content, parameter values or budgets were retrieved. The value here is as evidence that authored acoustic systems ship at AAA quality, not as a source of technique or numbers.
SpeedTree-based foliage (middleware reference)	IDV Inc. / NVIDIA	Direct3D 10, GeForce 8800 / 2007	·	GPU Gems 3 documents the production vegetation representation used by SpeedTree: leaf cards and silhouette fins extruded and traced with a height map for detailed silhouettes, plus relief mapping on trunks - i.e. the detail-per-triangle techniques that make a	This is a representation/middleware reference, not a game. The public page confirms the techniques; I quote no numbers from it. 2007-era D3D10 constraints (GeForce 8800) shaped the specific choices and are irrelevant to a modern target.
Split Fiction	Hazelight Studios	Unreal Engine 5 / 2025	·	Lead Programmer Jonas Mauritzsson: 'the graphics side is always a challenge, since we need to render both players' views at the same time, which imposes more limitations. This demands good communication between tech and art to find creative solutions.' The tea	No frame time, resolution or FPS figure is published. Hazelight's scale, art direction and UE5 customisations are their own; none of the cost transfers to another project.
Split Fiction	Hazelight Studios	Unreal Engine 5 / 2025	linear · coop	Hazelight states networking is always an extra challenge because many of the usual shortcuts do not work when the other player's screen is visible, and that the studio built a good foundation for networking its gameplay systems from the start.	Hazelight's networking is a bespoke, split-screen-first design with custom engine changes; the approach and any latency behaviour are not published and do not transfer to a dedicated-server shooter.

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Supreme Commander 2	Gas Powered Games	Supreme Commander 2 engine (in-house) / 2010	·	The flow field tile technique was built for and shipped in Supreme Commander 2: sector/portal world layout, cost/integration/flow fields, flow field cache, cost stamps for player-built structures, and a time-sliced rebuild priority queue.	The 50-70% clear-space ratio, the 10x10 sector grid, the 24/40-bit integration field and the 8-bit flow field are parameters of that RTS's terrain and memory budget. They are not performance numbers and do not transfer to a different world
Supreme Commander 2	Gas Powered Games	Supreme Commander 2 engine / 2010	·	Agents push each other and slide along walls using physics rather than a reciprocal-avoidance solver; this enabled explosions that push units back and super-large robots that push back a hundred tanks.	This is the physics-based alternative to RVO/ORCA, not an ORCA implementation. The specific gameplay scenarios (whirlwind structures, robots pushing a hundred tanks) depend on that game's physics integration and unit mass model; no avoidance
Team Fortress 2	Valve	Source / 2007	·	The reference case for art direction as an engineering specification: a peer-reviewed NPAR paper co-authored by engineers and the art director, in which the illustrative style ('early 20th century commercial illustration with 1960s industrial design elements')	A 2007 multiplayer shooter with nine character silhouettes and small maps. The readability requirement (identify a player at a glance across a map) does not transfer to a single-player game where the camera and the subject are authored, and
Team Fortress 2 / Counter-Strike (Source engine)	Valve Corporation	Source / 2004	·	Valve's shipped Source games run prediction at a default 15 ms timestep with <code>cl_predict</code> , smooth residual error over <code>cl_smoothtime</code> , and expose <code>cl_showerror</code> and <code>sv_showimpacts</code> so teams can see exactly when prediction and lag compensation disagree.	The default 15 ms timestep and 100 ms interpolation are 2004-era broadband assumptions (Valve's own rate recommendations top out at 10000 bytes/second). Reusing the tuning, as opposed to the technique, would be wrong today.
Team Fortress 2 / Counter-Strike (Source engine)	Valve Corporation	Source / 2004	·	The reference shipped implementation: one second of history, players-only by default, three selectable rewind scopes, an explicit 'use sparingly' warning for all-entity rewind, and shipped debug views (<code>sv_showimpacts</code> , <code>sv_showlagcompensation</code>) that let teams see	The one-second history is a 2004 latency assumption. On modern connections a shorter window is defensible; on a game with faster movement a longer one may be needed. The API shape and the debug tooling are the transferable parts.
Team Fortress 2 / Counter-Strike: Source (Source engine)	Valve Corporation	Source / 2004	·	Valve's counter-position: ship at 66 tick (15 ms), document that tickrate 100 costs ~1.5x CPU, and explicitly state that running higher than 66 is not suggested - while cs:go official servers settled at 64.	This is a 2004-era stance shaped by 2004 hosting economics and a 2004 player base. It is valuable as evidence that high tick rate is a choice with a documented cost, not as a recommendation for a 2026 competitive shooter.
Teardown	Tuxedo Labs (Voxagon)	In-house voxel engine / 2020	·	The studio's engine is voxel-based, so destruction is structural modification of the world rather than a pre-fractured, cached simulation - the contrasting architecture to a geometry cache.	This is the opposite approach, not an instance of geometry caching. A voxel world has different memory and authoring characteristics entirely; nothing about Teardown's performance or visual result transfers to a mesh-fracture pipeline. Also

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Teardown	Tuxedo Labs	In-house voxel engine / 2020	·	A shipped game whose world is a voxel grid, i.e. the scene representation that voxel-based GI techniques operate on directly.	Teardown does not implement voxel cone tracing; this evidences the data representation, not the algorithm.
Terrain Rendering in 'Far Cry 5'	Ubisoft Montreal	Dunia / 2018	·	A shipped AAA open-world terrain built on a GPU-driven pipeline where the CPU does streaming and the GPU does the terrain work (grid generation in compute, deferred texturing, cliff/rock shading) - the architecture lineage of the GPU clipmap approach.	I could not open the slides or video, so I assert no numbers from this talk at all - not a frame time, not a grid size, not a draw-call count. It is cited only as evidence that a shipped open-world game uses a GPU-driven terrain pipeline. A
The Tomorrow Children	Q-Games	Q-Games engine / 2016	·	The shipped PS4 title lit everything by tracing cones through six cascades of voxel 3D textures, occluding with screen-space directional occlusion and spherical occluders.	Do not copy the six-cascade voxel structure or the 16-direction cone set to hardware without the same memory/bandwidth budget.
Titanfall 2	Respawn Entertainment	Titanfall 2 engine (Source-derived) / 2016	·	Digital Foundry's breakdown observes a shadow 'filtering cascade' line on the floor on PS4/Xbox One and states console shadow quality is lower, with the PC retaining quality further out.	Do not copy console shadow-cascade split distances or the low-quality filtering cascade; PC/console budgets differ per platform.
Tom Clancy's Rainbow Six Siege	Ubisoft Montreal	AnvilNext 2 / 2015	·	The team published hard destruction budgets - roughly 6 ms of simulation for a wall, about 25 MB GPU memory, about 200 MB of data plus 150 MB engine overhead in RAM - and treat destruction as a designed, budgeted system with asynchronous event listeners.	These are Siege's own budgets for its destructible scope (mostly walls/hatches) on its target consoles. They do not transfer as targets: a game with whole-building destruction, a different fragmentation density, or a different platform has
Tom Clancy's Rainbow Six: Siege	Ubisoft Montreal	RealBlast / AnvilNext / 2015	·	The shipped game time-slices its procedural destruction, splitting functions into steps that are rescheduled if unfinished, to avoid 60 ms spikes.	Do not copy destruction time-slicing as a pathfinding technique; here it bounds destruction CPU, and the studio warns time-sliced multi-threaded code is hard to debug.
Tomb Raider	Crystal Dynamics	Crystal Engine (with AMD TressFX) / 2013	·	AMD's press release states the Crystal Dynamics collaboration delivered, with the launch of Tomb Raider, the world's first in-game implementation of a real-time, per-strand hair physics system.	This is a vendor press release: it confirms a per-strand hair system shipped in a major 2013 title, but it is not an independent measurement, gives no cost or strand count, and its 'world's first' framing is marketing. Nothing about the hair
Trials / RedLynx in-house engine	RedLynx (Ubisoft)	RedLynx in-house / 2015	·	Motivated by a modular in-game level editor where 'background [is] built from small objects' with high draw distance and no baked lighting, RedLynx pushed to a single draw call for the whole viewport and culls 250,000 moving objects with two DrawInstancedIndir	Achieved on a DirectX 11 path without Execute Indirect/MultiDrawIndirect, using 64-vertex strip clusters. The specific trick set (virtual deferred texturing, the 8xMSAA trick) exists to fit 2013-2015 console memory bandwidth budgets and is n
Trials-series in-house engine (RedLynx)	RedLynx (Ubisoft)	RedLynx in-house / 2015	·	Used virtual texturing primarily to eliminate texture-based batching, achieving a viewport rendered in a single draw call (x2) with a constant memory footprint, and built virtual deferred texturing and a virtual shadow map (128k ² , 256 ² pages) on the same pag	The 256k ² / 8k ² / 128 ² numbers are one team's configuration for a 2015 console memory budget on a specific engine. They are not recommended values. The measured 'up to 3.5x faster than SDSM' figure is for their virtual shadow map in comp

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Uncharted 4: A Thief's End	Naughty Dog	Naughty Dog engine / 2016	linear · single	Naughty Dog's GDC talk is documented as a deep dive into vertex shader pipelines and process using features implemented at Naughty Dog for Uncharted 4, covering wind systems, interactivity, ambient animations and optimisations.	Vertex-shader authoring for a linear, tightly authored game does not transfer to an open-world content pipeline with procedural placement.
Uncharted 4: A Thief's End	Naughty Dog	Naughty Dog engine / 2016	linear · single	Naughty Dog's GDC talk is documented as a deep dive into vertex shader pipelines used at Naughty Dog for Uncharted 4, covering wind systems, interactivity, ambient animations and optimisations.	Vertex-shader secondary animation is used to avoid CPU work in a linear, tightly authored game; it does not replace a general-purpose animation graph.
Unity DOTS production examples	Unity and partner studios	Unity DOTS / Entities / 2024	large-scale multiplayer and simulation examples · varies by project	Unity's official DOTS page lists production examples including V Rising, Megacity Metro and IXION.	Список showcase подтверждает применение, но не доказывает одинаковый выигрыш производительности для всех проектов.
Unity Netcode for Entities (DOTS sample stack)	Unity Technologies	Unity (ECS / DOTS) / 2022	·	Unity's DOTS netcode is a server-authoritative + client-prediction framework built directly on ECS, requiring Unity 2022.2.0f1 or higher, and marks predicted ghosts with a PredictedGhost component - i.e. the entire replication and prediction path operates on u	This is an engine capability, not a shipped game, so there is no player-facing result to copy. It also does not make allocation disappear - it moves it. Gameplay code that allocates per frame inside a system will still create garbage regard
Unreal Engine 5 (engine feature, adopted by LWC-enabled projects)	Epic Games	Unreal Engine 5.x / 2022	·	Epic shipped Large World Coordinates as the engine-level answer, raising default WORLD_MAX from 21 km to 88 million km and converting core engine types to doubles, while keeping the GPU path in floats through camera-relative translation.	This is an engine feature, not a shipped game's tuning. The 21 km / 88 million km figures are Epic's engine constants, not a recommendation for your world size. The fact that Epic flags LWC as Beta is the decision-relevant part.
Unreal Engine 5 Niagara-based titles (engine guidance)	Epic Games	Unreal Engine 5 (Niagara) / 2022	·	Epic publishes a dedicated Niagara scalability and best-practices page that treats GPU simulation as a scalability decision and warns that GPU particle cost is quantised (1 particle can cost the same as 64), i.e. GPU simulation is not a free win at low counts.	Engine guidance, not a shipped-game measurement. The 64-particle granularity figure is specific to a GPU dispatch model and is not a portable constant; it does not transfer as a budget to another engine or hardware generation.
Unreal Engine 5 projects using Niagara Fluids baked to flipbooks	Epic Games (engine feature)	Unreal Engine 5 (Niagara + Niagara Fluids plugin) / 2023	·	Epic's own worked example bakes the Grid 3D Gas Colored Smoke Niagara Fluids template - a 3D fluid effect - to a flipbook and re-applies it to a 2D sprite emitter, explicitly because the 3D effect cannot be run in real time on the target platform.	This is a documented engine workflow demonstrated on a template effect, not a shipped game. The 8x8/1024 defaults and the 30 fps default are starting points, not tuned values; the actual memory and performance result for another effect or p
Unreal Engine 5 sample 'Open World' / converted World Partition worlds	Epic Games	Unreal Engine 5.x / 2021	·	The Open World default map ships with World Partition, One File Per Actor, Data Layers and Hierarchical LOD all enabled by default, and contains a 2 km x 2 km sample Landscape - i.e. Epic's own template treats partitioning + OFPA + Data Layers + HLOD as one ma	This is a template, not a shipped game. The 2 km x 2 km landscape and the four-by-default features are a starting configuration for an Epic sample; they are not a validated budget for any real project and should be treated as a demonstratio

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
Unreal Engine 5 titles using Chaos Vehicles	Epic Games (engine feature)	Unreal Engine 5 (Chaos) / 2022	·	Unreal documents that Chaos Vehicles support the engine's Asynchronous Physics mode, and markets it primarily as improving determinism and predictability rather than raw throughput.	Engine capability, not a shipped-game result. No measured speedup, thread count, or body count transfers, and the determinism benefit is a property of that engine mode, not of async physics in general.
Unreal Engine City Sample	Epic Games	Unreal Engine 5 / 2021	4 km x 4 km city sample · technical demo	Официальная документация City Sample связывает World Partition, Nanite, Lumen, VSM, Mass AI, Chaos и MetaSounds.	Это демонстрационный проект; его polygon/asset counts и требования не являются минимальными требованиями для любой игры.
Unreal Engine Landscape (shipped engine terrain)	Epic Games	Unreal Engine 4/5 / 2014	·	Ships the simple, predictable option instead of an exotic codec: 16-bit height over a fixed range with a project Z scale, plus a component/section structure whose dimensions are constrained to (A*Quads+1, B*Quads+1) and capped at 1024 components - i.e. Epic tr	The 1024-component recommendation and the -256..255.992 range are Epic's constraints for its Landscape actor. They are not a compression scheme and have no bearing on a custom terrain codec beyond illustrating the trade-off between rate and
Unreal Engine MetaHuman grooms	Epic Games	Unreal Engine 5 / 2021	·	Unreal's own groom documentation illustrates a MetaHuman character with grooms rendered as strands (left) and as cards (right), i.e. Epic's reference character pipeline ships both representations of the same groom.	This confirms the dual-representation workflow is shipped by Epic, but it is an engine demo asset, not a game shipping on a specific platform. No cost, hair count, or swap distance transfers.
Unreal Engine Niagara sprite renderer (engine capability)	Epic Games	Unreal Engine 5 (Niagara) / 2023	·	Unreal's Sprite Renderer exposes Sub Image Size and works with a Sub UVAAnimation module, i.e. the engine has first-class support for tiled particle atlases and documents the correct sizing rules.	Engine capability, not a shipped-game result. No draw-call count, saving, or visual outcome transfers.
Unreal Engine groom samples / MetaHumans	Epic Games	Unreal Engine 5 / 2021	·	Unreal ships a documented strand-based groom pipeline with dedicated pages for enabling physics simulation on grooms, groom interpolation, and groom caches, and states strands require an allocated budget and are not supported on all platforms.	Engine capability, not a shipped game result. No cost, strand count, or platform outcome transfers to another project.
Unreal Engine shipped titles using Chaos Cloth	Epic Games (engine feature)	Unreal Engine 5 (Chaos) / 2020	·	The engine ships a documented, in-editor cloth authoring and simulation pipeline (Chaos Cloth solver, painted masks with Max Distance / Backstop / Anim Drive targets, Physics-Asset-driven collision) rather than each studio writing its own solver.	This confirms the technique is productionised in a shipping engine, but it is not a specific game, and no cost, particle count or visual result transfers to another title. The engine feature existing says nothing about whether a given proje
Unreal Engine third-person template worked example	Epic Games	Unreal Engine (Lightmass) / 2019	·	Epic's own documentation walks through deriving PlayAreaHeight from gameplay: measure the camera's highest rotational point (~395 units), add jump height (~210 units), and add buffer to reach 650, noting that higher values cost more runtime memory because more	The 395 / 210 / 650 numbers are properties of Epic's ThirdPerson template character rig and camera. They are not a recommended PlayAreaHeight; the transferable content is the derivation method (max camera height + max jump + buffer).
Unreal Engine titles using SaveGame (engine guidance)	Epic Games	Unreal Engine 5 / 2022	·	Unreal ships a documented save/load framework with an explicitly recommended async save path and a stated caveat that synchronous saving is only sufficient for small save formats.	Engine guidance, not a shipped-game measurement. No save size, duration, or hitch figure transfers; storage performance is platform-specific.

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
V Rising	Stunlock Studios	Unity (ECS for Unity / DOTS) / 2022	·	Unity documents that Stunlock Studios used ECS throughout development of V Rising, an open-world multiplayer survival game; ECS is the data-oriented alternative to per-agent throttling.	This is a vendor production page. It confirms ECS adoption, not that V Rising used a behaviour-tree update budget, and it reports no agent counts or framerate numbers.
V Rising	Stunlock Studios	Unity (ECS for Unity / DOTS) / 2022	·	Unity documents that Stunlock Studios used ECS throughout the development of V Rising, an open-world multiplayer survival game, including world building in the Editor with custom visual scripting and scalable open-world streaming.	Vendor production page. Confirms full-project ECS adoption in a shipped multiplayer survival game, but reports no entity counts, no framerate, and no evidence about crowd-specific use. Adoption by one studio is not evidence that ECS is the
VALORANT	Riot Games	собственная серверная инфраструктура / 2020	соревновательные арены · authoritative 128-tick server	Инженерный материал Riot разбирает серверную производительность и сетевую модель VALORANT.	128 tick - свойство описанного сервиса, а не универсальная рекомендация для каждой игры.
VALORANT	Riot Games	Unreal Engine 4 (not stated in the cited Riot article) / 2020	·	Riot hit the replication scan as a server CPU cost while chasing a 2.34 ms per-frame budget at 128 tick, and responded partly by moving state off replicated variables onto RPCs, reporting 100x-10000x improvements in many cases.	VALORANT is a 10-player fixed-arena game; it never had a relevancy problem in the battle-royale sense, only a per-frame scan problem. Their 'use an RPC instead' answer is a 10-client answer - it does not scale to 100 clients the way a relev
VALORANT	Riot Games	Unreal Engine 4 (not stated in the cited Riot article) / 2020	·	Riot published the full economics of their dedicated servers: 128 tick, a 2.34 ms per-frame budget derived from 3 games per core, 36-core hosts running 108 games, and a 50 ms -> sub-2 ms frame-time optimisation programme including NUMA binding, C-state limitin	Every number here is a function of Riot's game (10 players, fixed arena, 128 Hz), their hardware (36-core hosts) and their scale. Nothing about the 2.34 ms budget or 108 games/host transfers to a 100-player BR or a 32-player co-op game; wha
Valley of the Ancient	Epic Games	Unreal Engine / 2017	·	Epic's own shipped GAS sample is documented in the primary GAS page itself: '### Valley of the Ancient Sample' with a '### Walking Animation Example' and a '### Charge Attack Example', i.e. the reference implementation demonstrates abilities driving both locomo	It is a technology sample, not a shipped game with a measured ability count, and no performance figure is published for it. It proves the framework is complete enough to drive animation and combat; it says nothing about how many concurrent
Valley of the Ancient (Epic UE5 sample)	Epic Games	Unreal Engine 5 /	open_world · single	The Gameplay Ability System page states that Echo's charge-and-attack animation and the walking animation in the Valley of the Ancient sample are examples of a Gameplay Ability.	This is a first-party tech sample, not a shipped commercial title; its ability setup does not represent production content scale.
Valley of the Ancient (Epic sample project)	Epic Games	Unreal Engine 5 / 2022	·	Epic's own GAS/animation sample is the reference project for the shipped UE5 character pipeline, i.e. the GPUSkinVertexFactory / Skin Cache paths described in the rendering-paths documentation, rather than a bespoke skinner.	A sample project with one hero character says nothing about crowd-scale skin-cache VRAM pressure, which is the actual failure mode of this method.
Warhammer 40,000: Space Marine	Relic Entertainment	Relic in-house (Essence) / 2011	·	The RVO2 Library maintainers state the library 'has been licensed for the video game Warhammer 40,000: Space Marine from developer Relic', and quote a Daily Telegraph review noting that with many orks on screen 'the framerate never takes a hit'.	Confirms the library was licensed for a shipped title, but not which agent counts or subsystems used it, nor the integration cost. The newspaper quote is a review impression, not a measurement, and does not transfer as a performance number.

Кейс	Студия	Технология / год	Сценарий	Подтверждаемый факт / механизм	Ограничение переноса
World Partition HLOD reference workflow (Unreal Engine 5)	Epic Games	Unreal Engine 5.x / 2021	.	Epic's own shipped workflow demonstrates the class split: a merged-mesh default layer for architecture with baked lightmaps and materials, and a separate Instancing layer for a Blueprint tree class (HLODLayersForActorClasses maps Base_Tree_C to HLODLayer_Tree)	The LoadingRange=30000 and TargetLightMapResolution=256 in Epic's example are placeholder demonstration values. They encode nothing about a real game's draw distance or art density; treat the pattern (separate layers per content class on se

Кейс подтверждает факт применения, устройство или компромисс. Он не переносит FPS, количество активных сущностей, tick rate, latency, размер команды или требования к железу. /recommend показывает кейсы как практическую сверку со статусом not_calibrated.

17. Риски

Риск	Почему важен	Контроль
Evidence drift	A page or package changes while a claim stays published.	checked_at, version, locator, availability and review status; re-review before release.
False numerical precision	An expert score looks like FPS or benchmark data.	basis is explicit; measured/documented/derived are separated from expert_estimate and unknown.
Hidden dependency	A method appears available because an API, package, plugin or version was not modeled.	directed graph, transitive closure, unresolved edge list and version checks.
Late architecture change	A sound method is selected after content and tools are locked.	late_factor, stage warning, feasibility package and P50/P80 rework range.
Resource double counting	CPU/GPU/memory effects are multiplied or the same savings are counted twice.	one FrameModel; sequential/parallel CPU, raster/RT GPU and memory composition are separate.
Case over-transfer	A successful game is treated as a template with the same FPS or team size.	case evidence records relevance and transfer limits; no automatic numeric bonus.

18. Итоговый план внедрения

- Зафиксировать revision, входной профиль и опубликованный snapshot.
- Прогнать Alembic и проверить резервную копию SQLite.
- Выполнить seed доказательств; проверить claims без источника, локатора и версии.
- Ревизовать 40 функций, 124 методов, движки, инструменты, method-tool links, conflicts и hardware anchors.
- Для выбранной корзины закрыть prerequisites, version/API compatibility и hard conflicts.
- Запустить feasibility/prototype с трассами CPU/GPU/IO/network; сохранить условия измерения.
- Обновить raw benchmark anchors только вместе с контекстом и датой.
- Пересчитать P50/P80 и critical path по фактическим ролям команды.
- Проверить минимальную конфигурацию на Windows и Linux, затем пройти QA/regression/release gates.

19. Ограничения модели, критерии приёмки и аудит

19.1 Ограничения

Без исходного кода, ассетов и runtime-профиля невозможно честно вывести точный FPS, универсальную latency, точный размер RAM/VRAM или календарную дату. Публичная документация и книги подтверждают устройство механизма, но не обещают его результат в другом проекте. Нормализованные hardware indices пригодны для ранжирования внутри каталога и сценарного сравнения; это не замена измерению.

Статус калибровки: not_calibrated. Процент точности не показывается. Чтобы изменить статус, нужна реальная выборка: одинаковый профиль, версия движка, commit ассетов, target platform/API, measured frame-time percentiles, memory, IO and network metrics, а также правило train/test split.

19.2 Критерии приёмки

- одинаковый вход, algorithm version и catalog revision дают одинаковый результат;
- увеличение объекта/NPC count, resolution или quality не уменьшает соответствующую нагрузку;
- CPU main/parallel, GPU raster/RT, RAM/VRAM и storage/network не смешиваются без подписи;
- неизвестная зависимость не считается закрытой;
- hard conflict не попадает в рабочую корзину;
- complement не даёт числовой бонус без измерения;
- удаление prerequisite делает метод явно неприменимым;
- team size меняет календарь, но не person-days;
- critical path строится по DAG, P80 >= P50;
- отсутствие числовых данных получает unknown/expert_estimate;
- non-PC цели не попадают в PC quantitative hardware estimate;
- source title, authors/publisher, type, version and locator видны пользователю;
- Markdown и PDF проходят текстовый и визуальный QA.

19.3 Аудит покрытия каталога

Проверка	Значение	Интерпретация
methods: implementation / optimization	67 / 57	catalogue snapshot
methods: confidence < 0.70	20	manual review queue
methods: no explicit conditions	108	applicability gap
methods: no engine/platform/HW filter	122 / 114 / 111	transferability gap
method-tool links without own source_url	111 / 473	relationship provenance gap
engines/tools without docs_url	1 / 7	version/tooling gap
relations: alternative/complement/risk/unknown	41 / 183 / 75 / 8	conflict graph
distinct method source URLs	91	source reuse is not claim coverage

19.4 Глубокие исследовательские карточки

Потоковая загрузка мира и HLOD

Факт и источник: Epic описывает World Partition как единый persistent level, разделённый на grid cells, которые загружаются и выгружаются по streaming sources и runtime grid S01. HLOD заменяет дальние выгруженные ячейки proxy mesh/material и имеет отдельные режимы Instancing, Merged Mesh и Simplified Mesh S24. PCG может назначать сгенерированные актёры в Data Layers и HLOD Layers S25. В CryEngine документирован асинхронный StartRead: запрос проходит I/O, decompression threads, callback threads и затем main-thread completion; порядок запросов сортируется для эффективного чтения S28.

Синтез для DSS: Синтез: размер мира сам по себе не является числом для FPS. Управляемые переменные — размер ячейки, loading range, число источников, HLOD-представление, размер чанка, скорость чтения, декомпрессия и время активации. Поэтому в DSS они должны быть отдельными полями, а не одним scalar open_world=true. HLOD уменьшает видимую работу дальних объектов, но

добавляет cook/build и риск несоответствия проху исходной геометрии. PCG создаёт повторяемый content pipeline, но не гарантирует дешёвую runtime-сцену.

Проверка в проекте: Проверка: прогнать один маршрут и телепорт с cold/warm cache на Windows и Linux; сохранить active/loaded/visible cells, объём запрошенных байтов, I/O latency p50/p95/p99, decompression time, activation time, RAM/VRAM и frame-time spikes. Отдельно сравнить HLOD off/instancing/merged/simplified и PCG-generated/static content; в отчёте хранить commit ассетов и настройки grid/range.

Источники: S01, S24, S25, S28.

Рендеринг: геометрия, освещение, тени и API-возможности

Факт и источник: Nanite описан Epic как virtualized geometry с fine-grained streaming и автоматическим LOD; документация перечисляет поддерживаемые типы контента и практические ограничения S02. Lumen предназначен для динамического GI/reflections, но стоимость зависит от детализации, view distance, hardware/software path и числа экземпляров S03. Таблица rendering paths связывает Nanite/VSM/Lumen с RHI, Shader Model и RT-возможностями S04. VRS имеет tiered capability и ограничения по shading-rate S05, а mesh shaders заменяют традиционный VS/HS/DS/GS pipeline только на устройствах с нужной поддержкой S06.

Синтез для DSS: Синтез: GPU raster, GPU RT, VRAM, API и feature flags — независимые оси. Нельзя суммировать Nanite, VSM, Lumen, VRS и upscaling как проценты экономии: часть методов меняет объём работы, часть — способ её планирования, часть — качество/разрешение. Ветка capability должна предшествовать TOPSIS: D3D12 feature level задаёт функциональность, но не производительность S38; конкретные опциональные возможности запрашиваются через CheckFeatureSupport S39, а RT tier — отдельная проверка S40.

Проверка в проекте: Проверка: фиксировать сцену, camera path, draw resolution, internal resolution, upscaler, RT on/off, Nanite/VSM/Lumen/VRS flags и driver/API. Снять CPU main/render thread, GPU passes, raster/RT queue, shader/mesh preparation, VRAM residency and p99 frame time в packaged build. Критерий meets применять к frame-time budget, а не к одной средней частоте кадров.

Источники: S02, S03, S04, S05, S06, S19, S20, S21, S38, S39, S40.

Симуляция, ECS/Jobs и адаптивный AI

Факт и источник: Unity Job System запускает пользовательскую многопоточную работу на worker threads и строит dependency chains; документация отдельно предупреждает о главном потоке, Complete и маркере WaitForJobGroup S31, S46, S58. Unity перечисляет production cases DOTS: V Rising, IXION, Zenith: The Last City и Detonation Racing S32. В докладе Valve по Left 4 Dead описаны Survivor Intensity, tracking peak intensity, threat population и паузы между пиками S11. Godot рекомендует frustum/occlusion culling, LOD/HLOD, MultiMesh и измерение стоимости animation/physics/lighting S10.

Синтез для DSS: Синтез: ECS/Jobs может уменьшить последовательную часть и улучшить locality, но добавляет копирование, dependency barriers, ownership и миграцию данных. AI Director управляет пиками популяции и драматическим pacing; он не уменьшает автоматически стоимость одного NPC. В DSS это разные узлы: population_control, behaviour_update, navigation, perception, animation, physics. Нельзя переносить рекламное описание DOTS или поведение L4D в проценты ускорения другого проекта.

Проверка в проекте: Проверка: выбрать фиксированный seed и нагрузочные точки N=50/100/200/500 агентов; сравнить baseline, jobs/ECS и director on/off. Снимать main-thread, worker timeline, wait/barrier time, allocations, cache-sensitive data where available, navmesh queries, animation/physics, peak active NPC and frame-time p99. Для determinism/networked simulation добавить повторные прогоны и checksum состояния.

Источники: S07, S08, S09, S10, S11, S31, S32, S46, S58.

Сетевое моделирование и релевантность

Факт и источник: Valve объясняет в CS2 sub-tick updates как передачу точного момента движения, выстрела или броска внутри серверного такта S12. Riot фиксирует для VALORANT simulation timestep 128 Hz, работу prediction/correction и влияние client/server buffering S13. В отдельной статье Riot приводит расчёт: 128 Hz = 7.8125 ms, при трёх играх на ядро целевой бюджет 2.6 ms, после резерва 10% — 2.34 ms на игру S14. Epic показывает, что Replication Graph строит списки актёров для каждого соединения, а Actor Relevancy отсекает невливающие объекты S26, S27. Godot high-level multiplayer использует UDP/ENet и разные каналы для сообщений с разной надёжностью S30.

Синтез для DSS: Синтез: server tick, render FPS, input latency, buffering, relevance set, serialization, packet loss и bandwidth должны моделироваться раздельно. Sub-tick, prediction и rewind — дополняющие части протокола, а не три независимые скидки. Пример Epic с Fortnite (100 players/около 50 000 replicated actors) — документация механизма, а не лимит нового проекта. Число игроков без частоты обновлений и размера состояния не определяет traffic.

Проверка в проекте: Проверка: стенд 1/2/4/8/16 клиентов с controlled RTT, jitter, loss, reordering и bandwidth cap; сравнить fixed tick 30/60/128, relevance on/off и aggregation. Логировать server frame p50/p95/p99, simulation divergence, hit validation, snapshot/input bytes, packets, queueing/buffering и client input-to-photon proxy. Server-only costs не включать в PC client hardware row.

Источники: S12, S13, S14, S26, S27, S30, S53.

Аудио, окклюзия и читаемость

Факт и источник: CryEngine документирует режимы No Ray, SingleRay и MultipleRay; MultipleRay следует включать только если точности одного raycast недостаточно. Raycasts пропускаются для объектов без активного audio trigger S29. Там же описаны surface obstruction, отдельные occlusion/obstruction значения и отключение расчёта за s_OcclusionMaxDistance. Crytek в материале Hunt: Showdown связывает реалистичность с readability and consistency S17.

Синтез для DSS: Синтез: бюджет аудио зависит от числа активных источников, ray policy, частоты обновления, surface materials, DSP/middleware and mix graph. Один параметр spatial_audio=true скрывает важные trade-offs. Более точная окклюзия может улучшить физическую правдоподобность, но не обязательно UX; читаемость — отдельная quality goal. Поэтому audio_quality, audio_cpu, ray_queries и device/output нельзя объединять в один балл.

Проверка в проекте: Проверка: записать одинаковую сцену со статическими/динамическими преградами и 10/50/200 активными источниками; сравнить No Ray/Single/Multiple, distance cutoff и material classes. Снимать ray/query count, DSP time, CPU, memory, output latency и blinded listening/UX result. Хранить устройство вывода и микс, иначе сравнение не воспроизводится.

Источники: S17, S18, S29.

Сборка, cooking, чанки и I/O

Факт и источник: Unreal описывает packaging как последовательность Build, Cook, Stage, Package, с необязательными Deploy/Run; cooking конвертирует и оптимизирует ассеты под платформу, исключает неиспользуемые данные и создаёт Pak-файлы S23. Chunking разделяет контент для DLC/patch/streaming installation S23. Microsoft DirectStorage guidance требует проверять capability, compression path, staging and fallback, а GPU decompression конкурирует за GPU resources S52. Unreal texture metrics различают pool usage, wanted mips and streaming updates S54, S55.

Синтез для DSS: Синтез: build-time, install size, cold-start, streaming bandwidth, decompression CPU/GPU and runtime memory are different metrics. Chunking can improve delivery and patch granularity but creates dependency/packaging checks. A faster storage device cannot compensate for insufficient asset prioritization or activation work. DSS therefore keeps storage as an independent resource axis and marks DirectStorage as conditional capability, not universal optimization.

Проверка в проекте: Проверка: clean checkout and incremental cook; report code compile, shader compile, cook, stage, package, patch size, installed bytes, cold/warm startup, first-use hitch, read/decompress/activate latency, CPU/GPU queues and pool/wanted mips. Compare NVMe/SATA SSD/HDD only on the same build, OS cache policy and content commit; record fallback path.

Источники: S23, S28, S52, S54, S55.

Движки, инструменты, версии и переносимость

Факт и источник: Unity documents Jobs, Entities/DOTS, Addressables and Netcode as versioned packages/workflows; DOTS production examples show that the tool is used in different game classes, but not that one numeric gain transfers S31, S32. Unreal publishes a rendering-path matrix and separate hardware/software requirements S04, S41. Valve's developer resource shows that Hammer/Authoring Tools are branch/game-specific, with Source 2 tooling distributed through particular games S43. TGS now labels HeroEngine as legacy and its current site warns about official domains/status S44, S45.

Синтез для DSS: Синтез: поддержка движка — это не бинарный engine=yes. Она должна включать version, package/plugin, target platform/API, scope (runtime/editor/build/server), tool relation and evidence freshness. HeroEngine оставлен в каталоге из-за фиксированного охвата, но его public applicability — legacy/needs_review, а не подтверждённая текущая совместимость. Custom всегда требует project-owned design record. Branch-specific tools нельзя связывать с методом без source_url and locator.

Проверка в проекте: Проверка: для каждой пары method-tool зафиксировать engine version, tool/plugin version, platform, scope, docs URL, minimal reproduction and owner. Проверять upgrade/downgrade, clean install and package lock. Unknown/missing source должен дать unknown, а не supported; incompatible capability отфильтровывается до ranking.

Источники: S04, S31, S32, S41, S43, S44, S45.

Выбор по качеству, бенчмаркам и evidence hierarchy

Факт и источник: ISO/IEC 25010:2023 задаёт модель качества продукта с характеристиками и подхарактеристиками для specification, measurement and evaluation S33. ISO/IEC 20741 отдельно описывает purpose-oriented evaluation and selection of software engineering tools и предлагает учитывать capabilities и quality characteristics S34. PassMark предупреждает, что CPU Mark — агрегат тестов, single-thread полезнее для плохо распараллеливаемых приложений, а benchmark не воспроизводит каждую нагрузку S49, S50. 3DMark позволяет сравнивать связки CPU/GPU и смотреть frame rate/температуры, но не заменяет game-specific workload S51.

Синтез для DSS: Синтез: TOPSIS оправдан как прозрачное сравнение альтернатив при заданных весах, но его score — decision utility, не измерение. ISO-подход означает сначала определить quality goals и measurable acceptance criteria, затем выбрать метод/инструмент. Hardware benchmark anchor должен хранить raw value, тест, дату, API/resolution/RT context and normalization formula. Confidence — качество evidence, не процент FPS.

Проверка в проекте: Проверка: прогнать sensitivity weights $\pm 10\%$, добавить/убрать один критерий, сравнить ranking stability and reason codes. Для hardware собрать same-game or same-engine trace on target devices; до этого вывести class/range and gap list. Separate benchmark source from evidence claim and attach locator/context.

Источники: S33, S34, S49, S50, S51.

Оценка сроков, PERT и critical path

Факт и источник: PMI описывает three-point estimating и различает triangular $CE = (O+M+P)/3$ и beta/PERT $CE = (O+4M+P)/6$ S35. NASA описывает PERT/CPM через precedence network, три оценки activity, mean/variance and longest-path critical path, а также ограничения метода S36. NASA glossary определяет critical path как longest path and отдельно предупреждает о double counting uncertainty/risk S57. Amdahl формализует верхнюю границу ускорения как $S(N) = (Ts+Tp)/(Ts+Tp/N)$ и отмечает parallel overhead S37.

Синтез для DSS: Синтез: person-days и calendar duration — разные показатели. Команда может распараллелить независимые пакеты, но не отменяет dependencies, integration, QA или scarce roles. P50/P80 текущего DSS — scenario estimates; пока нет исторической выборки, нельзя называть их статистически калиброванными percentiles. Риски не должны второй раз прибавляться к уже расширенному P80.

Проверка в проекте: Проверка: хранить O/M/P, formula, assumptions and role capacity per package; строить DAG and longest path. Для примера из отчёта вычислить PERT mean/sigma, затем сопоставить фактическое завершение с prediction interval. После нескольких проектов заменить шаблонные коэффициенты empirical calibration, сохранив train/test split and revision.

Источники: S35, S36, S37, S57.

19.5 Воспроизводимые расчёты (сводка)

Target FPS	Frame budget, ms	Интерпретация
30	33.333	low/quality-first target
60	16.667	common responsiveness target
90	11.111	high-refresh baseline
120	8.333	high-refresh target
144	6.944	competitive/high-refresh case
240	4.167	very high-refresh case; content-dependent

Разрешение	Pixel ratio	База
1280x720	0.444	relative to 1920x1080
1920x1080	1.000	baseline
2560x1440	1.778	relative to 1080p
3840x2160	4.000	relative to 1080p; 2.25x 1440p

Workers	Speedup	Допущение
N=1	1.000	baseline
N=2	1.429	s=0.40, idealized
N=4	1.818	s=0.40, idealized
N=8	2.105	s=0.40, idealized
N=16	2.353	s=0.40, idealized
N→∞	2.500	serial ceiling 1/s

Inputs	Calculation	Статус
O=2, M=4, P=10 человеко-дней	$(2 + 4 \times 4 + 10) / 6 = 4.667$	beta/PERT expected duration
Тот же диапазон	$\sigma = (10 - 2) / 6 = 1.333$	approximate spread; not a calibrated percentile
P80 proxy, only if explicitly assumed normal	$4.667 + 0.8416 \times 1.333 = 5.789$	derived scenario, not a measurement

Task	Duration	Predecessors	Path result
A Design	3	—	A-B-D-E / A-C-D-E
B Prototype	5	A	A-B-D-E = 18
C Content/asset prep	8	A	A-C-D-E = 21; critical
D Integration	6	B,C	merge waits for max(B,C)
E QA/release gate	4	D	longest path ends at 21

Scenario	Inputs	Derived result	Limits
Server → 9 clients	C=9, f=20/s, payload=120 B, IPv4+UDP H=28 B	$9 \times 20 \times (120 + 28) = 26,640$ B/s ≈ 26.0 KiB/s ≈ 213 kbps	derived; excludes encryption, retransmits and other traffic
128 Hz server frame	1 / 128 s	7.8125 ms per tick	published VALORANT case S14; not universal target
Riot 3 games/core target	7.8125 / 3	2.6042 ms; $\times 0.90 = 2.3438$ ms	published case S14; host/game-specific

Component	Inputs	Result	Basis
Textures / geometry	2.4 + 1.1	3.5 GiB	scenario input
Render targets / transient	0.8 + 0.6	1.4 GiB	scenario input
Audio / other	0.2	0.2 GiB	scenario input
Subtotal	3.5 + 1.4 + 0.2	5.1 GiB	sum, no headroom
20% safety/headroom	5.1 × 1.20	6.12 GiB	derived planning allowance

Расчёты выше получены из явно указанных входов и формул. Они показывают структуру бюджета и границы сценария, но не являются измерением конкретной игры.

19.6 Публичный API

- GET /api/catalog/sources, /catalog/evidence, /catalog/evidence-summary;
- GET /api/catalog/cases, /catalog/cases/{code};

- GET /api/catalog/dependencies, /catalog/teams, /catalog/graph-checks;
- POST /api/schedule; POST /api/report-data, GET /api/report-data;
- расширенные /recommend и /hardware-estimate с evidence summary, cases, P50/P80, target assessments и unresolved items.

UI содержит экраны «Доказательства», «Кейсы игр», «Зависимости и конфликты», «Трудоёмкость и календарный план»; badges показывают documented, measured, derived, case_evidence, expert_estimate и unknown.

20. Полный список источников

Всего в реестре 935 источников. Ниже — полный перечень с типом, датой проверки, локатором и доступностью, затем активные ссылки.

Код	Название	Тип	Проверен	Локатор	Доступность
BOOK_GAME_ENGINE_ARCHITECTURE	Game Engine Architecture, Third Edition	book	2026-09-10	chapters 3-4, 16: engine architecture, runtime, resource sys	available
BOOK_GAME_PROGRAMMING_PATTERNS	Game Programming Patterns	book	2026-09-10	chapters: Data Locality, Object Pool, State	available
BOOK_GPU_GEMS_3	GPU Gems 3: Programming Techniques for High-Performance Graphics	book	2026-09-10	chapters on volumetric lighting, shadows and post-processing	available
BOOK_REAL_TIME_RENDERING	Real-Time Rendering, Fourth Edition	book	2026-09-10	chapters 2-3, 8: rendering pipeline, transformations, shadow	available
CLUSTERED_SHADING	Clustered Deferred and Forward Shading (Olsson, Billeter, Assarsson)	research	2026-09-10	overview page	available
COENEN_DOOM	Doom Eternal graphics study (Simon Coenen)	secondary	2026-09-10	overview page	available
CS2_SUBTICK	Counter-Strike 2: Moving Beyond Tick Rate	official_game_material	2026-09-10	section: Sub-tick updates	available
DF_ITTakesTWO	It Takes Two tech analysis (Digital Foundry)	secondary	2026-09-10	section: split-screen rendering and performance analysis	available
DOOM_ETERNAL	Rendering the Hellscape of Doom Eternal (SIGGRAPH 2020)	research	2026-09-10	slides: geometry caches, gore, material compositing, target	available
GAFFER_TIMESTEP	Fix Your Timestep! (Gaffer on Games)	secondary	2026-09-10	overview page	available
GAME_AI_FLOW_FIELDS	Crowd Pathfinding and Steering Using Flow Field Tiles — Elijah Emerson	secondary	2026-09-10	overview page	available
GODOT_DECALS	Godot Docs: Using Decals	official_documentation	2026-09-10	overview page	available
GODOT_LIGHTS	Godot Docs: Lights and Shadows	official_documentation	2026-09-10	overview page	available
GODOT_MESHLOD	Godot Docs: Mesh Level of Detail	official_documentation	2026-09-10	overview page	available
GODOT_MULTIMESH	Godot Docs: MultiMeshInstance3D	official_documentation	2026-09-10	overview page	available
GODOT_MULTIPLAYER	Godot Docs: High-level Multiplayer	official_documentation	2026-09-10	section: High-level multiplayer	available
GODOT_OCCLUSION	Godot Docs: Occlusion Culling	official_documentation	2026-09-10	overview page	available
GODOT_PARTICLES	Godot Docs: 3D Particles	official_documentation	2026-09-10	overview page	available
GODOT_PERF	Godot Docs: Optimizing 3D Performance	official_documentation	2026-09-10	section: Optimizing 3D Performance	available
GODOT_PHYSICS	Godot Docs: Physics Introduction	official_documentation	2026-09-10	overview page	available
GODOT_THREADS	Godot Docs: Using Multiple Threads	official_documentation	2026-09-10	overview page	available
GPP_DATALOCALITY	Game Programming Patterns: Data Locality	book	2026-09-10	overview page	available
GPP_OBJECTPOOL	Game Programming Patterns: Object Pool	book	2026-09-10	overview page	available
GPP_STATE	Game Programming Patterns: State	book	2026-09-10	overview page	available
GPU_GEMS_SHAFTS	GPU Gems 3, Ch.13: Volumetric Light Scattering as a Post-Process (Mitic)	engineering_article	2026-09-10	overview page	available

Код	Название	Тип	Проверен	Локатор	Доступность
HUNT_AUDIO	Hunt: Showdown — Audio readability, realism and consistency	engineering_article	2026-09-10	sections: Occlusion and Realism, Feedback and Readability	available
HUNT_AUDIO_2025	Dev Insight - A deep dive into 3D Audio in Hunt	studio_engineering_article	2026-09-10	section: CrySpatial	available
KHRONOS_ASYNC_COMPUTE	Khronos Vulkan Samples — Using async compute to saturate GPU	secondary	2026-09-10	overview page	available
L4D_AI_DIRECTOR	The AI Systems of Left 4 Dead	engineering_talk	2026-09-10	section: Adaptive Dramatic Pacing	available
MESHOPT	meshoptimizer: mesh optimization library	open_source	2026-09-10	overview page	available
MS_DIRECTSTORAGE_GUIDANCE	Microsoft DirectStorage — Developer Guidance	open_source	2026-09-10	overview page	available
MS_MESH_SHADER	DirectX mesh shader specification	official_documentation	2026-09-10	section: Required Support from the device	available
MS_VRS	Variable Rate Shading DirectX-Specs	official_documentation	2026-09-10	sections: Specifying Shading Rate, Feature Tiering	available
NVIDIA_NTC	NVIDIA RTX NTC SDK	engineering_article	2026-09-10	overview page	available
NVIDIA_VOXEL_CONES	Interactive Indirect Illumination Using Voxel Cone Tracing — Crassin e	engineering_article	2026-09-10	overview page	available
ORCA_RVO	Optimal Reciprocal Collision Avoidance (van den Berg et al., UNC Gamma	secondary	2026-09-10	overview page	available
RESEARCH_S23_PACKAGING	Unreal Engine: Packaging Your Project	official_documentation	2026-09-10	sections: Build, Cook, Stage, Package; Chunking	available
RESEARCH_S24_HLOD	Unreal Engine: World Partition HLOD	official_documentation	2026-09-10	sections: HLOD layers and proxy mesh methods	available
RESEARCH_S25_PCG	Using PCG with World Partition	official_documentation	2026-09-10	sections: PCG Data Layers and HLOD Layers	available
RESEARCH_S27_RELEVANCY	Unreal Engine: Actor Relevancy	official_documentation	2026-09-10	sections: actor relevancy and distance	available
RESEARCH_S28_CRY_STREAMING	CRYENGINE: Streaming System	official_documentation	2026-09-10	sections: asynchronous reads, decompression and main-thread	available
RESEARCH_S29_CRY_AUDIO	CRYENGINE: Audio & Occlusion	official_documentation	2026-09-10	sections: No Ray, SingleRay, MultipleRay and distance limits	available
RESEARCH_S31_UNITY_JOB_OVERVIEW	Unity Manual: Job System Overview	official_documentation	2026-09-10	sections: worker threads, cores, work stealing and safety	available
RESEARCH_S33_ISO_25010	ISO/IEC 25010:2023 Product Quality Model	standard	2026-09-10	product quality model and evaluation characteristics	available
RESEARCH_S34_ISO_20741	ISO/IEC 20741:2017 Software Engineering Tool Evaluation	standard	2026-09-10	purpose-oriented tool selection and quality characteristics	available
RESEARCH_S35_PMI PERT	Practice Standard for Scheduling - Second Edition	practice_standard	2026-09-10	three-point estimating: triangular and beta/PERT formulas	available
RESEARCH_S36_NASA_SCHEDULE	Analytical Technique for Schedule Risk Assessment	technical_report	2026-09-10	PERT/CPM precedence network and critical path sections	available
RESEARCH_S37_AMD AHL	Amdahl's Law & Parallel Speedup	conference_paper	2026-09-10	serial/parallel speedup and overhead	available
RESEARCH_S38_D3D_FEATURE_LEVELS	Direct3D Hardware Feature Levels	official_documentation	2026-09-10	feature-level functionality versus performance	available
RESEARCH_S39_D3D_CHECK_FEATURE	ID3D12Device::CheckFeatureSupport	official_documentation	2026-09-10	syntax, remarks and ray-tracing capability query	available
RESEARCH_S40_D3D_RT_TIER	D3D12 Raytracing Tier	official_documentation	2026-09-10	ray-tracing tier capability	available
RESEARCH_S41_UE_SPECS	Hardware and Software Specifications for Unreal Engine	official_documentation	2026-09-10	editor requirements and rendering-path constraints	available
RESEARCH_S42_RIOT_SCALABILITY	VALORANT: Scalability and Load Testing	studio_engineering	2026-09-10	sections: server scalability and 128-tick load test	available
RESEARCH_S43_HAMMER	Valve Hammer Editor / Source SDK	official_developer_resource	2026-09-10	branch-specific Hammer/Source SDK tooling	available

Код	Название	Тип	Проверен	Локатор	Доступность
RESEARCH_S44_HEROENGINE_LEGACY	HeroEngine Legacy	studio_information	2026-09-10	legacy status and current applicability warning	available
RESEARCH_S45_HEROENGINE_VIDEOS	HeroEngine Legacy Platform Videos	studio_engineering	2026-09-10	historical live collaboration and integration examples	available
RESEARCH_S46_UNITY_JOB_DEPENDENCIES	Unity Job Dependencies	official_documentation	2026-09-10	job dependencies and synchronization	available
RESEARCH_S49_PASSMARK_SINGLE	PassMark CPU Single Thread Chart	benchmark	2026-09-10	single-thread comparison chart and notes	available
RESEARCH_S50_PASSMARK_FAQ	PassMark PerformanceTest FAQ	benchmark_methodology	2026-09-10	benchmark limits and workload caveat	available
RESEARCH_S51_3DMARK	UL 3DMark	benchmark	2026-09-10	GPU/CPU comparison and frame-rate context	available
RESEARCH_S53_QUIC_RFC	RFC 9000: QUIC	standard	2026-09-10	packet format and transport assumptions	available
RESEARCH_S54_UNE_TEXTURE_METRICS	Unreal Engine: Texture Streaming Metrics	official_documentation	2026-09-10	wanted mips, pool usage and streaming metrics	available
RESEARCH_S55_UNE_TEXTURE_CONFIG	Unreal Engine: Texture Streaming Configuration	official_documentation	2026-09-10	pool sizing and update behavior	available
RESEARCH_S57_NASA_GLOSSARY	NASA PP&C Glossary: Critical Path and Double Counting	technical_guidance	2026-09-10	critical path, uncertainty and double-counting entries	available
RESEARCH_S58_UNITY_JOB_TROUBLESHOOTING	Unity Job System Troubleshooting	official_documentation	2026-09-10	WaitForJobGroup and Complete synchronization	available
RIOT_NETCODE	Peeking into VALORANT's Netcode	engineering_article	2026-09-10	sections: peeker's advantage and simulation divergence	available
RIOT_TICK	Valorant 128-tick servers (Riot Engineering)	engineering_article	2026-09-10	section: 128-tick servers	available
SAVE_PATTERNS	Save Systems & Persistence (Andrews Notebook)	secondary	2026-09-10	overview page	available
SHADOWGAMBIT_SAVE	Deep dive: save system of Shadow Gambit (GameDeveloper)	secondary	2026-09-10	overview page	available
SRC-AIS-001	The AI Systems of Left 4 Dead	conference_talk	2026-09-10	PDF p.64 (mob size/interval), p.65 (population categories),	verified_downloaded_text_extra
SRC-AIS-002	Crowd Pathfinding and Steering Using Flow Field Tiles (Game AI Pro, Ch	book	2026-09-10	pp.307-308 (intro/motivation), p.309 (integration field 24-b	verified_downloaded_text_extra
SRC-AIS-003	Optimal Reciprocal Collision Avoidance (ORCA)	academic_paper	2026-09-10	Description section: 'compute collision-free actions for all	verified
SRC-AIS-004	RVO2 Library: Reciprocal Collision Avoidance for Real-Time Multi-Agent	open_source	2026-09-10	News section: 'RVO2 Library has been licensed for the video	verified
SRC-AIS-005	Game Programming Patterns - Object Pool (Optimization Patterns)	book	2026-09-10	Sections 'The Curse of Fragmentation', 'Free List', 'O(1) co	verified
SRC-AIS-006	Game Programming Patterns - Component (Decoupling Patterns)	book	2026-09-10	Intent 'Allow a single entity to span multiple domains witho	verified
SRC-AIS-007	Gerstner wave (Trochoidal wave)	secondary	2026-09-10	Section 'In computer graphics': multi-component parametric f	verified
SRC-AIS-008	Rendering the Hellscape of Doom Eternal (Advances in Real-Time Renderi	conference_talk	2026-09-10	PDF p.54 (GORE SYSTEM: 'Separate mesh for each wound', 'Up t	verified_downloaded_text_extra
SRC-AIS-009	Hair Rendering and Simulation in Unreal Engine	official_documentation	2026-09-10	Intro: 'strand-based workflow ... simulate and render hundre	verified
SRC-AIS-010	Groom Scalability and Performance with Unreal Engine	official_documentation	2026-09-10	Sections 'Geometry Scalability' (strands need 'an allocated	verified
SRC-AIS-011	Recast Navigation (recastnav.com)	open_source	2026-09-10	Sections: 'How it Works' (voxel rasterization -> walkable fi	verified
SRC-AIS-012	RecastNavigation - Detour/Include/DetourNavMeshQuery.h (source)	open_source	2026-09-10	Lines 213-244: '@name Sliced Pathfinding Functions', initSli	verified_source_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-AIS-013	Godot Engine - Physics introduction	official_documentation	2026-09-10	Section 'Collision layers and masks' (32 layers, collision_l	verified
SRC-AIS-014	Saving and Loading Your Game in Unreal Engine	official_documentation	2026-09-10	Sections 'Asynchronous Saving' ('AsyncSaveGameToSlot is the	verified
SRC-AIS-015	Significance Manager in Unreal Engine	official_documentation	2026-09-10	Intro ('The Significance Manager itself does not actually im	verified
SRC-AIS-016	MassEntity in Unreal Engine	official_documentation	2026-09-10	'MassEntity is a gameplay-focused framework for data-orient	verified_index_page_only
SRC-AIS-017	Scalability and Best Practices for Niagara	official_documentation	2026-09-10	Sections 'GPU vs CPU' ('Particle scripts have the largest op	verified
SRC-AIS-018	Massive Crowd on Assassin's Creed Unity: AI Recycling (GDC 2015)	conference_talk	2026-09-10	Session Overview: 'With the limit of 40 real AIs and 120 hig	verified
SRC-AIS-019	The Art of Destruction in Rainbow Six: Siege (GDC 2016)	conference_talk	2026-09-10	PDF p.3 (procedural vs pre-fragmented definition), p.15 ('Fi	verified_downloaded_text_extra
SRC-AIS-020	Year summary (Voxagon Blog) - Teardown engine	engineering_blog	2026-09-10	Section 'Sparse voxel objects': 'Teardown uses a dense voxel	verified
SRC-AIS-021	Fix Your Timestep!	secondary	2026-09-10	Sections 'Fixed delta time' (dt=1/60), 'Variable delta time'	verified
SRC-AIS-022	Crest Ocean System (GitHub: wave-harmonic/crest)	open_source	2026-09-10	README: 'An advanced water system implemented in Unity'; Cre	verified
SRC-AIS-023	NVIDIA Blast (GitHub: NVIDIAGameWorks/Blast)	open_source	2026-09-10	README: 'A modular destruction SDK designed for performance	verified
SRC-AIS-024	Chaos Destruction in Unreal Engine	official_documentation	2026-09-10	'Chaos is Unreal Engine's high-performance physics and destr	verified
SRC-AIS-025	Clothing Tool in Unreal Engine (Chaos Cloth)	official_documentation	2026-09-10	'Unreal Engine uses Chaos Cloth solver which is a low-level	verified
SRC-AIS-026	Chaos Vehicles (Unreal Engine)	official_documentation	2026-09-10	'Chaos Vehicles is Unreal Engine's lightweight system for pe	verified
SRC-AIS-027	PxBroadPhaseType - PhysX SDK Documentation	official_documentation	2026-09-10	enum eSAP ('3-axes sweep-and-prune'), eMBP ('Multi box pruni	verified
SRC-AIS-028	GPU Gems 3, Chapter 32: Broad-Phase Collision Detection with CUDA	book	2026-09-10	Results: '30,720 objects' -> broad phase prunes ~450,000,000	verified
SRC-AIS-029	Advanced Graphics Techniques Tutorial: Wakes, Explosions and Lighting	conference_talk	2026-09-10	Session Overview: 'challenges ... during the implementation	verified
SRC-AIS-030	'Overwatch' Gameplay Architecture and Netcode (GDC 2017)	conference_talk	2026-09-10	Session Overview: 'Overwatch uses a cutting-edge Entity Comp	verified
SRC-AIS-031	Crowds in Hitman: Absolution (GDC Vault)	conference_talk	2026-09-10	Session Overview: 'the techniques and optimizations used to	verified
SRC-AIS-032	Godot Engine - Saving games	official_documentation	2026-09-10	Sections 'Identifying persistent objects' (Persist group), '	verified
SRC-AIS-033	Unity Manual - Cloth	official_documentation	2026-09-10	Page resolves to the Unity Manual shell; the article body wa	available_partial_cookie_wall
SRC-AIS-034	Unity Manual - Wheel collider component reference	official_documentation	2026-09-10	Page resolves to the Unity Manual shell; body NOT extracted	available_partial_cookie_wall
SRC-AIS-035	Unity Manual - Write multithreaded code with the job system	official_documentation	2026-09-10	Page resolves; only navigation chrome was returned by automa	available_partial_client_rende

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-AIS-036	AMD and Crystal Dynamics Collaboration ... With the Launch of 'Tomb Ra	vendor_press_releas e	2026-09-10	'TressFX Hair, the world's first in-game implementation of a	verified
SRC-AIS-037	Soft-body Physics - BeamNG.drive	official_documentati on	2026-09-10	Sections 'Node-Beam Structure' (nodes have mass; beams are s	verified
SRC-AIS-038	DOTS - Unity's Data-Oriented Technology Stack	vendor_press_releas e	2026-09-10	'DOTS in Production' section: Stunlock Studios used ECS thro	verified
SRC-AIS-039	Unity Manual - Introduction to GPU instancing	official_documentati on	2026-09-10	Page resolves to the Unity Manual shell; body NOT extracted	available_partial_ cookie_wall
SRC-AIS-040	Godot Engine - RigidBody3D class reference	official_documentati on	2026-09-10	Properties 'can_sleep' ('If true, the body can enter sleep m	verified
SRC-AIS-041	Physics in Unreal Engine (Chaos Physics overview)	official_documentati on	2026-09-10	Section 'Rigid Body Dynamics' ('Chaos Physics provides many	verified
SRC-AIS-042	Water System in Unreal Engine	official_documentati on	2026-09-10	Intro ('The Water system unifies the shading and mesh render	verified
SRC-AIS-043	AMD TressFX (GitHub: GPUOpen-Effects/TressFX)	open_source	2026-09-10	README 'New in TressFX 4.1' bullet list: 'Optimized physics	verified
SRC-AIS-044	Godot Engine - VehicleBody3D class reference	official_documentati on	2026-09-10	Description: 'It is based on the raycast vehicle system comm	verified
SRC-AIS-045	Godot Engine - GPUParticles3D class reference	official_documentati on	2026-09-10	Property 'amount' ('Higher values will increase GPU requirem	verified
SRC-AIS-046	Physics Sub-Stepping in Unreal Engine	official_documentati on	2026-09-10	Intro ('By using physics Sub-stepping you can get physics si	verified
SRC-AIS-047	Niagara Flipbook Baker Quick Start Guide in Unreal Engine	official_documentati on	2026-09-10	Intro ('At times, you may create a nice effect, but find it	verified
SRC-CHC-001	Animation Budget Allocator (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Animation Budget Allocator', '### Prerequisite	verified
SRC-CHC-002	Animation Optimization in Unreal Engine (Unreal Engine 5.8 Documentati	secondary	2026-09-10	Sections: '# Animation Optimization', '## Overview', '## Usi	verified
SRC-CHC-003	Skeletal Mesh LODs in Unreal Engine (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Skeletal Mesh LODs', '## Creating LODs', '## Re	verified
SRC-CHC-004	Animation Compression Library (acl) - repository README	secondary	2026-09-10	Sections: '# Animation Compression Library', '## Goals', '##	verified
SRC-CHC-005	ACL - Paragon database performance	secondary	2026-09-10	'# Paragon database performance' results table (v2.1.0 colum	verified
SRC-CHC-006	ACL - Carnegie-Mellon University database performance	secondary	2026-09-10	'# Carnegie-Mellon University database performance' results	verified
SRC-CHC-007	ACL - Decompression performance	secondary	2026-09-10	'# Decompression performance' -> '## Uniformly sampled algor	verified
SRC-CHC-008	Animation Compression Library in Unreal Engine (Unreal Engine 5.8 Docu	secondary	2026-09-10	Sections: plugin enable path, 'Anim Compress ACL' / 'Anim Co	verified
SRC-CHC-009	Animation Compression Codec Reference in Unreal Engine (Unreal Engine	secondary	2026-09-10	Sections: '# Animation Compression Codec Reference', '### Bo	verified
SRC-CHC-010	Motion Matching in Unreal Engine (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Motion Matching', '## Motion Matching Setup', '	verified
SRC-CHC-011	Motion Matching and The Road to Next-Gen Animation (GDC 2016 slides)	secondary	2026-09-10	Slides: 39-41 ('Introducing Motion Matching', 'A ridiculousl	verified
SRC-CHC-012	Learned Motion Matching (project page + abstract)	secondary	2026-09-10	Page body paragraph ('training three specialized neural netw	verified
SRC-CHC-013	Learned motion matching, ACM Transactions on Graphics 39(4)	secondary	2026-09-10	Abstract; ACM DL landing page. Full text was behind ACM's pa	unverified

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SRC-CHC-014	Skeletal Mesh Rendering Paths in Unreal Engine (Unreal Engine 5.8 Docu	secondary	2026-09-10	Sections: '# Skeletal Mesh Rendering Paths', '## How Skeleta	verified
SRC-CHC-015	GPU Gems, Chapter 4: Animation in the 'Dawn' Demo	secondary	2026-09-10	Sections: '## 4.4 Skinning', '#### 4.3.2 Morph Target Implem	verified
SRC-CHC-016	o3z-animation - open source C++ 3D skeletal animation library (documen	secondary	2026-09-10	Sections: '# What o3z-animation is', '# What it's not', '# L	verified
SRC-CHC-017	Blender Manual - Render Baking (Cycles)	secondary	2026-09-10	Sections: '# Render Baking', '## Setup', '## Settings' -> 'B	verified
SRC-CHC-018	MikkTSpace - a common standard for tangent space used in baking tools	secondary	2026-09-10	README.md: 'A common standard for tangent space used in baki	verified
SRC-CHC-019	Block Compression (Direct3D 10) - Microsoft Learn	secondary	2026-09-10	Sections: block-size statements, 'Compression Algorithms' ta	verified
SRC-CHC-020	Basis Universal GPU Texture Codec (repository README)	secondary	2026-09-10	Sections: intro ('.KTX2 ... rapid transcoding'), 'Supported	verified
SRC-CHC-021	Unity Manual - Sprite Atlas (class-SpriteAtlas)	secondary	2026-09-10	Intro paragraph: 'A Sprite Atlas is an Asset that consolidat	unverified
SRC-CHC-022	Unity Manual - Sprite Atlas reference	secondary	2026-09-10	'Allow Rotation' property description: 'Check this box to al	unverified
SRC-CHC-023	Unity Manual - Sprite Atlas workflow	secondary	2026-09-10	'Include in Build' paragraph: 'Enable or disable the Include	unverified
SRC-CHC-024	Unity Manual - Tilemap Renderer component reference	secondary	2026-09-10	'Mode' property description: 'Chunk: Renders multiple tiles	unverified
SRC-CHC-025	Unity Manual - Tilemap Renderer	secondary	2026-09-10	'Detect Chunk Culling Bounds' / 'Chunk Culling Bounds' descr	unverified
SRC-CHC-026	Tiled Documentation - JSON Map Format reference	secondary	2026-09-10	Sections: 'Layer' field table rows for 'chunks', 'compressio	verified
SRC-CHC-027	Tiled Documentation - Using Infinite Maps	secondary	2026-09-10	Sections: '# Using Infinite Maps', '## Creating an Infinite	verified
SRC-CHC-028	Spine User Guide - Mesh attachments	secondary	2026-09-10	Sections: 'Mesh attachments' intro and 'Setup', 'Deformation	verified
SRC-CHC-029	Spine User Guide - Metrics view	secondary	2026-09-10	Sections: '# Metrics view', '## Bones' (Nexus 4 ~2000 bones	verified
SRC-CHC-030	Unity 2D Animation package - Skinning Editor	secondary	2026-09-10	Sections: '# Skinning Editor' (intro: 'create the bones of y	verified
SRC-CHC-031	Unity 2D Animation package - Actor skinning and weighting workflow	secondary	2026-09-10	Workflow steps 1-7, in particular step 2 (Auto Geometry), st	verified
SRC-CHC-032	Gameplay Ability System for Unreal Engine (Unreal Engine 5.8 Documenta	secondary	2026-09-10	Sections: '# Gameplay Ability System' ('a framework for buil	verified
SRC-CHC-033	Understanding the Unreal Engine Gameplay Ability System (Unreal Engine	secondary	2026-09-10	Sections: '# Gameplay Ability System Overview', '## Componen	verified
SRC-CHC-034	Gameplay Ability Tasks in Unreal Engine (Unreal Engine 5.8 Documentati	secondary	2026-09-10	'# Ability Tasks' body: async work, EndTask, 'prevents phant	verified
SRC-CHC-035	Gameplay Ability System Component and Gameplay Attributes in Unreal En	secondary	2026-09-10	Intro: the Ability System Component (UAbilitySystemComponent	unverified
SRC-CHC-036	Scalability and Best Practices for Niagara (Unreal Engine 5.8 Document	secondary	2026-09-10	Sections: '# Scalability and Best Practices', '## Instance C	verified
SRC-CHC-037	Optimizing Niagara (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	'# Optimizing Niagara' hub page and its three child links: M	verified

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SRC-CHC-038	Niagara Systems as a Service (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Systems as a Service' -> 'Overview' (one-to-one)	verified
SRC-CHC-039	3D Toon Rendering in 'Hi-Fi RUSH' (GDC 2024 session)	secondary	2026-09-10	Session overview text: '60fps rhythm action game rendered in	verified
SRC-CHC-040	Illustrative Rendering in Team Fortress 2 (NPAR 2007)	secondary	2026-09-10	Page 1: Abstract ('a set of artistic choices and novel real-	verified
SRC-CHC-041	Rayman Legends: The Design Process Within the UbiArt Framework (GDC 20	secondary	2026-09-10	Session overview text: 'an engine originally developed for t	verified
SRC-CHC-042	Game Engine Architecture, 4th Edition (two-volume set)	book	2026-09-10	Publisher/author landing page: 'New to the Fourth Edition' -	verified
SRC-CHC-043	Game Programming Patterns	secondary	2026-09-10	Table of contents: '11. Bytecode', '12. Subclass Sandbox', '	verified
SRC-CHC-044	Unity Manual - Animation tab (AnimationClip import settings)	secondary	2026-09-10	'Anim. Compression' -> 'Off' description: 'This means that U	unverified
SRC-CHC-045	Unity Manual - Recommended, default, and supported texture compression	secondary	2026-09-10	Statement: 'For devices with DirectX 11 or better class GPUs	unverified
SRC-CHC-046	Cuphead - A Game Made with Unity (Unity customer story)	secondary	2026-09-10	Sections: '# Cuphead', '## They bet everything on a game', '	verified
SRC-DER-001	Steam Hardware & Software Survey	official_documentation	2026-09-10	sections 'Operating System Version' (Windows 11 64 bit 70.97	verified_fetched
SRC-DER-002	Steam Hardware & Software Survey - PC Video Card Usage Details	official_documentation	2026-09-10	section 'PC VIDEO CARD USAGE DETAILS -> OVERALL DISTRIBUTION	verified_fetched
SRC-DER-003	Source Multiplayer Networking	official_documentation	2026-09-10	sections 'Tickrate' (default timestep 15ms = 66.666 ticks/s;	verified_fetched
SRC-DER-004	What Every Programmer Needs To Know About Game Networking	engineering_blog	2026-09-10	sections 'Peer-to-Peer Lockstep', 'Client/Server', 'Client-S	verified_fetched
SRC-DER-005	Snapshot Interpolation	engineering_blog	2026-09-10	sections 'Snapshots' (CubeState = 225 bits / 28.1 bytes per	verified_fetched
SRC-DER-006	Direct3D 12 programming guide	official_documentation	2026-09-10	intro paragraph ('Direct3D 12 provides an API and platform .	verified_fetched
SRC-DER-007	Advanced Shader Delivery overview	official_documentation	2026-09-10	sections 'Advanced Shader Delivery' (on-device launch/JIT sh	verified_fetched
SRC-DER-008	Work submission in Direct3D 12 (command queues and command lists)	official_documentation	2026-09-10	intro: 'Direct3D 12 no longer supports an immediate context	verified_fetched
SRC-DER-009	Memory management in Direct3D 12	official_documentation	2026-09-10	intro: 'Moving to D3D12 involves doing proper synchronizatio	verified_fetched
SRC-DER-010	Engine class reference (Engine.max_fps, Engine.physics_ticks_per_second)	official_documentation	2026-09-10	Engine class properties: physics_ticks_per_second = 60 ('The	verified_fetched
SRC-DER-011	Upgrading from Godot 3 to Godot 4	official_documentation	2026-09-10	sections 'Should I upgrade to Godot 4?', 'List of automatica	verified_fetched
SRC-DER-012	Setting Up Dedicated Servers in Unreal Engine	official_documentation	2026-09-10	sections 'Overview' (client-server model, 'authoritative hos	verified_fetched
SRC-DER-013	Networking Overview for Unreal Engine	official_documentation	2026-09-10	sections on replication ('most actors do not replicate by de	verified_fetched
SRC-DER-014	Performance Guidelines for Mobile Devices in Unreal Engine	official_documentation	2026-09-10	section 'Draw Calls' ('Draw calls of the entire scene should	verified_fetched
SRC-DER-015	Common Memory and CPU Performance Considerations in Unreal Engine	official_documentation	2026-09-10	sections 'Garbage Collection' (processing spikes when the GC	verified_fetched
SRC-DER-016	Timing Insights in Unreal Engine	official_documentation	2026-09-10	intro ('displays per-frame performance data ... including se	verified_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-DER-017	Introduction to Performance Profiling and Configuration in Unreal Engi	official_documentati on	2026-09-10	statements: 'Generally, applications will target 30, 60, and	verified_fetched
SRC-DER-018	Saving and Loading Your Game in Unreal Engine	official_documentati on	2026-09-10	sections 'Creating a SaveGame Object' (USaveGame, Kismet/Gam	verified_fetched
SRC-DER-019	Unreal Engine Optimization Guide: Profiling Fundamentals	engineering_blog	2026-09-10	sections 'Frame Time (MPF)', 'Frame Budget' ('33.33ms for 30	verified_fetched
SRC-DER-020	What do 1% percentile statistics mean in PresentMon (issue #219)	open_source	2026-09-10	Intel maintainer reply (2024-04-05): 'In 2.0.0, the percenti	verified_fetched
SRC-DER-021	Vulkan 1.4.362 Specification (with all registered extensions)	secondary	2026-09-10	Preamble (version 1.4.362, generated 2026-09-04); section 2	verified_fetched
SRC-DER-022	Steam Deck OLED - Tech Specs	official_documentati on	2026-09-10	section 'Speeds and Feeds': CPU 'Zen 2 4c/8t, 2.4-3.5GHz (up	verified_fetched
SRC-DER-023	Steam Deck and Steam Machine Compatibility Review	official_documentati on	2026-09-10	'Performance' requirement: 'the game must ship with a default	verified_fetched
SRC-DER-024	Steam Networking	official_documentati on	2026-09-10	intro ('Our newest APIs relay packets through the Valve netw	verified_fetched
SRC-DER-025	Steam Datagram Relay	official_documentati on	2026-09-10	'Relaying the traffic protects your servers and players from	verified_fetched
SRC-DER-026	The Mythical Man-Month (encyclopedia article summarising Brooks, 1975)	secondary	2026-09-10	section 'The mythical man-month': 'Adding manpower to a late	verified_fetched
SRC-DER-027	Game Engine Architecture - official table of contents (3rd and 4th edi	book	2026-09-10	3rd ed. TOC: Ch.2 'Tools of the Trade' incl. 2.3 'Profiling	verified_fetched
SRC-DER-028	GeForce RTX 4060 / RTX 4060 Ti graphics cards - specifications	vendor_press_releas e	2026-09-10	specification block: '8 GB GDDR6' memory, 128-bit memory int	verified_fetched
SRC-DER-029	Quality project settings reference	official_documentati on	2026-09-10	sections 'Quality levels matrix' (per-platform quality level	verified_fetched
SRC-DER-030	Application.targetFrameRate	official_documentati on	2026-09-10	'the frame rate at which Unity tries to render your applicat	verified_fetched
SRC-DER-031	Developer Satisfaction Survey 2023 - Summary Report	academic_paper	2026-09-10	section 'Snapshot: Hours of Work': 'Across all respondents,	verified_fetched
SRC-DER-032	As Anthem shuts down, the game's executive producer has released a nea	secondary	2026-09-10	article reporting Mark Darrah's video post-mortem: Anthem 'r	verified_fetched
SRC-DER-033	VALORANT's 128-Tick Servers	engineering_blog	2026-09-10	'We need to be able to process an entire frame within 7.8125	verified_fetched
SRC-DER-034	Peeking into VALORANT's Netcode	engineering_blog	2026-09-10	'a smooth 128 server tickrate'; 'clients and servers always	verified_fetched
SRC-DER-035	Game Loop (Game Programming Patterns, chapter)	book	2026-09-10	sections 'Take a Little Nap' ('Say you want your game to run	verified_fetched
SRC-DER-036	Boehm Cost of Change Curve: What the 1981 Data Actually Shows	secondary	2026-09-10	table 'Relative cost to fix one defect by the phase it is ca	verified_fetched
SRC-DER-037	Unreal Build Tool in Unreal Engine	official_documentati on	2026-09-10	intro: 'UnrealBuildTool (UBT) is a custom tool that manages	verified_fetched
SRC-ENG-001	Nanite Virtualized Geometry Overview	official_documentati on	2026-09-10	sections 'Nanite Virtualized Geometry Overview', 'Benefits o	verified_fetched
SRC-ENG-002	Lumen Global Illumination and Reflections	official_documentati on	2026-09-10	sections 'Lumen Global Illumination and Reflections', 'Getti	verified_fetched
SRC-ENG-003	Virtual Shadow Maps	official_documentati on	2026-09-10	sections 'Goals of Virtual Shadow Maps', 'Soft Shadows with	verified_fetched
SRC-ENG-004	Virtual Texturing	official_documentati on	2026-09-10	sections 'Virtual Texturing Methods', 'Runtime Virtual Textu	verified_fetched

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SRC-ENG-005	World Partition	official_documentation	2026-09-10	sections 'World Partition', 'Enabling World Partition', 'Usi	verified_fetched
SRC-ENG-006	Hierarchical Level of Detail	official_documentation	2026-09-10	section 'Hierarchical Level of Detail' (overview paragraph)	verified_fetched
SRC-ENG-007	Large World Coordinates in Unreal Engine 5	official_documentation	2026-09-10	sections 'Large World Coordinates', 'Experimenting with Larg	verified_fetched
SRC-ENG-008	Instanced Static Mesh Component	official_documentation	2026-09-10	sections 'Instanced Static Mesh', 'Hierarchical Instanced St	verified_fetched
SRC-ENG-009	Static Mesh Automatic LOD Generation	official_documentation	2026-09-10	sections 'Static Mesh Automatic LOD Generation', 'Using LOD	verified_fetched
SRC-ENG-010	Niagara Overview	official_documentation	2026-09-10	sections 'Core Niagara Components', 'Emitters', 'Modules', '	verified_fetched
SRC-ENG-011	Chaos Physics Overview	official_documentation	2026-09-10	page 'Chaos Physics Overview'	verified_fetched
SRC-ENG-012	Chaos Vehicles	official_documentation	2026-09-10	page 'Chaos Vehicles' (overview paragraph)	verified_fetched
SRC-ENG-013	Gameplay Ability System	official_documentation	2026-09-10	sections 'Gameplay Ability System', 'Valley of the Ancient S	verified_fetched
SRC-ENG-014	MassEntity Overview	official_documentation	2026-09-10	sections 'Basic Concepts', 'Processing the Entities', 'Subsy	verified_fetched
SRC-ENG-015	Behavior Trees	official_documentation	2026-09-10	section 'Behavior Trees' (overview paragraph)	verified_fetched
SRC-ENG-016	Navigation System	official_documentation	2026-09-10	section 'Navigation System' (overview paragraph)	verified_fetched
SRC-ENG-017	Replication Graph	official_documentation	2026-09-10	sections 'Replication Graph', 'Structure', 'Enabling The Sys	verified_fetched
SRC-ENG-018	Significance Manager	official_documentation	2026-09-10	sections 'Significance Manager', 'Setup', 'Significance Mana	verified_fetched
SRC-ENG-019	Animation Budget Allocator	official_documentation	2026-09-10	sections 'Animation Budget Allocator', 'Prerequisites', 'Set	verified_fetched
SRC-ENG-020	Unreal Insights	official_documentation	2026-09-10	sections 'Unreal Insights', 'Setting Up Unreal Insights', 'C	verified_fetched
SRC-ENG-021	Unreal Engine 5.0 Release Notes	official_documentation	2026-09-10	sections 'Rendering' (Nanite, Lumen, Virtual Shadow Maps, Vi	verified_fetched
SRC-ENG-022	Unreal Engine 5.4 Release Notes	official_documentation	2026-09-10	sections 'Rendering', 'World Building', 'Platforms'	verified_fetched
SRC-ENG-023	Physics in Unreal Engine	official_documentation	2026-09-10	page 'Physics in Unreal Engine' (Chaos overview sections)	verified_fetched
SRC-ENG-024	Chaos Destruction	official_documentation	2026-09-10	page 'Chaos Destruction' (overview)	verified_fetched
SRC-ENG-025	Universal Render Pipeline overview	official_documentation	2026-09-10	section 'Universal Render Pipeline overview'	verified_fetched
SRC-ENG-026	High Definition Render Pipeline overview	official_documentation	2026-09-10	section 'High Definition Render Pipeline overview'	verified_fetched
SRC-ENG-027	Scriptable Render Pipeline Batcher in URP	official_documentation	2026-09-10	section 'Scriptable Render Pipeline Batcher in URP'	verified_fetched
SRC-ENG-028	Introduction to GPU instancing	official_documentation	2026-09-10	sections 'Introduction to GPU instancing', 'Render pipeline	verified_fetched
SRC-ENG-029	LOD Group component reference	official_documentation	2026-09-10	section 'LOD Group component reference'	verified_fetched
SRC-ENG-030	Excluding hidden objects with occlusion culling	official_documentation	2026-09-10	section 'Occlusion culling'	verified_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-ENG-031	Precalculating indirect light with Light Probes	official_documentation	2026-09-10	section 'Light Probes'	verified_fetched
SRC-ENG-032	Choose a light baking backend	official_documentation	2026-09-10	section 'Choose a light baking backend'	verified_fetched
SRC-ENG-033	Job system overview	official_documentation	2026-09-10	sections 'Job system overview', 'Multithreading'	verified_fetched
SRC-ENG-034	Addressables package overview	official_documentation	2026-09-10	section 'Addressables package'	verified_fetched
SRC-ENG-035	Optimizing GPU texture memory with mipmap streaming	official_documentation	2026-09-10	sections 'Introduction to mipmap streaming', 'Configure mipm	verified_fetched
SRC-ENG-036	Navigation System in Unity	official_documentation	2026-09-10	section 'Navigation System in Unity'	verified_fetched
SRC-ENG-037	Visual Effect Graph	official_documentation	2026-09-10	section 'Visual Effect Graph'	verified_fetched
SRC-ENG-038	Unity Profiler	official_documentation	2026-09-10	section 'Unity Profiler'	verified_fetched
SRC-ENG-039	Quality project settings reference	official_documentation	2026-09-10	sections 'Quality project settings reference', 'Quality leve	verified_fetched
SRC-ENG-040	Entities overview (DOTS)	official_documentation	2026-09-10	sections 'Entities overview', 'Package installation'	verified_fetched
SRC-ENG-041	Unity Netcode for Entities	official_documentation	2026-09-10	sections 'Unity Netcode for Entities', 'Requirements'	verified_fetched
SRC-ENG-042	AI Navigation package	official_documentation	2026-09-10	page 'AI Navigation' (package listing)	verified_fetched
SRC-ENG-043	DOTS - Unity's Data-Oriented Technology Stack	engineering_blog	2026-09-10	sections 'DOTS in Production' (V Rising / Zenith / Detonatio	verified_fetched
SRC-ENG-044	Addressables: Planning and best practices	engineering_blog	2026-09-10	sections 'Addressables: Planning and best practices', 'Intro	verified_fetched
SRC-ENG-045	V Rising: Behind the vampire realm built on 1,600 ECS systems	postmortem	2026-09-10	sections 'The challenge', 'The results'	verified_fetched
SRC-ENG-046	Unity render pipeline features (URP / HDRP)	vendor_press_release	2026-09-10	sections 'Universal Render Pipeline (URP)', 'High-Definition	verified_fetched
SRC-ENG-047	Unity Engine product page	vendor_press_release	2026-09-10	sections 'Deploy - Multiplatform', 'Frequently asked questio	verified_fetched
SRC-ENG-048	Unity plans and pricing	vendor_press_release	2026-09-10	plan cards 'Personal', 'Pro', 'Enterprise'	verified_fetched
SRC-ENG-049	Unity (game engine) - Wikipedia	secondary	2026-09-10	infobox 'Written in' / 'License'	verified_fetched
SRC-ENG-050	Unity-Technologies/EntityComponentSystemSamples	open_source	2026-09-10	repository README (Entities samples, Netcode samples, Physic	verified_url_only
SRC-ENG-052	Unreal Engine - Wikipedia	secondary	2026-09-10	infobox ('Written in C++', 'License', 'Operating system') an	verified_fetched
SRC-ENG-053	Godot Engine home	official_documentation	2026-09-10	sections 'Script with an object-oriented API', 'Exemplary XR	verified_fetched
SRC-ENG-054	Godot Engine documentation (stable)	official_documentation	2026-09-10	table of contents (Rendering, 2D/3D graphics, 2D/3D physics,	verified_fetched
SRC-ENG-055	Godot (game engine) - Wikipedia	secondary	2026-09-10	infobox ('License: MIT', 'Written in C++') and section 'Nota	verified_fetched
SRC-ENG-056	godotengine/godot	open_source	2026-09-10	repository header (86,319 commits on the default branch)	verified_url_only
SRC-ENG-057	CRYENGINE home	official_documentation	2026-09-10	page 'Features' / 'CRYENGINE Documentation'	verified_fetched
SRC-ENG-058	CRYENGINE Documentation	official_documentation	2026-09-10	sections 'CRYENGINE 3', 'CRYENGINE V', 'CRYENGINE V Manual',	verified_fetched
SRC-ENG-059	CryEngine - Wikipedia	secondary	2026-09-10	infobox ('Written in', 'License', 'Platform', 'Stable releas	verified_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-ENG-060	Source (game engine) - Wikipedia	secondary	2026-09-10	infobox ('Written in C++', 'License: Source-available', 'Rel	verified_fetched
SRC-ENG-061	Source 2 - Wikipedia	secondary	2026-09-10	infobox ('Release September 2015', 'License: Proprietary') a	verified_fetched
SRC-ENG-062	HeroEngine home	vendor_press_release	2026-09-10	landing page / HeroCloud login	verified_fetched
SRC-ENG-063	HeroEngine - Wikipedia	secondary	2026-09-10	infobox ('Written in', 'License', 'Stable release') and sect	verified_fetched
SRC-ENG-064	bevyengine/bevy	open_source	2026-09-10	repository header (12,166 commits on the default branch)	verified_url_only
SRC-ENG-065	Rockstar Advanced Game Engine - Wikipedia	secondary	2026-09-10	infobox ('License: Proprietary') and lead	verified_fetched
SRC-ENG-066	CD Projekt - Wikipedia	secondary	2026-09-10	section 'Games developed' (The Witcher 2/3, Cyberpunk 2077)	verified_fetched
SRC-ENG-067	Black Myth: Wukong wows with UE5 early access visuals	interview	2026-09-10	Q&A 'Does the team have any favorite Unreal Engine 5 feature	verified_fetched
SRC-ENG-068	Drop into the Next Generation of Fortnite Battle Royale, Powered by Un	vendor_press_release	2026-09-10	sections 'Nanite', 'Lumen', 'Virtual Shadow Maps', 'What are	verified_fetched
SRC-ENG-069	Battle-testing Unreal Engine 5.1's new features on Fortnite Battle Roy	engineering_blog	2026-09-10	feature list (Lumen, Nanite, Virtual Shadow Maps, Niagara, W	verified_fetched
SRC-ENG-070	The making of Senua's Saga: Hellblade 2 - the big tech interview	interview	2026-09-10	Q&A on moving from UE4 to UE5 (Mark Slater-Tunstall, Dan Atw	verified_fetched
SRC-ENG-071	Fortnite Battle Royale - Wikipedia	secondary	2026-09-10	infobox 'Engine: Unreal Engine 5'	verified_fetched
SRC-ENG-072	Netcode for GameObjects - About	official_documentation	2026-09-10	sections 'Netcode for GameObjects', 'Before you begin'	verified_fetched
SRC-ENG-073	Black Myth: Wukong - the PC tech review	secondary	2026-09-10	sections on Nanite, foliage LOD, Virtual Shadow Maps and sof	verified_fetched
SRC-ENG-074	Brilliant visuals and growing pains: examining the first generation of	secondary	2026-09-10	sections on Lumen, Nanite and Virtual Shadow Maps across UE5	verified_fetched
SRC-ENG-075	Lords of the Fallen is a stunning UE5 Soulslike with ongoing tech issu	secondary	2026-09-10	paragraphs on the game's heavy use of Nanite and Lumen	verified_fetched
SRC-FUNC-001	Lumen Global Illumination and Reflections in Unreal Engine	official_documentation	2026-09-10	section 'Lumen Global Illumination and Reflections' (page in	verified_fetched
SRC-FUNC-002	Lumen Technical Details in Unreal Engine	official_documentation	2026-09-10	section 'Lumen Technical Details' intro + 'Surface Cache'	verified_fetched
SRC-FUNC-003	Lumen Performance Guide for Unreal Engine	official_documentation	2026-09-10	page intro + section 'Scalability Settings'	verified_fetched
SRC-FUNC-004	World Partition in Unreal Engine	official_documentation	2026-09-10	section 'World Partition' (page intro)	verified_fetched
SRC-FUNC-005	Nanite Virtualized Geometry Overview	official_documentation	2026-09-10	section 'Nanite Virtualized Geometry Overview' + 'Benefits o	verified_fetched
SRC-FUNC-006	Virtual Shadow Maps in Unreal Engine	official_documentation	2026-09-10	sections 'Goals of Virtual Shadow Maps' and 'Soft Shadows wi	verified_fetched
SRC-FUNC-007	Render Dependency Graph in Unreal Engine	official_documentation	2026-09-10	section 'Render Dependency Graph' (page intro) + feature lis	verified_fetched
SRC-FUNC-008	Temporal Super Resolution in Unreal Engine	official_documentation	2026-09-10	section 'Temporal Super Resolution' (page intro)	verified_fetched
SRC-FUNC-009	Dynamic Resolution in Unreal Engine	official_documentation	2026-09-10	section 'Dynamic Resolution' (page intro)	verified_fetched
SRC-FUNC-010	Path Tracer in Unreal Engine	official_documentation	2026-09-10	section 'Path Tracer' (page intro) + 'Benefits of the Path T	verified_fetched
SRC-FUNC-011	Lightmass Basics in Unreal Engine	official_documentation	2026-09-10	section 'Lightmass Basics' + 'Lightmass Importance Volume'	verified_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-FUNC-012	Foliage Mode in Unreal Engine	official_documentation	2026-09-10	sections 'Foliage Types', 'Culling Settings', 'Foliage Scala	verified_fetched
SRC-FUNC-013	Landscape Technical Guide	official_documentation	2026-09-10	sections 'Landscape Components' and 'Landscape Dimensions'	verified_fetched
SRC-FUNC-014	Mesh Drawing Pipeline	official_documentation	2026-09-10	sections 'Mesh Drawing Pipeline' intro and 'FPrimitiveSceneP	verified_fetched
SRC-FUNC-015	Skeletal Mesh LODs	official_documentation	2026-09-10	section 'Creating LODs'	verified_fetched
SRC-FUNC-016	Memory Insights	official_documentation	2026-09-10	sections 'Investigation - Allocation Queries' and 'Call Stac	verified_fetched
SRC-FUNC-017	Audio Mixer Overview	official_documentation	2026-09-10	section 'Audio Mixer Overview' (page intro)	verified_fetched
SRC-FUNC-018	Post Process Effects	official_documentation	2026-09-10	section 'Post Process Effects' (page intro)	verified_fetched
SRC-FUNC-019	Volumetric Fog	official_documentation	2026-09-10	section 'Volumetric Fog' (page intro)	verified_fetched
SRC-FUNC-020	Large World Coordinates	official_documentation	2026-09-10	section 'Large World Coordinates' (page intro)	verified_fetched
SRC-FUNC-021	Saving and Loading Your Game	official_documentation	2026-09-10	section 'Saving and Loading Your Game' (page intro)	verified_fetched
SRC-FUNC-022	UnrealBuildTool	official_documentation	2026-09-10	section 'UnrealBuildTool' + 'Modular Architecture'	verified_fetched
SRC-FUNC-023	Multiplayer Programming Quick Start	official_documentation	2026-09-10	sections 'Replicating Variables' and 'Remote Procedure Calls	verified_fetched
SRC-FUNC-024	Modeling Mode Overview	official_documentation	2026-09-10	section 'Modeling Mode Overview' (page intro)	verified_fetched
SRC-FUNC-025	Hair Rendering and Simulation	official_documentation	2026-09-10	section 'Hair Rendering and Simulation' (page intro)	verified_fetched
SRC-FUNC-026	Water System	official_documentation	2026-09-10	section 'Water System' (page intro)	verified_fetched
SRC-FUNC-027	MassEntity Overview	official_documentation	2026-09-10	section 'MassEntity Overview' (page intro)	verified_fetched
SRC-FUNC-028	Gameplay Ability System	official_documentation	2026-09-10	section 'Gameplay Ability System' (page intro)	verified_fetched
SRC-FUNC-029	Split Screen (Unreal Engine 4.27)	official_documentation	2026-09-10	sections 'Detail Mode Overview' and 'How do I determine Draw	verified_fetched
SRC-FUNC-030	Stat Commands	official_documentation	2026-09-10	section 'Stat Commands' command reference	verified_fetched
SRC-FUNC-031	Unreal Insights	official_documentation	2026-09-10	sections 'Recording a Session' and 'Timing View'	verified_fetched
SRC-FUNC-032	Chaos Destruction	official_documentation	2026-09-10	section 'Chaos Destruction' (page intro)	verified_fetched
SRC-FUNC-033	Geometry Collections User Guide	official_documentation	2026-09-10	section 'Geometry Collections User Guide' (page intro)	verified_fetched
SRC-FUNC-034	Quality project settings reference (Unity Manual)	official_documentation	2026-09-10	section 'Quality' project settings reference	verified_fetched
SRC-FUNC-035	Cloth (Unity Manual)	official_documentation	2026-09-10	section 'Cloth' (page intro)	verified_fetched
SRC-FUNC-036	Terrain (Unity Manual)	official_documentation	2026-09-10	section 'Creating and editing Terrains'	verified_fetched
SRC-FUNC-037	Lightmapping (Unity Manual)	official_documentation	2026-09-10	section 'Lightmapping' (page intro)	verified_fetched
SRC-FUNC-038	Animator (Unity Manual)	official_documentation	2026-09-10	section 'Animator' (page intro)	verified_fetched
SRC-FUNC-039	Physics Overview (Unity Manual)	official_documentation	2026-09-10	section 'Physics' overview	verified_fetched
SRC-FUNC-040	Wheel Collider (Unity Manual)	official_documentation	2026-09-10	section 'Wheel Collider' (page intro)	verified_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-FUNC-041	Introduction to AssetBundles (Unity Manual)	official_documentation	2026-09-10	section 'Introduction to AssetBundles' (page intro)	verified_fetched
SRC-FUNC-042	Memory in Unity: managed memory (Unity Manual)	official_documentation	2026-09-10	section 'Memory in Unity introduction'	verified_fetched
SRC-FUNC-043	LOD Group (Unity Manual)	official_documentation	2026-09-10	section 'LOD Group' (page intro)	verified_fetched
SRC-FUNC-044	Post-processing (Unity Manual)	official_documentation	2026-09-10	section 'Post-processing' overview	verified_fetched
SRC-FUNC-045	Render pipelines overview (Unity Manual)	official_documentation	2026-09-10	section 'Render pipelines' overview	verified_fetched
SRC-FUNC-046	Particle System (Unity Manual)	official_documentation	2026-09-10	section 'Particle System' (page intro)	verified_fetched
SRC-FUNC-047	Build Settings (Unity Manual)	official_documentation	2026-09-10	section 'Build Settings' (page intro)	verified_fetched
SRC-FUNC-048	Skinned Mesh Renderer (Unity Manual)	official_documentation	2026-09-10	section 'Skinned Mesh Renderer' (page intro)	verified_fetched
SRC-FUNC-049	Trees (Unity Terrain, Unity Manual)	official_documentation	2026-09-10	section 'Trees' (page intro)	verified_fetched
SRC-FUNC-050	Multi-scene editing (Unity Manual)	official_documentation	2026-09-10	section 'Multi-scene editing' (page intro)	verified_fetched
SRC-FUNC-051	The AI Systems of Left 4 Dead	conference_talk	2026-09-10	slides 'AI Director', 'Relax period', 'Peak Fade', 'Two conc	verified_fetched
SRC-FUNC-052	Three States and a Plan: The A.I. of F.E.A.R.	conference_talk	2026-09-10	pp.1-14 (three states, GOAP, cost per action, world-state ar	verified_fetched
SRC-FUNC-053	Crowd Pathfinding and Steering Using Flow Field Tiles (Game AI Pro, ch	book	2026-09-10	pp.307+ (sections 23.1-23.4: cost fields, integration fields	verified_fetched
SRC-FUNC-054	Rendering 'DOOM Eternal' (Advances in Real-Time Rendering, SIGGRAPH 20	conference_talk	2026-09-10	slides on forward rendering, light binning budget and GPU-dr	verified_fetched
SRC-FUNC-055	Streaming the World of Horizon Zero Dawn	engineering_blog	2026-09-10	talk abstract, 'Description' block	verified_fetched
SRC-FUNC-056	Massive Crowd on Assassin's Creed Unity: AI Recycling	conference_talk	2026-09-10	session 'Overview' text	verified_fetched
SRC-FUNC-057	Continuous World Generation in 'No Man's Sky'	conference_talk	2026-09-10	session 'Overview' text	verified_fetched
SRC-FUNC-058	'Overwatch' Gameplay Architecture and Netcode	conference_talk	2026-09-10	session 'Overview' text	verified_fetched
SRC-FUNC-059	NVIDIA DLSS: Your Questions, Answered	engineering_blog	2026-09-10	Q&A on how DLSS works, where it helps and 4K vs 1080p input	verified_fetched
SRC-FUNC-060	DirectStorage Overview (Microsoft Game Development Kit)	official_documentation	2026-09-10	sections 'High CPU usage', 'Insufficient maximum bandwidth',	verified_fetched
SRC-FUNC-061	Generating Complex Procedural Terrains Using the GPU (GPU Gems 3, ch.1	book	2026-09-10	sections on marching cubes case tables and geometry-shader p	verified_fetched
SRC-FUNC-062	Summed-Area Variance Shadow Maps (GPU Gems 3, ch.8)	book	2026-09-10	chapter sections on variance shadow maps and SAT filtering	verified_fetched
SRC-FUNC-063	Portals and Mirrors: Simple, Fast Evaluation of Potentially Visible Se	academic_paper	2026-09-10	Abstract + 'Introduction' (cells, portals, PVS)	verified_fetched
SRC-FUNC-064	How Northlight makes Alan Wake 2 shine	engineering_blog	2026-09-10	sections 'Core engine', 'Graphics and rendering', 'Ray traci	verified_fetched
SRC-FUNC-065	Multiplayer Level Design in Red Faction Guerrilla	conference_talk	2026-09-10	session 'Overview' text	verified_fetched
SRC-FUNC-066	AMD TressFX (GPUOpen)	official_documentation	2026-09-10	sections 'Hair rendering and simulation', 'Features', 'Meet	verified_fetched
SRC-FUNC-067	Visual Effects Summit: Can We Do It with Particles? VFX Learnings from	conference_talk	2026-09-10	session 'Overview' text	verified_fetched
SRC-FUNC-068	Uncharted Animation: An In-depth Look at the Character Animation Workf	conference_talk	2026-09-10	session listing (title, speakers, track, GDC 2008)	verified_url_only
SRC-FUNC-069	Technical Art Techniques of Naughty Dog: Vertex Shaders and Beyond	conference_talk	2026-09-10	session 'Overview' text	verified_fetched
SRC-FUNC-070	Making and Using Non-Standard Textures	conference_talk	2026-09-10	slides 2-4 (Portal 2 water rendering, flow data generation,	verified_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-FUNC-071	Vehicle Physics and Tire Dynamics in 'Just Cause 4'	conference_talk	2026-09-10	session 'Overview' text	verified_fetched
SRC-FUNC-072	NVIDIA PhysX SDK	official_documentation	2026-09-10	sections 'NVIDIA PhysX' intro and 'Key Benefits'	verified_fetched
SRC-FUNC-073	Component (Game Programming Patterns)	book	2026-09-10	chapter 'Component'	verified_fetched
SRC-FUNC-074	Object Pool (Game Programming Patterns)	book	2026-09-10	chapter 'Object Pool'	verified_fetched
SRC-FUNC-075	Spatial Partition (Game Programming Patterns)	book	2026-09-10	chapter 'Spatial Partition'	verified_fetched
SRC-FUNC-076	Game Loop (Game Programming Patterns)	book	2026-09-10	chapter 'Game Loop'	verified_fetched
SRC-FUNC-077	Cyberpunk 2077: Technology Preview Of New Ray Tracing Overdrive Mode O	engineering_blog	2026-09-10	article body (Ray Tracing: Overdrive Mode, path tracing, DLS	verified_fetched
SRC-FUNC-078	It Just Works: Ray-Traced Reflections in 'Battlefield V'	conference_talk	2026-09-10	session 'Overview' text	verified_fetched
SRC-FUNC-079	'It Just Works': Ray-Traced Reflections in 'Battlefield V' (GTC 2019)	conference_talk	2026-09-10	slides on ray divergence, DXR primer and Battlefield V relea	verified_fetched
SRC-FUNC-080	Virtual Shadow Maps in 'Fortnite Battle Royale' Chapter 4	engineering_blog	2026-09-10	sections on VSM-Nanite coupling, caching challenges, coarse	verified_fetched
SRC-FUNC-081	Visibility Preprocessing for Interactive Walkthroughs	academic_paper	2026-09-10	Abstract and the interactive walkthrough phase description (	verified_fetched
SRC-FUNC-082	Anti-cheat middleware (PCGamingWiki)	secondary	2026-09-10	sections 'Anti-cheat middleware' intro and 'nProtect GameGua	verified_fetched
SRC-FUNC-083	Set up split-screen rendering in URP (Unity Manual)	official_documentation	2026-09-10	section 'Set up split-screen rendering in URP'	verified_fetched
SRC-FUNC-084	Sci-fi and fantasy worlds collide in UE5-powered co-op adventure Split	interview	2026-09-10	Q&A on split-screen co-op technical challenges and the custo	verified_fetched
SRC-GCS-001	Background loading	official_documentation	2026-09-10	Sections 'Background loading', 'Using ResourceLoader'	verified_fetched
SRC-GCS-002	Global illumination	official_documentation	2026-09-10	Page TOC listing 'Using Voxel global illumination', 'Signed	verified_fetched
SRC-GCS-003	Using Voxel global illumination	official_documentation	2026-09-10	Sections 'Using Voxel global illumination', 'VoxelGI node pr	verified_fetched
SRC-GCS-004	Signed distance field global illumination (SDFGI)	official_documentation	2026-09-10	Sections 'Signed distance field global illumination (SDFGI)'	verified_fetched
SRC-GCS-005	Using Lightmap global illumination	official_documentation	2026-09-10	Sections 'Using Lightmap global illumination' (intro), 'Sett	verified_fetched
SRC-GCS-006	Particle systems (3D)	official_documentation	2026-09-10	Section 'Node overview' (GPUParticles3D vs CPUParticles3D, a	verified_fetched
SRC-GCS-007	Optimization using MultiMeshes	official_documentation	2026-09-10	Sections 'MultiMeshes', example 'extends MultiMeshInstance3D	verified_fetched
SRC-GCS-008	High-level multiplayer	official_documentation	2026-09-10	Sections 'Mid-level abstraction', 'Hosting considerations',	verified_fetched
SRC-GCS-009	Occlusion culling	official_documentation	2026-09-10	Sections 'Why use occlusion culling', 'How occlusion culling	verified_fetched
SRC-GCS-010	Mesh level of detail (LOD)	official_documentation	2026-09-10	Sections 'Introduction', 'Generating mesh LOD', 'Configuring	verified_fetched
SRC-GCS-011	Visibility ranges (HLOD)	official_documentation	2026-09-10	Sections 'How it works', 'Visibility range properties', 'Fad	verified_fetched
SRC-GCS-012	Optimization using Servers	official_documentation	2026-09-10	Sections 'Optimization using Servers', 'Servers', 'RIDs' (in	verified_fetched

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SRC-GCS-013	The Profiler	official_documentation	2026-09-10	Sections 'An overview of the profiler', 'The measured data'	verified_fetched
SRC-GCS-014	3D lights and shadows	official_documentation	2026-09-10	Sections 'Shadow mapping', 'Directional light', 'Directional	verified_fetched
SRC-GCS-015	Using NavigationServer	official_documentation	2026-09-10	Sections 'Threading and Synchronization', 'Waiting for synch	verified_fetched
SRC-GCS-016	Using multiple threads	official_documentation	2026-09-10	Sections 'Threads', 'Creating a Thread' (warning about slow	verified_fetched
SRC-GCS-017	Thread-safe APIs	official_documentation	2026-09-10	Sections 'Global scope', 'Scene tree', 'Rendering', 'Physics	verified_fetched
SRC-GCS-018	GPUParticles3D class reference	secondary	2026-09-10	Class description and Property Descriptions for amount, amou	verified_fetched
SRC-GCS-019	MultiMeshInstance3D class reference	secondary	2026-09-10	Class description ('Fastest way to render the same mesh in l	verified_fetched
SRC-GCS-020	WorkerThreadPool class reference	secondary	2026-09-10	Class description; Methods add_task / add_group, task (tasks_	verified_fetched
SRC-GCS-021	PhysicsServer3D class reference	secondary	2026-09-10	Class description (space / shape / body / area / joint model	verified_fetched
SRC-GCS-022	NavigationServer3D class reference	secondary	2026-09-10	Class description (Experimental marker; edge_connection_marg	verified_fetched
SRC-GCS-023	ResourceLoader class reference	secondary	2026-09-10	Methods load_threaded_request (use_sub_threads, cache_mode),	verified_fetched
SRC-GCS-024	Using physics interpolation	official_documentation	2026-09-10	Sections 'Turn on the physics interpolation setting', 'Move	verified_fetched
SRC-GCS-025	Performance class reference	secondary	2026-09-10	Class description and Notes (debug-only monitors, up to 1 se	verified_fetched
SRC-GCS-026	DirectionalLight3D class reference	secondary	2026-09-10	ShadowMode enum (SHADOW_ORTHOGONAL / SHADOW_PARALLEL_2_S PLIT	verified_fetched
SRC-GCS-027	MultiplayerAPI class reference	secondary	2026-09-10	Class description; inherited by SceneMultiplayer and Multipl	verified_fetched
SRC-GCS-028	OccluderInstance3D class reference	secondary	2026-09-10	Class description (CPU parallel rasterisation with Embree, m	verified_fetched
SRC-GCS-029	ENetMultiplayerPeer class reference	secondary	2026-09-10	Class description; create_server(port, max_clients = 32, max	verified_fetched
SRC-GCS-030	Controlling thousands of fish with Particles (godot-docs source, RST)	official_documentation	2026-09-10	Section 'Controlling thousands of fish with Particles' (part	verified_fetched
SRC-GCS-031	Godot Engine Showcase	vendor_press_release	2026-09-10	Showcase listing 'Made with Godot' (e.g. 'Slay the Spire 2 -	verified_fetched
SRC-GCS-032	Godot 4.0 sets sail: All aboard for new horizons	vendor_press_release	2026-09-10	Sections 'Global illumination' (SDFGI, VoxelGI replaces GIPr	verified_fetched
SRC-GCS-033	Voxel-Based Global Illumination (SVOGI)	official_documentation	2026-09-10	Sections 'Overview', 'How It Works', 'Performance' (3-4 ms X	verified_fetched
SRC-GCS-034	Volumetric Fog	official_documentation	2026-09-10	Sections 'Overview' (voxel buffer, ray-marching vs analytica	verified_fetched
SRC-GCS-035	Vegetation 02 Grass (Merged Meshes) - CRYENGINE	official_documentation	2026-09-10	Sections 'Asset Setup in CRYENGINE', 'Material Settings' (Ve	verified_fetched
SRC-GCS-036	Bending Setup (CRYENGINE 3 Manual, Vegetation)	official_documentation	2026-09-10	Sections 'Overview' (Touch / Detail / Automerged bending), '	verified_fetched

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SRC-GCS-037	CRYENGINE 5.7 Documentation mirror - Vegetation 02 Grass (Merged Meshe	secondary	2026-09-10	Section 'Overview' (MM combines vegetation/physics/explosion	verified_fetched
SRC-GCS-038	Audio & Occlusion	official_documentation	2026-09-10	Sections 'Overview' (SoundObstructionType: Ignore / SingleRa	verified_fetched
SRC-GCS-039	CRYENGINE 5.7 Documentation mirror - Procedural Volumetric Clouds	secondary	2026-09-10	Sections 'Overview', 'Usage' (r_VolumetricClouds, e_Clouds),	verified_fetched
SRC-GCS-040	Touch Bending (CRYENGINE 3 Manual, Vegetation)	official_documentation	2026-09-10	Sections 'Overview' (rope setup, merged mesh deform alternat	verified_fetched
SRC-GCS-041	Voxel-Based Global Illumination (CRYENGINE 3 Manual)	official_documentation	2026-09-10	Sections 'Performance' (4-5 ms Xbox One, 2-3 ms on GTX 780,	verified_fetched
SRC-GCS-042	CRYENGINE 5.7 Documentation mirror - Audio CVars & Console Commands	secondary	2026-09-10	Sections 's_OcclusionAccumulate', 's_OcclusionGlobalType' (0	verified_fetched
SRC-GCS-043	CryEngine (encyclopedia article)	secondary	2026-09-10	Infobox 'latest release version 5.7.1'; body text on Warhors	verified_fetched
SRC-GCS-044	Valve Vulkan Session (GDC 2015, Khronos session deck)	conference_talk	2026-09-10	Slide 18 'Vulkan is here' ('Source 2 supports Vulkan alpha t	verified_fetched
SRC-GCS-045	Counter-Strike 2 update notes (Steam news API for app 730)	vendor_press_release	2026-09-10	News item gid 1811772772259703 (2025-09-26): 'sv_subtick_mov	verified_fetched
SRC-GCS-046	source-sdk-2013 - src/public/bspfile.h	open_source	2026-09-10	Lump enum (LUMP_VISIBILITY=4, LUMP_NODES=5, LUMP_LEAFS=10, L	verified_fetched
SRC-GCS-047	source-sdk-2013 - src/public/vphysics_interface.h	open_source	2026-09-10	VPHYSICS_INTERFACE_VERSION 'VPhysics031'; abstract_class IPH	verified_fetched
SRC-GCS-048	source-sdk-2013 - src/public/tier0/vprof.h	open_source	2026-09-10	Header comment 'Real-Time Hierarchical Profiling'; '#define	verified_fetched
SRC-GCS-049	source-sdk-2013 - src/game/client/prediction.cpp	open_source	2026-09-10	Top-of-file ConVars: cl_predictweapons ('1'), cl_lagcompensa	verified_fetched
SRC-GCS-050	source-sdk-2013 - src/game/client/clientleafsystem.cpp	open_source	2026-09-10	ConVars r_PortalTestEnts ('1', 'Clip entities against portal	verified_fetched
SRC-GCS-051	source-sdk-2013 - src/game/client/vgui_netgraphpanel.cpp	open_source	2026-09-10	ConVars net_graph, net_graphmsecs ('400', 'The latency graph	verified_fetched
SRC-GCS-052	source-sdk-2013 - src/game/shared/usercmd.h	open_source	2026-09-10	class CUserCmd fields command_number, tick_count ('the tick	verified_fetched
SRC-GCS-053	source-sdk-2013 - src/public/dt_common.h	open_source	2026-09-10	Defines MAX_DATATABLES 1024, MAX_DATATABLE_PROPS 4096, MAX_A	verified_fetched
SRC-GCS-054	source-sdk-2013 - src/game/shared/physics_shared.cpp	open_source	2026-09-10	g_PhysDefaultObjectParams initialiser (mass 1.0, inertia 1.0	verified_fetched
SRC-GCS-055	Cassette Beasts Steam news (news API for app 1321440)	vendor_press_release	2026-09-10	News item gid 5747233970433069427 'Multiplayer Update - Now	verified_fetched
SRC-GCS-056	Brotato Steam news (news API for app 1942280)	vendor_press_release	2026-09-10	News item gid 6212245217911850965 'Update 1.1' - 'Local mult	verified_fetched
SRC-GCS-057	Source 2 (encyclopedia article)	secondary	2026-09-10	Infobox 'Release September 2015'; body text: 'the first game	verified_fetched
SRC-GCS-058	Tracy Profiler README	open_source	2026-09-10	README body: 'A real time, nanosecond resolution, remote tel	verified_fetched
SRC-GCS-059	Bevy README	open_source	2026-09-10	Sections 'What is Bevy?', 'Design Goals' ('Data Focused: Dat	verified_fetched

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SRC-GCS-060	Hazel README	open_source	2026-09-10	README body: 'Hazel is primarily an early-stage interactive	verified_fetched
SRC-GCS-061	The Forge README	open_source	2026-09-10	Sections 'Game Layer / App / Renderer - Scene (not provided)	verified_fetched
SRC-GCS-062	EnTT README	open_source	2026-09-10	README intro: 'header-only, tiny and easy to use library for	verified_fetched
SRC-GCS-063	FrameGraph README	open_source	2026-09-10	Sections 'Features' (multithreaded command buffer building a	verified_fetched
SRC-GCS-064	FrameGraph - docs/Multithreading.md	open_source	2026-09-10	Sections 'Validation' (FG_ENABLE_DATA_RACE_CHECK), 'CPU thre	verified_fetched
SRC-GCS-065	FrameGraph - docs/Introduction.md	open_source	2026-09-10	Sections 'Initialization' (I FrameGraph::CreateFrameGraph, Cr	verified_fetched
SRC-GCS-066	Bevy ECS README	open_source	2026-09-10	Sections 'What is Bevy ECS?' ('aims to be simple to use, erg	verified_fetched
SRC-GCS-067	Bevy Tasks README	open_source	2026-09-10	README body: 'simple threadpool with minimal dependencies. T	verified_fetched
SRC-GCS-068	mimalloc readme	open_source	2026-09-10	README body: 'free list sharding' (per mimalloc page, 'usual	verified_fetched
SRC-HE-001	HeroEngine (official product homepage)	official_documentation	2026-09-11	Footer line: 'Last Updated: 09.27.2024 12:20 PM Version: 3.0	verified_fetched
SRC-HE-002	HeroEngine (encyclopedia article)	secondary	2026-09-11	Infobox 'Stable release 2.074 / January 19, 2024'; sections	verified_fetched
SRC-HE-003	BioWare Licenses HeroEngine For Star Wars: The Old Republic MMO	secondary	2026-09-11	Body paragraph beginning 'MMO technology company Simutronics	verified_fetched
SRC-HE-004	HeroEngine - The Cloud-Based MMO Game Engine Behind SWTOR	secondary	2026-09-11	Sections 'Licensing and Pricing', 'Is HeroEngine Still Used	verified_fetched
SRC-HE-005	HeroEngine homepage (Internet Archive snapshot, 2013)	official_documentation	2026-09-11	Homepage banner block 'GET THE HEROENGINE, NOW FOR ONLY \$99	verified_fetched
SRC-HE-006	HeroBlade HeroEngine (Internet Archive snapshot, 2014)	official_documentation	2026-09-11	Sections 'All-in-one Development Environment', 'Massively Co	verified_fetched
SRC-HE-007	World Building HeroEngine (Internet Archive snapshot, 2013-12-27)	official_documentation	2026-09-11	Sections 'Real-time Collaborative World Building', 'Seamless	verified_fetched
SRC-HE-008	Server Systems HeroEngine (Internet Archive snapshot, 2013)	official_documentation	2026-09-11	Intro paragraphs ('Segregate your geography into individual	verified_fetched
SRC-HE-009	Game Systems HeroEngine (Internet Archive snapshot, 2013)	official_documentation	2026-09-11	Sections 'Game Systems Built for Real-Time Environments', 'H	verified_fetched
SRC-HE-010	Licensing Options HeroEngine (Internet Archive snapshot, 2013)	official_documentation	2026-09-11	Sections 'HeroCloud: HeroEngine in the Cloud, ONLY \$99 a yea	verified_fetched
SRC-HE-011	HEWIKI Main Page (Internet Archive snapshot, 2014; page last modified	official_documentation	2026-09-11	Navigation sections 'World Building', 'Scripting', 'Engine A	verified_fetched
SRC-HE-012	Scalability and Building For Massive Multiplayer Audiences (HEWIKI, In	official_documentation	2026-09-11	Sections 'Terminology', 'Per Physical Server', 'Per Area Ins	verified_fetched
SRC-HE-013	HeroCloud Tech Features HeroEngine (Internet Archive snapshot, 2014)	official_documentation	2026-09-11	Blocks 'HeroCloud gets you a license to the HeroEngine for j	verified_fetched
SRC-HE-015	SWTOR DirectX 12 Spring 2026 Update (official developer blog)	engineering_blog	2026-09-11	Section 'Next Steps', paragraph beginning 'SWTOR was origina	verified_fetched
SRC-HE-016	Idea Fabrik, the company behind Hero Engine and The Repopulation, disa	secondary	2026-09-11	Opening paragraphs naming Idea Fabrik as holder of the Hero	verified_fetched

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SRC-HE-017	Software:Faxion Online (encyclopedia article, Wikipedia mirror)	secondary	2026-09-11	Infobox field 'Engine: HeroEngine'; sentence 'It makes use o	verified_fetched
SRC-HE-018	Laniatus Cloud Tech (official company site)	vendor_press_releas e	2026-09-11	Sections 'We build the engine beneath the game', 'The platfo	verified_fetched
SRC-HE-019	PlayM2M on Steam (store page)	secondary	2026-09-11	Store metadata block: Developer 'Laniatus LLC', Publisher 'L	verified_fetched
SRC-HE-020	Idea Fabrik PLC (official corporate site)	official_documentati on	2026-09-11	Blocks 'Agile Development', 'Disruptive Development', 'Game	verified_fetched
SRC-MPR-001	Rendering the Hellscape of Doom Eternal (Advances in Real-Time Renderi	conference_talk	2026-09-10	slides 3, 4, 9, 14, 24, 44, 45, 49, 53, 57, 62, 66, 72, 76	verified_fetched
SRC-MPR-002	Exploring Ray Traced Future in Metro Exodus (NVIDIA GTC 2019)	conference_talk	2026-09-10	slides 8, 9, 22, 26, 27, 28, 32, 40, 48, 73, 76, 87	verified_fetched
SRC-MPR-003	How Northlight makes Alan Wake 2 shine	engineering_blog	2026-09-10	sections 'GPU-driven rendering and meshlets', 'Character-sty	verified_fetched
SRC-MPR-004	Moving Gears to Tier 2 Variable Rate Shading (DirectX Developer Blog,	engineering_blog	2026-09-10	sections 'Moving to Tier 2', 'VRS Texture Generation', 'Work	verified_fetched
SRC-MPR-005	Face-Off: Titanfall 2 (Digital Foundry)	secondary	2026-09-10	section 'Shadow quality', 'PC vs console'	verified_fetched
SRC-MPR-006	Brilliant visuals and growing pains: analysing Unreal Engine 5 first-g	secondary	2026-09-10	sections on Lumen/Nanite/VSM adoption, 'RT quality levels',	verified_fetched
SRC-MPR-007	Black Myth: Wukong - the PC tech review (Digital Foundry)	secondary	2026-09-10	sections on Nanite, Lumen, virtual shadow maps, full RT / Re	verified_fetched
SRC-MPR-008	Lords of the Fallen is a stunning UE5 soulslike with ongoing tech issu	secondary	2026-09-10	sections on virtual shadow maps, dynamic resolution, volumet	verified_fetched
SRC-MPR-009	Senua's Saga: Hellblade 2 is a defining moment in the evolution of rea	secondary	2026-09-10	sections on Nanite, virtual shadow maps, Lumen, volumetric f	verified_fetched
SRC-MPR-010	Senua's Saga: Hellblade 2 PC tech review (Digital Foundry)	secondary	2026-09-10	sections on upscaling/TSR, dynamic resolution, frame generat	verified_fetched
SRC-MPR-011	The AI Systems of Left 4 Dead (GDC 2009)	conference_talk	2026-09-10	slides 53-56, 59, 64-75, 78-82	verified_fetched
SRC-MPR-012	Three States and a Plan: The A.I. of F.E.A.R. (GDC 2006)	conference_talk	2026-09-10	pages 1-4, sections 'Managing Complexity', 'FSMs vs Planning	verified_fetched
SRC-MPR-013	Crowd Pathfinding and Steering Using Flow Field Tiles (Game AI Pro, Ch	book	2026-09-10	sections 23.3-23.8, 23.14, 23.15	verified_fetched
SRC-MPR-014	The Art of Destruction in Rainbow Six: Siege (GDC 2016)	conference_talk	2026-09-10	slides 54-55, 57-58, 61-68, 73-74	verified_fetched
SRC-MPR-015	Massive Crowd on Assassin's Creed Unity: AI Recycling (GDC 2015, sessi	conference_talk	2026-09-10	session abstract	verified_fetched
SRC-MPR-016	Continuous World Generation in 'No Man's Sky' (GDC 2017, session page)	conference_talk	2026-09-10	session abstract	verified_fetched
SRC-MPR-017	Graphics Deep Dive: Cascaded voxel cone tracing in The Tomorrow Childr	interview	2026-09-10	sections 'What: Cascaded voxel cone tracing', 'Why: A totall	verified_fetched
SRC-MPR-018	Interactive Indirect Illumination Using Voxel Cone Tracing (Pacific Gr	academic_paper	2026-09-10	abstract	verified_fetched
SRC-MPR-019	GPU Gems 3, Chapter 13: Volumetric Light Scattering as a Post-Process	book	2026-09-10	section 13.2 'Crepuscular Rays', 13.6 'Caveats'	verified_fetched
SRC-MPR-020	Decima Engine: Advances in Lighting and AA (Guerrilla, SIGGRAPH 2017 t	conference_talk	2026-09-10	abstract	verified_fetched
SRC-MPR-021	Streaming the World of Horizon Zero Dawn (Guerrilla talk page)	conference_talk	2026-09-10	abstract	verified_fetched
SRC-MPR-022	GPU-Based Procedural Placement in Horizon Zero Dawn (Guerrilla talk pa	conference_talk	2026-09-10	abstract	verified_fetched
SRC-MPR-023	Virtual Shadow Maps in Unreal Engine (Epic official documentation)	official_documentati on	2026-09-10	sections 'Shadow Cache Invalidation Behavior', 'Contact Shad	verified_fetched

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SRC-MPR-024	Lumen Technical Details in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Surface Cache', 'Radiance Cache', 'Distance Field'	verified_fetched
SRC-MPR-025	Lightmass Basics in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Lightmass Importance Volume', 'Lightmaps'	verified_fetched
SRC-MPR-026	Optimizing Rendering with PSO Caches in Unreal Engine (Epic official d	official_documentation	2026-09-10	sections 'Terminology and Supported PSO Types', 'Generate PS	verified_fetched
SRC-MPR-027	Scalability in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Scalability Levels', 'Quality Levels'	verified_fetched
SRC-MPR-028	Dynamic Resolution in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'How Dynamic Resolution Works', 'FAQ'	verified_fetched
SRC-MPR-029	Hierarchical Level of Detail in Unreal Engine (Epic official documenta	official_documentation	2026-09-10	section 'Hierarchical Level of Detail'	verified_fetched
SRC-MPR-030	Hair Rendering and Simulation in Unreal Engine (Epic official document	official_documentation	2026-09-10	sections 'Groom', 'Card-based hair', 'Level of Detail'	verified_fetched
SRC-MPR-031	Water System in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Water Bodies', 'Simulating Waves Using the Water W	verified_fetched
SRC-MPR-032	Scriptable Render Pipeline Batcher in URP (Unity official documentatio	official_documentation	2026-09-10	sections 'How the SRP Batcher works', 'Requirements and comp	verified_fetched
SRC-MPR-033	Quality Settings (Unity official documentation)	official_documentation	2026-09-10	sections 'Shadow Cascades', 'Cascade splits', 'Shadowmask mo	verified_fetched
SRC-MPR-034	Lightmapping (Unity official documentation)	official_documentation	2026-09-10	sections 'Precalculating surface lighting with lightmaps', '	verified_fetched
SRC-MPR-035	The Progressive Lightmapper (Unity official documentation)	official_documentation	2026-09-10	sections 'Progressive CPU Lightmapper', 'Progressive GPU Lig	verified_fetched
SRC-MPR-036	Light Probes (Unity official documentation)	official_documentation	2026-09-10	sections 'Precalculating indirect light with Light Probes',	verified_fetched
SRC-MPR-037	V Rising: Behind the vampire realm built on 1,600 ECS systems (Unity c	engineering_blog	2026-09-10	sections 'The challenge', 'Scaling up operations with DOTS',	verified_fetched
SRC-MPR-038	Using SDFGI (Godot official documentation)	official_documentation	2026-09-10	sections 'Cascades', 'Cascade 0 Distance', 'SDFGI caveats'	verified_fetched
SRC-MPR-039	2D lights and shadows (Godot official documentation)	official_documentation	2026-09-10	sections 'LightOccluder2D', 'Item Cull Mask', 'Limitations'	verified_fetched
SRC-MPR-040	Using LightmapGI (Godot official documentation)	official_documentation	2026-09-10	sections 'LightmapGI', 'Shadowmasking'	verified_fetched
SRC-MPR-041	Variable Rate Shading (Microsoft Direct3D 12 documentation)	secondary	2026-09-10	sections 'Variable Rate Shading Tiers', 'Shading Rate Image'	verified_fetched
SRC-MPR-042	DirectStorage overview (Microsoft Game Development Kit documentation)	official_documentation	2026-09-10	sections 'DirectStorage overview', 'Decompression'	verified_fetched
SRC-MPR-043	GPU Lightmass Global Illumination in Unreal Engine (Epic official docu	official_documentation	2026-09-10	sections 'GPU Lightmass', 'Enabling GPU Lightmass'	verified_fetched
SRC-MPR-044	Distance Field Soft Shadows in Unreal Engine (Epic official documentat	official_documentation	2026-09-10	sections 'Distance Field Soft Shadows', 'Distance Field Ambi	verified_fetched
SRC-MPR-045	Saving and Loading Your Game in Unreal Engine (Epic official documenta	official_documentation	2026-09-10	sections 'SaveGame', 'SaveGameToSlot', 'Async Saving'	verified_fetched
SRC-MPR-046	Audio Mixer Overview in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Audio Mixer', 'Sound Waves'	verified_fetched
SRC-MPR-047	Gameplay Ability System in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Gameplay Abilities', 'Gameplay Effects', 'Gameplay	verified_fetched
SRC-MPR-048	The creators of the Havok engine discouraged the authors of Red Fatio	secondary	2026-09-10	article body, paragraphs on Geo-Mod 2.0 and its rendering/AI	verified_fetched
SRC-MPR-049	Using Light Shafts in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Occlusion Method', 'Bloom Method', 'GPU Cost'	verified_fetched

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SRC-MPR-050	Volumetric Fog in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Volumetric Fog', 'Froxel Grid', 'Performance'	verified_fetched
SRC-MPR-051	Portal developer commentary (Portal Wiki transcription of Valve in-gam	secondary	2026-09-10	commentary entries on portal rendering and recursive portal	verified_fetched
SRC-MPR-052	Occlusion Culling (Unity official documentation)	official_documentation	2026-09-10	sections 'Occlusion Culling', 'Occlusion Culling data'	verified_fetched
SRC-MPR-053	Occlusion culling (Godot official documentation)	official_documentation	2026-09-10	sections 'Why use occlusion culling', 'How occlusion culling	verified_fetched
SRC-MPR-054	Mesh level of detail (LOD) (Godot official documentation)	official_documentation	2026-09-10	sections 'Automatic mesh LOD', 'Visibility ranges (HLOD)'	verified_fetched
SRC-MPR-055	Using physics interpolation (Godot official documentation)	official_documentation	2026-09-10	sections 'What is physics interpolation', 'Setup'	verified_fetched
SRC-MPR-056	Reflections Captures in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Scene Capture', 'Planar Reflections', 'Reflection	verified_fetched
SRC-MPR-057	Overview of Niagara Effects in Unreal Engine (Epic official documentat	official_documentation	2026-09-10	sections 'Niagara', 'GPU Particles', 'SubUV / Flipbook'	verified_fetched
SRC-MPR-058	Cloth (Unity official documentation)	official_documentation	2026-09-10	sections 'Cloth component', 'Constraints'	verified_fetched
SRC-MPR-059	Job System overview (Unity official documentation)	official_documentation	2026-09-10	sections 'Job System', 'Safety system'	verified_fetched
SRC-MPR-060	Sprite Atlas (Unity official documentation)	official_documentation	2026-09-10	sections 'Sprite Atlas', 'Sprite packing'	verified_fetched
SRC-MPR-061	True Impostors (GPU Gems 3, Chapter 21)	book	2026-09-10	chapter body, section on impostor rendering	verified_fetched
SRC-MPR-062	Baking Normal Maps on the GPU (GPU Gems 3, Chapter 22)	book	2026-09-10	chapter body	verified_fetched
SRC-MPR-063	Chaos Physics Overview in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Chaos Physics', 'Scene Queries'	verified_fetched
SRC-MPR-064	Motion Matching in Unreal Engine (Epic official documentation)	official_documentation	2026-09-10	sections 'Motion Matching', 'Pose Search'	verified_fetched
SRC-MPR-065	3D lights and shadows (Godot official documentation)	official_documentation	2026-09-10	sections 'Shadow mapping', 'Shadow bias'	verified_fetched
SRC-MPR-066	Illuminating Roco's Legacy: Scalable Global Illumination from Theory t	conference_talk	2026-09-10	slides 5, 9, 15-17, 23-24, 28-31, 33-34, 36, 52	verified_fetched
SRC-NTA-001	Source Multiplayer Networking (Valve Developer Community wiki)	secondary	2026-09-10	Sections: '## Basic networking', '## Servers for these games	verified
SRC-NTA-002	Lag Compensation (Valve Developer Community wiki)	secondary	2026-09-10	Sections: '# Lag Compensation' (intro paragraph), '## Config	verified
SRC-NTA-003	Latency Compensating Methods in Client/Server In-game Protocol Design	secondary	2026-09-10	Linked from the Lag Compensation page ('You may be looking f	available
SRC-NTA-004	'Overwatch' Gameplay Architecture and Netcode (GDC 2017)	conference_talk	2026-09-10	Session page: 'Session Name: 'Overwatch' Gameplay Architectu	verified
SRC-NTA-005	Replay Technology in 'Overwatch': Kill Cam, Gameplay, and Highlights (	conference_talk	2026-09-10	Session page title + abstract ('kill cams ... Highlights and	available
SRC-NTA-006	VALORANT's 128-Tick Servers (Riot Games tech blog)	secondary	2026-09-10	Body sections: '# VALORANT's 128-Tick Servers', frame-budget	verified
SRC-NTA-007	Peeking into VALORANT's Netcode (Riot Games tech blog)	secondary	2026-09-10	Body sections: server-authoritative note, fixed-timestep par	verified
SRC-NTA-008	1500 Archers on a 28.8: Network Programming in Age of Empires and Beyo	secondary	2026-09-10	Sections: '## Simultaneous Simulations', '## "Speed Control"	verified
SRC-NTA-009	1500 Archers on a 28.8 (PDF mirror with tables and diagrams)	secondary	2026-09-10	Same section titles as the HTML version; the PDF carries the	available
SRC-NTA-010	Fast-Paced Multiplayer (Part I): Client-Server Game Architecture (PDF)	secondary	2026-09-10	Article I body: authoritative server / dumb client section a	available

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SRC-NTA-011	Client-Side Prediction and Server Reconciliation (Fast-Paced Multiplay	secondary	2026-09-10	Article body: 'Client-Side Prediction and Server Reconciliat	available
SRC-NTA-012	Counter-Strike 2 official feature page ('What you see is what you get'	secondary	2026-09-10	Section: 'Counter-Strike 2: What you see is what you get' -	verified
SRC-NTA-013	Sub-tick (Counter-Strike community wiki)	secondary	2026-09-10	Sections: 'Mechanics of the Sub-tick System', 'Implementatio	available
SRC-NTA-014	Replication Graph in Unreal Engine (UE 5.8 Documentation)	secondary	2026-09-10	Sections: '# Replication Graph' (intro with the Fortnite 100	verified
SRC-NTA-015	Setting Up Dedicated Servers in Unreal Engine (UE 5.8 Documentation)	secondary	2026-09-10	Sections: '## Overview' -> '### Dedicated Server' (headless	verified
SRC-NTA-016	Iris Replication System in Unreal Engine (UE 5.8 Documentation)	secondary	2026-09-10	Page: '# Iris Replication System' - 'The Iris Replication Sy	available
SRC-NTA-017	Unity Netcode for Entities (package manual 1.0.17)	secondary	2026-09-10	Sections: '# Unity Netcode for Entities' (intro: 'server aut	verified
SRC-NTA-018	Introduction to prediction (Netcode for Entities 7.0 manual)	secondary	2026-09-10	Section: '# Introduction to prediction' - 'Client prediction	available
SRC-NTA-019	Unity's netcode packages (Unity Multiplayer docs)	secondary	2026-09-10	Page: 'Unity has two netcode frameworks for creating real-ti	available
SRC-NTA-020	Steam Audio - Programmer's Guide (C API)	secondary	2026-09-10	Sections: 'Initialization' -> 'Memory Allocation', 'HRTF', '	verified
SRC-NTA-021	Steam Audio core documentation (GitHub, index.rst)	secondary	2026-09-10	Top of document: overview bullets including 'Steam Audio can	available
SRC-NTA-022	Using Features: Occlusion (Wwise documentation)	secondary	2026-09-10	Section: 'Obstruction and Occlusion' - describes how the ser	available
SRC-NTA-023	Wwise Acoustics Concepts (Spatial Audio concepts)	secondary	2026-09-10	Section: acoustic concepts - 'Wwise maps obstruction and occ	available
SRC-NTA-024	Obstruction and Occlusion (Wwise Unity documentation mirror)	secondary	2026-09-10	Sections: '# Obstruction and Occlusion', '## A. Emitter Obst	verified
SRC-NTA-025	Wwise Spatial Audio (product page)	secondary	2026-09-10	Product overview: 'Use the Unity or Unreal editors to design	available
SRC-NTA-026	FMOD - Core API: Using DSP Effects	secondary	2026-09-10	Page: DSP effects can be inserted into the graph with functi	available
SRC-NTA-027	FMOD API - FMOD_DSP_CONVOLUTION_REVERB	secondary	2026-09-10	Page: FMOD_DSP_CONVOLUTION_REVERB parameter list (wet/dry mi	available
SRC-NTA-028	Steam Datagram Relay (Steamworks Documentation)	secondary	2026-09-10	Page: Steam Datagram Relay overview and identity requirement	available
SRC-NTA-029	Steam Datagram Relay (Valve Developer Community wiki)	secondary	2026-09-10	Page intro: 'Using our APIs, you can not only carry your gam	available
SRC-NTA-030	Using the Anti-Cheat Interfaces (Epic Online Services / Easy Anti-Chea	secondary	2026-09-10	Sections: '# Using the Anti-Cheat Interfaces', '### Client M	verified
SRC-NTA-031	Anti-Cheat Interfaces (Epic Online Services, Trust & Safety)	secondary	2026-09-10	Page: 'Integrate the Anti-Cheat Interfaces (also known as "E	available
SRC-NTA-032	Vanguard On-Demand - Anti-Cheat Update (Riot Games)	secondary	2026-09-10	Article: 'Starting later today, ... Vanguard, will begin to	available
SRC-NTA-033	Riot Vanguard FAQ (Riot Games Support)	secondary	2026-09-10	Page: 'Riot Vanguard is Riot Games' custom game security sof	available
SRC-NTA-034	Garbage collection modes (Unity Manual, Unity 6 / 6000.0)	secondary	2026-09-10	Page: '# Garbage collection modes' - 'Incremental garbage co	partial
SRC-NTA-035	Garbage Collection Settings in the Unreal Engine Project Settings (UE	secondary	2026-09-10	Reference rows including 'Minimum GC Cluster Size' and 'Maxi	available

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-NTA-036	Migrate to Iris in Unreal Engine (UE 5.8 Documentation)	secondary	2026-09-10	Page: 'Learn what has changed between Unreal Engine's existi	available
SRC-NTA-037	Quake 3 Network Protocol	secondary	2026-09-10	Page: protocol overview (snapshot / delta against a baseline	available
SRC-NTA-038	Quake-III-Arena / code/server/sv_snapshot.c (id Software GPL source re	secondary	2026-09-10	Source file: snapshot generation / delta logic (SV_BuildClie	available
SRC-NTA-039	GameNetworkingSockets (ValveSoftware)	secondary	2026-09-10	Repository README: 'Reliable ...' message-oriented connectio	available
SRC-NTA-040	Ambience Design and Acoustic Systems in 'Senua's Saga: Hellblade II' (	conference_talk	2026-09-10	Session page: 'Sound Designer Pablo Canas explains the proce	available
SRC-NTA-041	How Ninja Theory created Hellblade II's unsettling soundscape (Game De	secondary	2026-09-10	Article lede: 'During the 2025 Game Developers Conference, N	available
SRC-NTA-042	Developer Insight - Did you hear that? (Hunt: Showdown, Crytek)	secondary	2026-09-10	Post intro: 'With the help of our audio engineers, we have c	available
SRC-NTA-043	Dedicated Server (Unity Manual)	secondary	2026-09-10	Page: 'Unity provides support for development of games and a	available
SRC-NTA-044	Build your application for Dedicated Server (Unity Manual)	secondary	2026-09-10	Page: 'When you build for the dedicated server platform, Uni	available
SRC-NTA-045	Why Bullets "Miss": An Analysis of the CS2 Subtick System	secondary	2026-09-10	Sections: 'What Is Tickrate and Why It Matters', 'Why CS2 Se	available
SRC-NTA-046	Lag compensation (Official Team Fortress Wiki)	secondary	2026-09-10	Page: 'Team Fortress 2, being a Source Engine game, is gener	available
SRC-NTA-047	VAN: Limiting and Closing the Vanguard Pre-Boot Motherboard Security G	secondary	2026-09-10	Article: 'In the near future, Vanguard will enforce stricter	available
SRC-NTA-048	Fan translation/summary of 'Overwatch Gameplay Architecture and Netcod	secondary	2026-09-10	Sections covering the command-frame timing ('16ms ... 7ms'),	verified
SRC-NTA-049	'Overwatch' Gameplay Architecture and Netcode (full talk video)	secondary	2026-09-10	Linked as the original talk from SRC-NTA-048; no timestamp v	unverified
SRC-NTA-050	Audio file compression in Unity (Unity Manual)	secondary	2026-09-10	Page: '# Audio file compression in Unity' - format sections	partial
SRC-NTA-051	Conversion Settings Editor (Wwise documentation)	secondary	2026-09-10	Page: 'In each ShareSet you can define various conversion se	partial
SRC-NTA-052	Prediction (Netcode for Entities documentation, GitHub mirror)	secondary	2026-09-10	Document: 'Unity adds the PredictedGhost component to all pr	available
SRC-RND-001	Lumen Global Illumination and Reflections (Unreal Engine 5.8 Documenta	secondary	2026-09-10	Sections: '# Lumen Global Illumination and Reflections', '##	available
SRC-RND-002	Virtual Shadow Maps (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Virtual Shadow Maps', '## Clipmaps for Directio	available
SRC-RND-003	Nanite Virtualized Geometry Overview (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '## How does Nanite work?', '## Benefits of Nanite	available
SRC-RND-004	Temporal Super Resolution (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Temporal Super Resolution', '### Understanding	available
SRC-RND-005	Render Dependency Graph (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Render Dependency Graph', '#### Transient Resou	available
SRC-RND-006	Dynamic Resolution (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Dynamic Resolution', '## Dynamic Resolution Cru	available
SRC-RND-007	Path Tracer (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Path Tracer', '### Path Tracer Post Process Vol	available
SRC-RND-008	Scalability (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# Scalability'; linked 'Scalability Reference' (s	available

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SRC-RND-009	PSO Caches (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# PSO Caches', '## Terminology and Supported PSO	available
SRC-RND-010	NVIDIA DLSS (developer landing page)	secondary	2026-09-10	Sections: '## DLSS AI Technologies', '## Key Benefits' -> '#	available
SRC-RND-011	Accelerating Ultra-Realistic Game Development with NVIDIA DLSS 3 and N	secondary	2026-09-10	Sections: '## A revolution in neural graphics' (Figure 1 cap	available
SRC-RND-012	FidelityFX Super Resolution 2 (FSR 2.2.1) README	secondary	2026-09-10	Sections: 'Scaling modes' (per-dimension factors table), 'Ca	available
SRC-RND-013	Intel XeSS Super Resolution (XeSS-SR) Developer Guide 2.0	secondary	2026-09-10	Sections: quality-preset table (fixed resolution scaling), '	available
SRC-RND-014	DirectX Raytracing (DXR) Functional Spec, v1.48	api_specification	2026-09-10	Sections: '# Walkthrough' -> '## Geometry and acceleration s	available
SRC-RND-015	Ray Tracing (Vulkan Documentation Project / Khronos guide)	secondary	2026-09-10	Sections: extension list ('A set of five interrelated extens	available
SRC-RND-016	VK_EXT_mesh_shader proposal (Vulkan Documentation Project)	secondary	2026-09-10	Sections: '2. Solution Space', '3.1.1' (mesh shaders), '3.1.	available
SRC-RND-017	Variable Rate Shading (VRS) Functional Spec	api_specification	2026-09-10	Sections: '## Feature Tiering', '### Tier 1', '### Tier 2',	available
SRC-RND-018	Interactive Indirect Illumination Using Voxel Cone Tracing	secondary	2026-09-10	Abstract (quoted verbatim): 'Our approach can manage two lig	available
SRC-RND-019	Volumetric Fog: Unified, compute shader based solution to atmospheric	secondary	2026-09-10	PDF pages: p.25 (frustum-aligned froxel volume, exponential	available
SRC-RND-020	Clustered Deferred and Forward Shading	secondary	2026-09-10	p.1 Figure 1 caption ('~2400 light sources ... rendered in 1	available
SRC-RND-021	FrameGraph: Extensible Rendering Architecture in Frostbite	secondary	2026-09-10	Session page: 'Session Name: FrameGraph: Extensible Renderin	available
SRC-RND-022	Rendering the Hellscape of Doom Eternal	conference_talk	2026-09-10	PDF pages: p.1 (title/authors), p.3 ('IDTECH 7 - Fully forwa	available
SRC-RND-023	How Northlight makes Alan Wake 2 shine	secondary	2026-09-10	Sections: '## New GPU-driven rendering pipeline' (mesh shad	available
SRC-RND-024	Path Tracing & Overdrive Mode - Requirements & How-To (Cyberpunk 2077	secondary	2026-09-10	Body text: 'With Patch 1.62 we've introduced Ray Tracing: Ov	available
SRC-RND-025	Quake II RTX: Re-Engineering a Classic with Ray Tracing Effects on Vul	secondary	2026-09-10	Body text: 'Quake II RTX is a pure ray-traced game. That mea	available
SRC-RND-026	Global Illumination in Metro Exodus: An Artist's Point of View	secondary	2026-09-10	Article body discussing ray-traced GI in a shipped game from	available
SRC-RND-027	Exploring Ray Traced Future in Metro Exodus (GTC 2019)	conference_talk	2026-09-10	Slide agenda items: 'Added buffer to cache raytrace data for	available
SRC-RND-028	Decima Engine: Advances in Lighting and AA	conference_talk	2026-09-10	Abstract: 'our 2-frame temporal anti-aliasing solution for 1	available
SRC-RND-029	2D lights and shadows (Godot Engine documentation)	secondary	2026-09-10	Sections: '# 2D lights and shadows', '## Nodes', '## Point I	available
SRC-RND-030	"A Dive into Render Graphs" (SIGGRAPH 2023) - NOT LOCATED	secondary	2026-09-10	Full-text scan of the SIGGRAPH Advances in Real-Time Renderi	unavailable
SRC-RND-031	Real-Time Rendering, 4th edition	book	2026-09-10	Standard reference chapters on shadow mapping / cascaded sha	available (publisher page reac
SRC-RND-032	Shadow mapping (Wikipedia)	secondary	2026-09-10	Article body on shadow map aliasing and cascaded shadow maps	available

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-RND-033	Ambient occlusion (Wikipedia)	secondary	2026-09-10	Article body on screen-space ambient occlusion and derived s	available
SRC-RND-034	Lightmap (Wikipedia)	secondary	2026-09-10	Article body on precomputed lightmaps and their memory/light	available
SRC-RND-035	Progressive Lightmapper (Unity Manual)	secondary	2026-09-10	n/a - request redirected to 'Choose a light baking backend'	unavailable (consent-gated)
SRC-RND-036	Scriptable Render Pipeline Batcher (Unity Manual)	secondary	2026-09-10	n/a - returned only the OneTrust consent banner	unavailable (consent-gated)
SRC-RND-037	Scene Capture 2D (Unreal Engine documentation)	secondary	2026-09-10	n/a - URL redirected to the documentation index (page not fo	unavailable (404 / redirect)
SRC-RND-038	GPU Lightmass (Unreal Engine documentation)	secondary	2026-09-10	n/a - URL redirected to the documentation index (page not fo	unavailable (404 / redirect)
SRC-RND-039	SRP Batcher: Speed up your rendering!	engineering_blog	2026-09-10	Sections: 'How SRP Batcher works', 'How to enable SRP Batche	available
SRC-RND-040	Hierarchical-Z map based occlusion culling	engineering_blog	2026-09-10	Sections: 'Introduction', 'Motivation', 'The algorithm' -> '	available
SRC-RND-041	Leveraging Asynchronous Queues for Concurrent Execution	secondary	2026-09-10	Sections: 'Why concurrency is important', 'Improved performa	available
SRC-RND-042	Sci-fi and fantasy worlds collide in UE5-powered co-op adventure Split	interview	2026-09-10	Section: 'Are there particular technical challenges you face	available
SRC-RND-043	Portals Part 6 - Portal Recursion	engineering_blog	2026-09-10	Sections: 'Portal recursion', 'Recursive portal shader', 'Re	available
SRC-RND-044	Mesh Shader Functional Spec (D3D12)	api_specification	2026-09-10	Sections: 'Intro', 'Motivation for adding Mesh Shader', 'Con	available
SRC-RND-045	DOOM Eternal - Graphics Study	secondary	2026-09-10	Sections: 'Introduction', '# A frame in Doom Eternal', '## O	available
SRC-RND-046	Using async compute to saturate GPU (Vulkan Samples)	secondary	2026-09-10	Sections: 'Overview', 'Compute all the things - a post proce	available
SRC-RND-047	HLSL Dynamic Resources (Shader Model 6.6)	api_specification	2026-09-10	Sections: 'HLSL Changes' -> 'ResourceD escriptorHeap and Samp	available
SRC-RND-048	It Takes Two tech analysis (Digital Foundry)	secondary	2026-09-10	n/a - server returned '404 Not Found'	verified_fetched
SRC-TN-001	DirectX 12 programming guide (Windows Win32)	official_documentati on	2026-09-11	Landing page of the Win32 Direct3D 12 programming guide sect	verified_fetched
SRC-TN-002	Command queues and command lists (Direct3D 12)	official_documentati on	2026-09-11	Section overview describing ID3D12CommandQueue / ID3D12Comma	verified_fetched
SRC-TN-003	Direct3D 11 graphics (table of contents)	official_documentati on	2026-09-11	Table of contents of the Direct3D 11 graphics documentation	verified_fetched
SRC-TN-004	Direct3D 11 programming reference (Win32 API index)	secondary	2026-09-11	API index page for the ID3D11Device / ID3D11DeviceContext in	verified_fetched
SRC-TN-005	Vulkan Specification (latest, HTML)	secondary	2026-09-11	Full normative specification document (vkspec.html)	verified_fetched
SRC-TN-006	Vulkan-Docs (Khronos Group repository)	secondary	2026-09-11	Repository root README describing the spec/refpages/XML sour	verified_fetched
SRC-TN-007	DirectX Raytracing (DXR) functional specification overview	official_documentati on	2026-09-11	Overview section introducing raytracing pipelines, accelerat	verified_fetched
SRC-TN-008	HLSL reference for DirectX Raytracing	secondary	2026-09-11	Reference listing of raytracing-specific HLSL intrinsics and	verified_fetched
SRC-TN-009	acl - Animation Compression Library	secondary	2026-09-11	Repository README: purpose, feature list and design goals	verified_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-TN-010	Animation Compression Library in Unreal Engine	official_documentation	2026-09-11	Engine documentation page for the ACL plugin and its compress	verified_fetched
SRC-TN-011	meshoptimizer - mesh optimization library	secondary	2026-09-11	Repository root: algorithm list (index/vertex-cache/overdraw	verified_fetched
SRC-TN-012	meshoptimizer README	official_documentation	2026-09-11	README body describing each algorithm, its purpose and its e	verified_fetched
SRC-TN-013	RVO2 Library: documentation (2.0)	official_documentation	2026-09-11	Documentation index and API reference for RVO2 2.0 (Agent, S	verified_fetched
SRC-TN-014	RVO2 Library - project page	secondary	2026-09-11	Project landing page with the abstract of the underlying ORC	verified_fetched
SRC-TN-015	Tracy Profiler	secondary	2026-09-11	Repository root describing the hybrid instrumentation + samp	verified_fetched
SRC-TN-016	Tracy Profiler README	official_documentation	2026-09-11	README describing features, supported GPU APIs and integrati	verified_fetched
SRC-TN-017	Opus Codec - official site	official_documentation	2026-09-11	Landing page stating the intended application range (VoIP, v	verified_fetched
SRC-TN-018	RFC 6716 - Definition of the Opus Audio Codec	secondary	2026-09-11	Normative RFC defining the codec, its modes (SILK/CELT hybri	verified_fetched
SRC-TN-019	PhysX (NVIDIA-Omniverse) repository	secondary	2026-09-11	Repository root: SDK scope, build instructions, feature summ	verified_fetched
SRC-TN-020	NVIDIA PhysX SDK developer page	secondary	2026-09-11	Product page describing the SDK and its platform/feature cov	verified_fetched
SRC-TN-021	Unity DOTS product page	secondary	2026-09-11	Product page defining DOTS as the combination of Entities, B	verified_fetched
SRC-TN-022	Unity Entities package manual	official_documentation	2026-09-11	Package documentation landing page (ECS concepts, worlds, sy	verified_fetched
SRC-TN-023	Unity Netcode for Entities manual	official_documentation	2026-09-11	Package documentation landing page (client/server prediction	verified_fetched
SRC-TN-024	Unity Transport package manual	official_documentation	2026-09-11	Package documentation landing page (connections, pipelines,	verified_fetched
SRC-TN-025	Unity Addressables package manual	official_documentation	2026-09-11	Package documentation landing page (groups, labels, build an	verified_fetched
SRC-TN-026	Unity Burst compiler package manual	official_documentation	2026-09-11	Package documentation landing page (LLVM backend, [BurstComp	verified_fetched
SRC-TN-027	Universal Render Pipeline package manual	official_documentation	2026-09-11	Package documentation landing page for the Scriptable Render	verified_fetched
SRC-TN-028	World Partition in Unreal Engine	official_documentation	2026-09-11	Documentation page describing the grid-based partitioning, s	verified_fetched
SRC-TN-029	Replication Graph in Unreal Engine	official_documentation	2026-09-11	Documentation page describing the replication graph node cla	verified_fetched
SRC-TN-030	Steamworks SDK documentation	official_documentation	2026-09-11	SDK landing page listing the API surface and integration req	verified_fetched
SRC-TN-031	Steamworks Features overview	official_documentation	2026-09-11	Feature index page enumerating Steamworks subsystems	verified_fetched
SRC-TN-032	Microsoft DirectStorage repository	secondary	2026-09-11	Repository root: runtime + tools, GDeflate codec, staging bu	verified_fetched

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-TN-033	DirectStorage README	official_documentation	2026-09-11	README describing requirements, supported decompression code	verified_fetched
SRC-TN-040	Ashes of the Singularity (Wikipedia)	secondary	2026-09-11	Game article: release, developer, and engine/API notes	verified_fetched
SRC-TN-041	Civilization VI (Wikipedia)	secondary	2026-09-11	Game article: release, developer, platforms	verified_fetched
SRC-TN-042	The Witcher 3: Wild Hunt (Wikipedia)	secondary	2026-09-11	Game article: release, developer, engine and platform notes	verified_fetched
SRC-TN-043	Grand Theft Auto V (Wikipedia)	secondary	2026-09-11	Game article: PC release and engine notes	verified_fetched
SRC-TN-044	Doom (2016 video game)	secondary	2026-09-11	Game article including the development/technology notes on t	verified_fetched
SRC-TN-045	Doom Eternal (Wikipedia)	secondary	2026-09-11	Game article including technology notes on the id Tech 7 ren	verified_fetched
SRC-TN-046	Battlefield V (Wikipedia)	secondary	2026-09-11	Game article including the ray tracing / DXR section	verified_fetched
SRC-TN-047	Metro Exodus (Wikipedia)	secondary	2026-09-11	Game article including the ray tracing technology section	verified_fetched
SRC-TN-048	Batman: Arkham Asylum (Wikipedia)	secondary	2026-09-11	Game article including the PhysX technology notes	verified_fetched
SRC-TN-049	Mafia II (Wikipedia)	secondary	2026-09-11	Game article including the PhysX technology notes	verified_fetched
SRC-TN-050	Team Fortress 2 (Wikipedia)	secondary	2026-09-11	Game article including engine (Source) and platform notes	verified_fetched
SRC-TN-051	Counter-Strike 2 (Wikipedia)	secondary	2026-09-11	Game article including engine (Source 2) and release/platfor	verified_fetched
SRC-TN-052	Fortnite (Wikipedia)	secondary	2026-09-11	Game article including engine (Unreal Engine) and mode (Batt	verified_fetched
SRC-TN-053	V Rising (Wikipedia)	secondary	2026-09-11	Game article: developer (Stunlock Studios), engine (Unity),	verified_fetched
SRC-TN-054	Ratchet & Clank: Rift Apart (Wikipedia)	secondary	2026-09-11	Game article including the PC port technology notes	verified_fetched
SRC-TN-055	Forspoken (Wikipedia)	secondary	2026-09-11	Game article including the PC technology/streaming notes	verified_fetched
SRC-TN-056	Portal 2 (Wikipedia)	secondary	2026-09-11	Game article: release, developer, engine and portal gameplay	verified_fetched
SRC-WRS-001	World Partition (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '# World Partition', '## Enabling World Partition'	verified
SRC-WRS-002	World Partition - Data Layers in Unreal Engine	secondary	2026-09-10	Sections: '# Data Layers', '## Creating Data Layers', '## ..	verified
SRC-WRS-003	One File Per Actor in Unreal Engine	secondary	2026-09-10	Sections: '# One File Per Actor', '## Enabling One File Per	verified
SRC-WRS-004	World Partition - Hierarchical Level of Detail in Unreal Engine	secondary	2026-09-10	Sections: '# World Partition - Hierarchical Level of Detail'	verified
SRC-WRS-005	Large World Coordinates in Unreal Engine 5	secondary	2026-09-10	Sections: '# Large World Coordinates' (marked Beta), '## Upg	verified
SRC-WRS-006	Large World Coordinates Rendering in Unreal Engine 5	secondary	2026-09-10	Sections: '# Large World Coordinates Rendering', '## ...HLSL	verified
SRC-WRS-007	Nanite Virtualized Geometry Overview (Unreal Engine 5.8 Documentation)	secondary	2026-09-10	Sections: '## How does Nanite work?', '## Benefits of Nanite	verified

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-WRS-008	Virtual Texturing in Unreal Engine	secondary	2026-09-10	Sections: '# Virtual Texturing', '## Virtual Texturing Metho	verified
SRC-WRS-009	Streaming Virtual Texturing in Unreal Engine	secondary	2026-09-10	Sections: '# Streaming Virtual Texturing', '## Enabling Stre	verified
SRC-WRS-010	Runtime Virtual Texturing in Unreal Engine	secondary	2026-09-10	Sections: '# Runtime Virtual Texturing', '## Setting Up Run	verified
SRC-WRS-011	Virtual Texture Memory Pools in Unreal Engine	secondary	2026-09-10	Sections: '# Virtual Texture Memory Pools', '## Physical Mem	verified
SRC-WRS-012	Texture Streaming in Unreal Engine	secondary	2026-09-10	Sections: '# Texture Streaming', '## ...Mip-based streaming'	verified
SRC-WRS-013	Mesh Distance Fields in Unreal Engine	secondary	2026-09-10	Sections: '# Mesh Distance Fields', '## How does it work?'	verified
SRC-WRS-014	Distance Field Soft Shadows in Unreal Engine	secondary	2026-09-10	Sections: '# Distance Field Soft Shadows', '## Setting Up Di	verified
SRC-WRS-015	Precomputed Visibility Volumes in Unreal Engine	secondary	2026-09-10	Sections: '# Precomputed Visibility Volumes', '## Setup and	verified
SRC-WRS-016	Visibility and Occlusion Culling in Unreal Engine	secondary	2026-09-10	Sections: '# Visibility and Occlusion Culling', '## Culling	verified
SRC-WRS-017	Landscape Technical Guide in Unreal Engine	secondary	2026-09-10	Sections: '# Landscape Technical Guide', '## Landscape Compo	verified
SRC-WRS-018	Asynchronous Asset Loading in Unreal Engine	secondary	2026-09-10	Sections: '# Asynchronous Asset Loading', '## FSoftObjectPat	verified
SRC-WRS-019	Oodle Data (Unreal Engine Documentation)	secondary	2026-09-10	Sections: '# Oodle Data', '## Key Concepts for Oodle Data'	verified
SRC-WRS-020	Cooking and Chunking in Unreal Engine	secondary	2026-09-10	Sections: '# Cooking and Chunking', '## ...Chunking', '## ..	verified
SRC-WRS-021	Foliage Mode in Unreal Engine	secondary	2026-09-10	Sections: '# Foliage Mode', '## Painting Foliage', '## Folia	verified
SRC-WRS-022	Instanced Static Mesh in Unreal Engine	secondary	2026-09-10	Sections: '# Instanced Static Mesh', '## ...Creating an ISM'	verified
SRC-WRS-023	Hierarchical Instanced Static Mesh in Unreal Engine	secondary	2026-09-10	Sections: '# Hierarchical Instanced Static Mesh', '## ...Clu	verified
SRC-WRS-024	Automatic Static Mesh LOD Generation in Unreal Engine	secondary	2026-09-10	Sections: '# Automatic Static Mesh LOD Generation', '## ...R	verified
SRC-WRS-025	Procedural Content Generation Overview (Unreal Engine)	secondary	2026-09-10	Sections: '# Procedural Content Generation Overview', '## PC	verified
SRC-WRS-026	Nanite: A Deep Dive (SIGGRAPH 2021, Advances in Real-Time Rendering in	secondary	2026-09-10	Slides: p.16 'Triangle cluster culling'; p.17 'Occlusion cul	verified
SRC-WRS-027	GPU-Driven Rendering Pipelines (SIGGRAPH 2015, Advances in Real-Time R	secondary	2026-09-10	Slides: p.9 AC Unity '~10x instances compared to previous As	verified
SRC-WRS-028	GPU Gems 2, Chapter 2: Terrain Rendering Using GPU-Based Geometry Clip	secondary	2026-09-10	Sections: '2.1 Review of Geometry Clipmaps', '2.2 Overview o	verified
SRC-WRS-029	GPU Gems 3, Chapter 4: Next-Generation SpeedTree Rendering	secondary	2026-09-10	Sections: 'Chapter 4. Next-Generation SpeedTree Rendering',	verified
SRC-WRS-030	Chunked LOD (SIGGRAPH 2002 'Super-size it! Scaling up to Massive Virtu	secondary	2026-09-10	Sections: 'Chunked LOD' overview, 'News' dated entries (23 J	verified
SRC-WRS-031	C-BDAM - Compressed Batched Dynamic Adaptive Meshes for Terrain Render	secondary	2026-09-10	p.1 abstract and 'Contribution' / 'Advantages' sections; Fig	verified

Код	Название	Тип	Проверен	Локатор	Доступность
SRC-WRS-032	Sparse Virtual Textures (GDC 2008 talk, slides + public-domain demo so	secondary	2026-09-10	Page sections: '# Sparse Virtual Textures' (technique descri	verified
SRC-WRS-033	Software Virtual Textures	secondary	2026-09-10	Paper: page table and physical page cache architecture, virt	verified
SRC-WRS-034	DirectStorage (Win32 apps) - Microsoft Learn	secondary	2026-09-10	Sections: '# DirectStorage' ('intended to allow games to mak	verified
SRC-WRS-035	Using DirectStorage - Microsoft Learn	secondary	2026-09-10	Sections: 'Using DirectStorage' walkthrough: queue creation,	verified
SRC-WRS-036	microsoft/DirectStorage (GitHub) - GDeflate README	secondary	2026-09-10	Sections: '# GDeflate' description of the bitstream ('design	verified
SRC-WRS-037	Tiled resources (Direct3D 11.2) - Microsoft Learn	api_specification	2026-09-10	Sections: '# Tiled resources', '## Why use tiled resources?'	verified
SRC-WRS-038	Terrain Rendering in 'Far Cry 5' (GDC 2018)	secondary	2026-09-10	GDC Vault session page abstract; the talk's described struct	verified
SRC-WRS-039	GPU-Based Procedural Placement in Horizon Zero Dawn (GDC 2017)	secondary	2026-09-10	Studio page abstract: 'Jaap van Muijden describes the GPU ba	verified
SRC-WRS-040	Continuous World Generation in 'No Man's Sky' (GDC 2017)	secondary	2026-09-10	GDC Vault session page abstract ('Giving insight into the ch	verified
SRC-WRS-041	Chunk format - Minecraft Wiki	secondary	2026-09-10	Sections: '# Chunk format', '## NBT structure' (chunk = 16x3	verified
SRC-WRS-042	Chunk - Minecraft Wiki	secondary	2026-09-10	Sections: '# Chunk' (definition: 16x16 columns of blocks ext	verified
SRC-WRS-043	Simulation distance - Minecraft Wiki	secondary	2026-09-10	Sections: '# Simulation distance' (definition and its separa	verified
SRC-WRS-044	meshoptimizer (GitHub README)	secondary	2026-09-10	README sections: '# Vertex cache optimization' ('The worst-c	verified
SRC-WRS-045	Random-Access Neural Compression of Material Textures (project page, S	secondary	2026-09-10	Page sections: title/authors ('Accepted to Siggraph 2023'),	verified
SRC-WRS-046	Random-Access Neural Compression of Material Textures (author's versio	secondary	2026-09-10	Paper sections: abstract, network architecture and per-mater	verified
SRC-WRS-047	NVIDIA-RTX/RTXNTC: NVIDIA Neural Texture Compression SDK	secondary	2026-09-10	README sections: '# Neural Texture Compression' ('NTC is an	verified
SRC-WRS-048	Octahedral Impostors	secondary	2026-09-10	Article: octahedral mapping of view direction to atlas locat	verified
SRC-WRS-049	Large world coordinates - Godot Engine (4.4) documentation	secondary	2026-09-10	Sections: '# Large world coordinates', '## Why use large wor	verified
SRC-WRS-050	Updating Game Build (Steamworks Documentation)	secondary	2026-09-10	Sections: 'Updating Game Build' ('To update your game or sof	verified (page returns 200; fu
SRC-WRS-051	Building Worlds Using Math(s) (GDC 2017, No Man's Sky)	secondary	2026-09-10	GDC Vault session page abstract (procedural generation of a	verified
SRC-WRS-052	DirectStorage GpuDecompressionBenchmark sample (GitHub)	secondary	2026-09-10	README: 'This sample provides a quick way to see the DirectS	verified
SRC-WRS-053	SteamPipe / Uploading to Steam (Steamworks Documentation)	secondary	2026-09-10	Sections: 'SteamPipe' ('SteamPipe initially splits each file	verified (page returns 200; fu
SRC-WRS-054	DirectStorage API Now Available on PC (DirectX Developer Blog)	secondary	2026-09-10	'# DirectStorage API Now Available on PC': 'Starting today,	verified
SRC-WRS-055	meshoptimizer users (maintainer's list of commercial games and engines	secondary	2026-09-10	'Games' list (Alan Wake 2, Baldur's Gate 3, Counter-Strike 2	verified

Код	Название	Тип	Проверен	Локатор	Доступность
STANDARD_VULKAN_SPEC	Vulkan 1.3 Extensions Specification	standard	2026-09-10	chapters: device features, queues, synchronization	available
TECH_CAPTURE	Epic: Scene Capture	official_documentation	2026-09-10	overview page	available
TECH_CLOTH	Unity 6: Cloth	official_documentation	2026-09-10	overview page	available
TECH_DESTRUCTION	Epic: Destruction Overview	official_documentation	2026-09-10	overview page	available
TECH_HAIR	Epic: Hair Rendering and Simulation	official_documentation	2026-09-10	overview page	available
TECH_REVERB	Epic: Convolution Reverb	official_documentation	2026-09-10	overview page	available
TECH_SECURITY	Epic: Anti-Cheat Interfaces	official_documentation	2026-09-10	overview page	available
UE_ANIMBUDGET	Unreal Engine: Animation Budget Allocator	official_documentation	2026-09-10	overview page	available
UE_CHAOS	Unreal Engine: Chaos Physics	official_documentation	2026-09-10	overview page	available
UE_CITY_SAMPLE	City Sample Project Unreal Engine Demonstration	official_case_study	2026-09-10	sections: World Partition, Nanite Virtualized Geometry, Mass	available
UE_CITY_SAMPLE_PCG	City Sample PCG for Unreal Engine	official_documentation	2026-09-10	section: procedural city and PCG graph examples	available
UE_DISTANCE_SHADOWS	Unreal Engine: Using Distance Field Shadows	official_documentation	2026-09-10	overview page	available
UE_FEATURE_MATRIX	Supported Features by Rendering Path for Desktop	official_documentation	2026-09-10	table: Supported Features by Rendering Path	available
UE_HLOD	Unreal Engine: Hierarchical Level of Detail	official_documentation	2026-09-10	overview page	available
UE_INSIGHTS	Unreal Engine: Unreal Insights	official_documentation	2026-09-10	overview page	available
UE_ISM	Unreal Engine: Instanced Static Mesh	official_documentation	2026-09-10	overview page	available
UE_LOD	Unreal Engine: Level of Detail	official_documentation	2026-09-10	overview page	available
UE_LUMEN	Unreal Engine: Lumen Global Illumination and Reflections	official_documentation	2026-09-10	section: Lumen Global Illumination and Reflections	available
UE_LWC	Unreal Engine: Large World Coordinates	official_documentation	2026-09-10	overview page	available
UE_MASS	Unreal Engine: Mass Entity	official_documentation	2026-09-10	overview page	available
UE_MOTION_MATCHING	Motion Matching in Unreal Engine	official_documentation	2026-09-10	overview page	available
UE_NANITE	Unreal Engine: Nanite Virtualized Geometry	official_documentation	2026-09-10	section: Nanite Virtualized Geometry overview	available
UE_NAVMESH	Unreal Engine: Navigation Mesh	official_documentation	2026-09-10	overview page	available
UE_NETWORKING	Unreal Engine: Networking and Multiplayer	official_documentation	2026-09-10	overview page	available
UE_NIAGARA	Unreal Engine: Niagara Visual Effects	official_documentation	2026-09-10	overview page	available
UE_REPGRAPH	Unreal Engine: Replication Graph	official_documentation	2026-09-10	overview page	available
UE_SAVEGAME	Unreal Engine: Saving and Loading Your Game	official_documentation	2026-09-10	overview page	available
UE_SCALABILITY	Unreal Engine: Scalability Reference	official_documentation	2026-09-10	overview page	available
UE_SIGNIFICANCE	Unreal Engine: Significance Manager	official_documentation	2026-09-10	overview page	available
UE_VIRTUALTEXTURING	Unreal Engine: Virtual Texturing	official_documentation	2026-09-10	overview page	available
UE_VSM	Unreal Engine: Virtual Shadow Maps	official_documentation	2026-09-10	section: Virtual Shadow Maps	available
UE_WORLDPARTITION	Unreal Engine: World Partition	official_documentation	2026-09-10	sections: World Partition, Streaming Sources	available

Код	Название	Тип	Проверен	Локатор	Доступность
UNITY_ADDRESSABLES	Unity Manual: Addressables	official_documentation	2026-09-10	section: Addressables overview	available
UNITY_DOTS_PRODUCTION	Unity DOTS - Data-Oriented Technology Stack	official_case_index	2026-09-10	section: DOTS in Production	available
UNITY_DRAW_CALLS	Unity Manual: Choose a method for optimizing draw calls	official_documentation	2026-09-10	overview page	available
UNITY_ENTITIES	Unity Manual: Entities (DOTS)	official_documentation	2026-09-10	section: Entities overview	available
UNITY_GC_BEST_PRACTICES	Unity Manual: Garbage collection best practices	official_documentation	2026-09-10	overview page	available
UNITY_GFX_PERF	Unity Manual: Optimizing Graphics Performance	official_documentation	2026-09-10	overview page	available
UNITY_GPU_BUDGETS	Unity — Managing GPU usage for PC and console games	official_documentation	2026-09-10	overview page	available
UNITY_INSTANCING	Unity Manual: GPU Instancing	official_documentation	2026-09-10	overview page	available
UNITY_JOBS	Unity Manual: Job System	official_documentation	2026-09-10	section: Job System overview	available
UNITY_LIGHTMAPPER	Unity Manual: Progressive Lightmapper	official_documentation	2026-09-10	overview page	available
UNITY_LIGHTMAPUV	Unity Manual: Generating Lightmapping UVs	official_documentation	2026-09-10	overview page	available
UNITY_LIGHTPROBES	Unity Manual: Light Probes	official_documentation	2026-09-10	overview page	available
UNITY_NETCODE	Unity Manual: Netcode	official_documentation	2026-09-10	section: Netcode for Entities overview	available
UNITY_OCCLUSION	Unity Manual: Occlusion Culling	official_documentation	2026-09-10	overview page	available
UNITY_PROFILER	Unity Manual: Profiler	official_documentation	2026-09-10	overview page	available
UNITY_QUALITY	Unity Manual: Quality Settings	official_documentation	2026-09-10	overview page	available
UNITY_SHADERLOAD	Unity Manual: Optimizing Shader Load Time	official_documentation	2026-09-10	overview page	available
UNITY_SRP_BATCHER	Unity Manual: SRP Batchers	official_documentation	2026-09-10	overview page	available
UNITY_TEXTURE_STREAMING	Unity Manual: Texture Streaming	official_documentation	2026-09-10	overview page	available
VALVE_DIRECTOR	The AI Systems of Left 4 Dead — Michael Booth, Valve	secondary	2026-09-10	overview page	available
VALVE_REWIND	Valve Source SDK — player_lagcompensation.cpp	open_source	2026-09-10	overview page	available
VALVE_SUBTICK	Counter-Strike 2 — sub-tick updates, Valve	engineering_article	2026-09-10	overview page	available
WIKI_AO	Ambient occlusion	secondary	2026-09-10	overview page	available
WIKI_ATLAS	Texture atlas	secondary	2026-09-10	overview page	available
WIKI_BEHAVIOR_TREE	Behavior tree (artificial intelligence, robotics and control)	secondary	2026-09-10	overview page	available
WIKI_BVH	Bounding volume hierarchy	secondary	2026-09-10	overview page	available
WIKI_DEFERRED	Deferred shading	secondary	2026-09-10	overview page	available
WIKI_DELTA	Delta encoding	secondary	2026-09-10	overview page	available
WIKI_DLSS	Deep learning super sampling	secondary	2026-09-10	overview page	available
WIKI_DOD	Data-oriented design	secondary	2026-09-10	overview page	available
WIKI_ECS	Entity component system	secondary	2026-09-10	overview page	available
WIKI_FFT	Fast Fourier transform	secondary	2026-09-10	overview page	available
WIKI_FSM	Finite-state machine	secondary	2026-09-10	overview page	available
WIKI_FSR	FidelityFX Super Resolution	secondary	2026-09-10	overview page	available
WIKI_GAME_AI	Video game artificial intelligence	secondary	2026-09-10	overview page	available
WIKI_GERSTNER	Gerstner wave	secondary	2026-09-10	overview page	available
WIKI_GLOBAL_ILLUMINATION	Global illumination	secondary	2026-09-10	overview page	available
WIKI_HSR	Hidden-surface determination	secondary	2026-09-10	overview page	available
WIKI_IK	Inverse kinematics	secondary	2026-09-10	overview page	available
WIKI_IMPOSTOR	Impostor (computer graphics)	secondary	2026-09-10	overview page	available
WIKI_LIGHTMAP	Lightmap	secondary	2026-09-10	overview page	available

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WIKI_LOD	Level of detail (computer graphics)	secondary	2026-09-10	overview page	available
WIKI_MIPMAP	Mipmap	secondary	2026-09-10	overview page	available
WIKI_NAVMESH	Navigation mesh	secondary	2026-09-10	overview page	available
WIKI_OBJECTPOOL	Object pool pattern	secondary	2026-09-10	overview page	available
WIKI_PARTICLES	Particle system	secondary	2026-09-10	overview page	available
WIKI_PATH_TRACING	Path tracing	secondary	2026-09-10	overview page	available
WIKI_PREDICTION	Client-side prediction	secondary	2026-09-10	overview page	available
WIKI_PROCEDURAL	Procedural generation	secondary	2026-09-10	overview page	available
WIKI_RAGDOLL	Ragdoll physics	secondary	2026-09-10	overview page	available
WIKI_RAY_TRACING	Ray tracing (graphics)	secondary	2026-09-10	overview page	available
WIKI_RIGID_BODY	Rigid body dynamics	secondary	2026-09-10	overview page	available
WIKI_SDF	Signed distance function	secondary	2026-09-10	overview page	available
WIKI_SHADOWMAP	Shadow mapping	secondary	2026-09-10	overview page	available
WIKI_SKINNING	Skinning	secondary	2026-09-10	overview page	available
WIKI_SVO	Sparse voxel octree	secondary	2026-09-10	overview page	available
WIKI_TAA	Temporal anti-aliasing	secondary	2026-09-10	overview page	available
WIKI_VEHICLE_DYNAMICS	Vehicle dynamics	secondary	2026-09-10	overview page	available
WIKI_VOLUMETRIC	Volumetric rendering	secondary	2026-09-10	overview page	available
hardware:00acf849ec56	PassMark Video Card Benchmarks — Radeon RX 7900 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:04891bb1e6c7	PassMark Intel Core i7-11800H Benchmark — CPU Mark 19596, Single 3007	hardware_benchmark	2026-09-10	benchmark page	available
hardware:05e7e544759d	PassMark Video Card Benchmarks — GeForce RTX 3050 6GB Laptop GPU	hardware_benchmark	2026-09-10	benchmark page	available
hardware:064fe7ca5e68	PassMark CPU Benchmarks — AMD Ryzen 5 5500	hardware_benchmark	2026-09-10	benchmark page	available
hardware:06d672d982fc	PassMark CPU Benchmarks — Intel Core i7-9700K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:080bb8b4f6d1	PassMark Intel Core i7-13700H Benchmark — CPU Mark 25860, Single 3552	hardware_benchmark	2026-09-10	benchmark page	available
hardware:0ae6626d2565	PassMark Video Card Benchmarks — GeForce RTX 5050 Laptop GPU	hardware_benchmark	2026-09-10	benchmark page	available
hardware:0b39637c4ec8	PassMark Video Card Benchmarks — GeForce RTX 2060	hardware_benchmark	2026-09-10	benchmark page	available
hardware:0c09a16c68f9	PassMark AMD Ryzen 5 7500F Benchmark — CPU Mark 26537, Single 3825	hardware_benchmark	2026-09-10	benchmark page	available
hardware:0ff0fa934e0c	PassMark CPU Benchmarks — Intel Core i3-10100	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1059383223be	PassMark Intel Core i5-11400F Benchmark — CPU Mark 16869, Single 2979	hardware_benchmark	2026-09-10	benchmark page	available
hardware:13a9cace366f	PassMark - Radeon Vega 3 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1440e3fdc3aa	PassMark Video Card Benchmarks — GeForce GTX 1650	hardware_benchmark	2026-09-10	benchmark page	available
hardware:15733d23becf	PassMark CPU Benchmarks — Intel Core i5-10400F	hardware_benchmark	2026-09-10	benchmark page	available
hardware:15b098956b50	PassMark AMD Ryzen 9 5950X Benchmark — CPU Mark 45259, Single 3476	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1655dac07397	PassMark Video Card Benchmarks — GeForce GTX 1050 Ti	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1663f825c7f5	PassMark Video Card Benchmarks — GeForce RTX 3070	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1997146c058f	PassMark Video Card Benchmarks — GeForce RTX 2070 SUPER	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1adf7bbcc286	PassMark CPU Benchmarks — AMD Ryzen 5 3600	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1c641b1616ea	PassMark CPU Benchmarks — AMD Ryzen 3 1200	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1cd7069270e3	PassMark Video Card Benchmarks — GeForce RTX 3050 8GB	hardware_benchmark	2026-09-10	benchmark page	available
hardware:1f99ebff4ca8	PassMark CPU Benchmarks — AMD Ryzen 9 5900X	hardware_benchmark	2026-09-10	benchmark page	available

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hardware:213472decf76	PassMark - Radeon 780M - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:22f84720ecb7	PassMark Video Card Benchmarks — GeForce GTX 1660 Ti	hardware_benchmark	2026-09-10	benchmark page	available
hardware:23eb35a6cdef	PassMark Video Card Benchmarks — Radeon RX 580	hardware_benchmark	2026-09-10	benchmark page	available
hardware:24823f308139	PassMark - Radeon RX 7700 XT - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:2e59fdf3fded	PassMark - GeForce GT 1030 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:2f0ce84b7851	PassMark - GeForce GTX 1050 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:2f4c6f21f764	PassMark Intel Core i7-10700K Benchmark — CPU Mark 18501, Single 3036	hardware_benchmark	2026-09-10	benchmark page	available
hardware:32c454082048	PassMark - GeForce RTX 2080 SUPER - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:32c984630655	PassMark - Intel UHD Graphics 630 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:39023f2f2854	PassMark CPU Benchmarks — Intel Core i7-13700K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:3981b71516d5	PassMark - GeForce RTX 2080 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:3a1abc3aed60	PassMark Video Card Benchmarks — Radeon RX 6650 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:3a9a732e872b	PassMark Video Card Benchmarks — GeForce RTX 5060	hardware_benchmark	2026-09-10	benchmark page	available
hardware:3c9600f567cf	PassMark Video Card Benchmarks — GeForce RTX 5060 Ti 16GB	hardware_benchmark	2026-09-10	benchmark page	available
hardware:4089384ecd15	PassMark Video Card Benchmarks — GeForce RTX 5070 Ti	hardware_benchmark	2026-09-10	benchmark page	available
hardware:485542379b36	PassMark CPU Benchmarks — Intel Core i5-8400	hardware_benchmark	2026-09-10	benchmark page	available
hardware:497f54760950	PassMark Intel Core i5-13400F Benchmark — CPU Mark 24891, Single 3628	hardware_benchmark	2026-09-10	benchmark page	available
hardware:509fd48b901b	PassMark Video Card Benchmarks — GeForce RTX 3090	hardware_benchmark	2026-09-10	benchmark page	available
hardware:549aac87930e	PassMark - Radeon RX 550 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:54a1d0ca2092	PassMark Video Card Benchmarks — GeForce GTX 1660	hardware_benchmark	2026-09-10	benchmark page	available
hardware:5530b216b0c2	PassMark Video Card Benchmarks — Radeon RX 7800 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:560fb8a216cf	PassMark CPU Benchmarks — AMD Ryzen 7 3700X	hardware_benchmark	2026-09-10	benchmark page	available
hardware:56cc05a182df	PassMark AMD Ryzen 5 5600 Benchmark — CPU Mark 21494, Single 3253	hardware_benchmark	2026-09-10	benchmark page	available
hardware:572e0cbae2d3	PassMark - Intel HD 4600 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:5837988cc683	PassMark Video Card Benchmarks — GeForce RTX 4060	hardware_benchmark	2026-09-10	benchmark page	available
hardware:5946562912cb	PassMark Video Card Benchmarks — GeForce RTX 5080	hardware_benchmark	2026-09-10	benchmark page	available
hardware:5a14b3ed7af2	PassMark - GeForce GTX 750 Ti - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:62699f3b35fb	PassMark Video Card Benchmarks — GeForce GTX 1060	hardware_benchmark	2026-09-10	benchmark page	available
hardware:67dc4c02a2fb	PassMark CPU Benchmarks — AMD Ryzen 9 3900X	hardware_benchmark	2026-09-10	benchmark page	available
hardware:67de7e601b40	PassMark - GeForce RTX 2080 Ti - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:6aa408b0f708	PassMark Intel Core i7-12700H Benchmark — CPU Mark 24995, Single 3484	hardware_benchmark	2026-09-10	benchmark page	available
hardware:6d93777fa7453	PassMark Video Card Benchmarks — GeForce RTX 4050 Laptop GPU	hardware_benchmark	2026-09-10	benchmark page	available
hardware:6f7108f11cd3	PassMark Video Card Benchmarks — Radeon RX 470/570	hardware_benchmark	2026-09-10	benchmark page	available

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hardware:70abaad3db6f	PassMark - GeForce GTX 970 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:70c608c01ca0	PassMark Video Card Benchmarks — Radeon RX 9070	hardware_benchmark	2026-09-10	benchmark page	available
hardware:7591010c1bc0	PassMark CPU Benchmarks — Intel Core i9-14900K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:77f7921c463e	PassMark CPU Benchmarks — AMD Ryzen 7 7700X	hardware_benchmark	2026-09-10	benchmark page	available
hardware:7888397167eb	PassMark Video Card Benchmarks — GeForce RTX 5090	hardware_benchmark	2026-09-10	benchmark page	available
hardware:79f489f9e9ec	PassMark CPU Benchmarks — Intel Core i7-8700K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:7e2d8d5bf5f3	PassMark Intel Core i5-14400F Benchmark — CPU Mark 25440, Single 3700	hardware_benchmark	2026-09-10	benchmark page	available
hardware:81c4287af43b	PassMark - Intel Iris Xe - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:836a9fba84c4	PassMark CPU Benchmarks — AMD Ryzen 9 7900X	hardware_benchmark	2026-09-10	benchmark page	available
hardware:840ef9ea050b	PassMark - GeForce GT 730 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:848d487dc667	PassMark AMD Custom APU 0932 (Steam Deck) Benchmark — CPU Mark 9411, S	hardware_benchmark	2026-09-10	benchmark page	available
hardware:8660240c2073	PassMark Video Card Benchmarks — GeForce RTX 4090	hardware_benchmark	2026-09-10	benchmark page	available
hardware:890bf8c8bc53	PassMark Video Card Benchmarks — Radeon RX 6800 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:89cbc8278eca	PassMark CPU Benchmarks — AMD Ryzen 5 5600X	hardware_benchmark	2026-09-10	benchmark page	available
hardware:8ab47fd159d4	PassMark CPU Benchmarks — Intel Core i9-12900K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:8bacc05d7fc8	PassMark Video Card Benchmarks — GeForce RTX 4070	hardware_benchmark	2026-09-10	benchmark page	available
hardware:8d936bf26c17	PassMark Video Card Benchmarks — GeForce RTX 5070 Ti Laptop GPU	hardware_benchmark	2026-09-10	benchmark page	available
hardware:8e7744a9be93	PassMark CPU Benchmarks — AMD Ryzen 9 7950X	hardware_benchmark	2026-09-10	benchmark page	available
hardware:93986f7bde58	PassMark Video Card Benchmarks — GeForce RTX 3060	hardware_benchmark	2026-09-10	benchmark page	available
hardware:9a24b670ca22	PassMark - Intel HD Graphics 620 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:9b457cb5c826	PassMark CPU Benchmarks — AMD Ryzen 7 5800X3D	hardware_benchmark	2026-09-10	benchmark page	available
hardware:9b567a14b843	PassMark Video Card Benchmarks — GeForce RTX 2050	hardware_benchmark	2026-09-10	benchmark page	available
hardware:9b7841f454f7	PassMark AMD Ryzen 7 9800X3D Benchmark — CPU Mark 39927, Single 4421	hardware_benchmark	2026-09-10	benchmark page	available
hardware:9c5901396f3d	PassMark Video Card Benchmarks — GeForce GTX 1660 SUPER	hardware_benchmark	2026-09-10	benchmark page	available
hardware:9f4761331ae8	PassMark Video Card Benchmarks — GeForce RTX 5070 Laptop GPU	hardware_benchmark	2026-09-10	benchmark page	available
hardware:9fd08f6b5c8a	PassMark AMD Ryzen 7 8745H Benchmark — CPU Mark 29058, Single 3675	hardware_benchmark	2026-09-10	benchmark page	available
hardware:a195c699bec6	PassMark CPU Benchmarks — Intel Core i5-9400F	hardware_benchmark	2026-09-10	benchmark page	available
hardware:a3ccb503c8b3	PassMark CPU Benchmarks — Intel Core i9-13900K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:a4f5c1213fb9	PassMark - Intel HD 520 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:a565eaa9a35f	PassMark CPU Benchmarks — Intel Core i7-12700K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:a9dda230b9dd	PassMark CPU Benchmarks — AMD Ryzen 5 2600	hardware_benchmark	2026-09-10	benchmark page	available
hardware:a9f7d96a856b	PassMark - GeForce GTX 960 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:ab9c0bf794b3	PassMark Video Card Benchmarks — GeForce RTX 3050 Laptop GPU	hardware_benchmark	2026-09-10	benchmark page	available

Код	Название	Тип	Проверен	Локатор	Доступность
hardware:ae2fbd7314d	PassMark Video Card Benchmarks — Radeon RX 7900 XTX	hardware_benchmark	2026-09-10	benchmark page	available
hardware:b01112378bcb	PassMark Video Card Benchmarks — GeForce RTX 5070	hardware_benchmark	2026-09-10	benchmark page	available
hardware:b4f43214ca49	PassMark - Radeon RX 6750 XT - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:b8eca4bb9b58	PassMark Video Card Benchmarks — GeForce RTX 5060 Laptop GPU	hardware_benchmark	2026-09-10	benchmark page	available
hardware:b95413f9cceb	PassMark Video Card Benchmarks — Radeon RX 7600	hardware_benchmark	2026-09-10	benchmark page	available
hardware:bbbc1eae1809	PassMark CPU Benchmarks — Intel Core i7-11700K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:bc7ca35ed973	PassMark Video Card Benchmarks — Intel Arc A750	hardware_benchmark	2026-09-10	benchmark page	available
hardware:bcf377fe5b92	PassMark AMD Ryzen 7 5700X Benchmark — CPU Mark 26561, Single 3386	hardware_benchmark	2026-09-10	benchmark page	available
hardware:bf9c50614fe8	PassMark - Radeon Vega 8 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:c04777fc31fd	PassMark Video Card Benchmarks — Intel Arc A770	hardware_benchmark	2026-09-10	benchmark page	available
hardware:c430a7d5ed8e	PassMark Video Card Benchmarks — Radeon RX 6600	hardware_benchmark	2026-09-10	benchmark page	available
hardware:c4f6951eccff	PassMark AMD Ryzen 5 7600 Benchmark — CPU Mark 26975, Single 3908	hardware_benchmark	2026-09-10	benchmark page	available
hardware:c6cc8afb6ebc	PassMark CPU Benchmarks — Intel Core i5-13600K	hardware_benchmark	2026-09-10	benchmark page	available
hardware:c7b836ca6f4e	PassMark - GeForce GTX 1080 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:c9938a31b45b	PassMark Video Card Benchmarks — GeForce RTX 4080	hardware_benchmark	2026-09-10	benchmark page	available
hardware:cd4a51d804a1	PassMark Intel Core i7-7700 Benchmark — CPU Mark 8640, Single 2441	hardware_benchmark	2026-09-10	benchmark page	available
hardware:ce50b64b1067	PassMark Video Card Benchmarks — GeForce RTX 5050	hardware_benchmark	2026-09-10	benchmark page	available
hardware:ceefabb71294	PassMark - Intel UHD Graphics 620 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d1496a7a3bae	PassMark Video Card Benchmarks — Radeon RX 9060 XT 16GB	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d1758b433170	PassMark - GeForce GTX 1070 Ti - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d23f22993b28	PassMark Video Card Benchmarks — Radeon RX 6700 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d271088a60c5	PassMark - GeForce GTX 1070 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d39e7ced2250	PassMark CPU Benchmarks — AMD Ryzen 5 7600X	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d4075fe02279	PassMark Intel Core i5-14600K Benchmark — CPU Mark 38402, Single 4267	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d55d08a9d426	PassMark AMD Ryzen 7 5700X3D Benchmark — CPU Mark 26302, Single 2968	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d5b349a295a2	PassMark Video Card Benchmarks — Radeon RX 5600 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:d91278fceb3	PassMark AMD Ryzen 7 7800X3D Benchmark — CPU Mark 34277, Single 3759	hardware_benchmark	2026-09-10	benchmark page	available
hardware:de567ffb48ae	PassMark Video Card Benchmarks — GeForce GTX 1080 Ti	hardware_benchmark	2026-09-10	benchmark page	available
hardware:df2b757155c9	PassMark Intel Core i5-12600K Benchmark — CPU Mark 27512, Single 3917	hardware_benchmark	2026-09-10	benchmark page	available
hardware:e97887202ea5	PassMark Video Card Benchmarks — Radeon RX 6500 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:eb15eee5c39e	PassMark CPU Benchmarks — Intel Core i5-12400F	hardware_benchmark	2026-09-10	benchmark page	available
hardware:ed4189e76f39	PassMark Video Card Benchmarks — GeForce RTX 3080	hardware_benchmark	2026-09-10	benchmark page	available

Код	Название	Тип	Проверен	Локатор	Доступность
hardware:ef05e429b7a4	PassMark CPU Benchmarks — AMD Ryzen 3 3100	hardware_benchmark	2026-09-10	benchmark page	available
hardware:ef3fed8afcb2	PassMark Video Card Benchmarks — Radeon RX 6900 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:f2d3732fd926	PassMark Intel Core i7-14700K Benchmark — CPU Mark 51958, Single 4456	hardware_benchmark	2026-09-10	benchmark page	available
hardware:f3d9708ed9f4	PassMark CPU Benchmarks — AMD Ryzen 5 1600	hardware_benchmark	2026-09-10	benchmark page	available
hardware:f3e9327afe75	PassMark CPU Benchmarks — Intel Core i3-8100	hardware_benchmark	2026-09-10	benchmark page	available
hardware:f5e8e9acc2ec	PassMark - Radeon RX 6800 - Price performance comparison	hardware_benchmark	2026-09-10	benchmark page	available
hardware:f84c1215e206	PassMark Video Card Benchmarks — Radeon RX 9070 XT	hardware_benchmark	2026-09-10	benchmark page	available
hardware:f991632d00d5	PassMark Video Card Benchmarks — GeForce RTX 3050 Ti Laptop GPU	hardware_benchmark	2026-09-10	benchmark page	available
hardware:fb999a0301a9	PassMark Video Card Benchmarks — GeForce RTX 3060 Ti	hardware_benchmark	2026-09-10	benchmark page	available

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